

Title: Technical Specification for MINI640 IDDCA

(with 15um pixel pitch and XD-2 Stirling cooler)

Reference: TS-MINI640-IDDCA-1801

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

TECHNICAL DATA PACKAGE

MINI640-XD-2

& COOLER DRIVE

ISSUES	EVENTS	DATES
1	Creation of the document	01/2018
2	Modification of Annex 7	04/2018
3	Modification of Annex 5	12/2018

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1. INTRODUCTION

1.1 Scope

This document is the Technical Data Package (TDP) of the Integrated Detector Dewar Cooler Assembly (IDDCA) referenced MINI640-XD-2 + COOLER DRIVE.

It provides all information necessary to handle, operate and maintain this IDDCA.



All the people who have to handle the IDDCA should know the cautions for handling it and the recommendations to operate it. Therefore it is strongly recommend that the users read this document before operating the IDDCA.

2. REFERENCED DOCUMENTS

Customer:

Country:

2.1 Contractual documents

- Purchase Order:
- IDDCA Technical specification: MINI640-XD-2
- COOLER DRIVE specification
- Acceptance Test Report of MINI640-XD-2
- Acceptance Test Report of COOLER DRIVE

2.2 NCRIO referenced documents

[REF 1]	General handling procedure of NCRIO detectors	See Annex 1
[REF 2]	Optical interfaces of MW IDDCAs	See Annex 2
[REF 3]	Electrical interface of the FPA for MINI640 products with ROIC	See Annex 3
[REF 4]	XD-2 Stirling-cycle microcooler ICD	See Annex 4
[REF 5]	MINI640 Electrical Interface Critical Document	See Annex 5
[REF 6]	Digital External Controller for XD-2 cooler, Interfaces Control Document	See Annex 6
[REF 7]	IDDCA drawings	See Annex 7

3. LIST OF ACRONYMS USE IN THIS DOCUMENT

ATR	Acceptance Test Report
MCT	Mercury Cadmium
Telluride ESD	Electro Static Discharge
FPA	Focal Plane Array
HgCdTe	Mercury Cadmium Telluride
I/O	Inputs / Outputs
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
ITR	Integration Then Read
IWR	Integration While Read
MC	Master Clock
MCT	Mercury Cadmium Telluride
MW	Mid Wave band = 3-5 μm
PV	Photovoltaic
ROIC	Read Out Integrated Circuit
TDP	Technical Data Package

Note:

The yellow warning sign  highlights an important note to be respected.

The red warning sign  highlights an important note that could lead to a permanent failure if not respected.

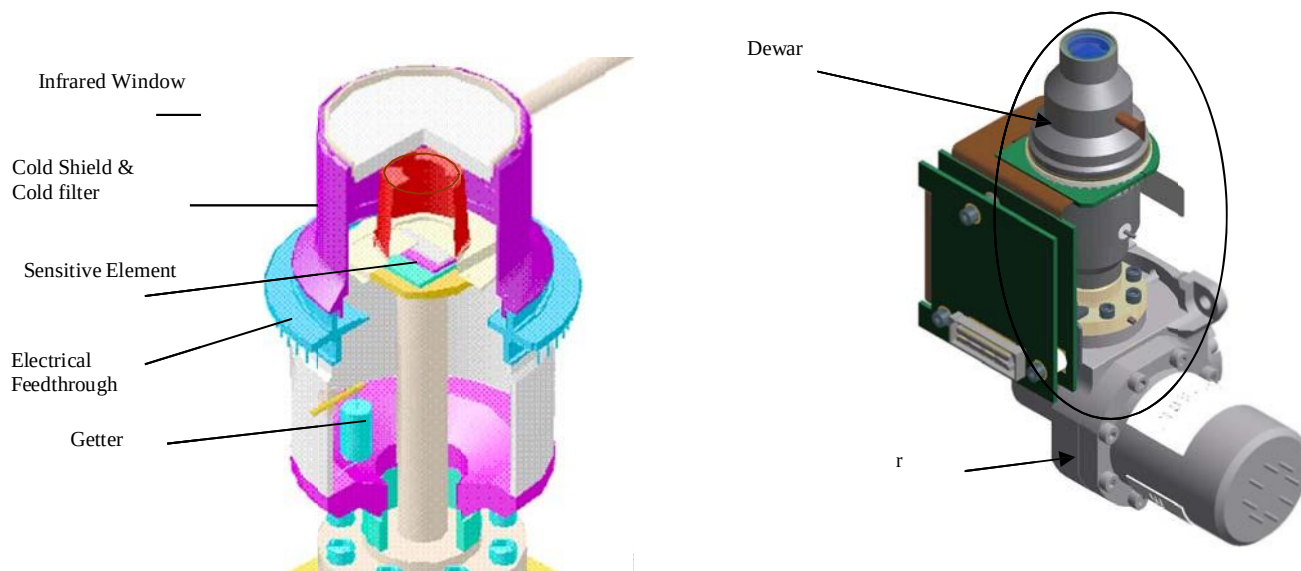
4. DESCRIPTION OF THE IDDCA

4.1 IDDCA general overview

The IDDCA is composed of the following main items:

- 640×512 IRCMOS IRFPA, 15 μm pixel pitch, with HgCdTe sensitive array,
- sealed metal outer dewar,
- getter device designed for long vacuum life,
- window to allow infrared radiation to penetrate into the vacuum sealed dewar
- cold filter to precisely define the incident radiation flux on the IRFPA,
- high efficiency cold shield system to define the field of view of the IRFPA and limit parasitic fluxes,
- temperature sensors at the level of the IRFPA, for monitoring and feed back to the cooler,
- a cold plate and a cold finger which is the thermal interface between the cryocooler and the dewar,
- A cryogenic Stirling-cycle microcooler referenced XD-2.

A general overview of the main parts of the IDDCA is presented in the following drawing:



Drawing not contractual: the actual shape of the delivered product may differ from this schematic drawing. Refer to the mechanical interface drawings for the guaranteed shape of the IDDCA

Figure 2.1: General overview of the IDDCA

4.2 Description of the dewar

Refer to NCRIEO generic document [REF 1] for the general description of the MINI640 Dewar.

4.3 Description of the FPA

Refer to NCRIEO generic document [REF 3] for the general description of the MINI640 or MINI640 FPA.

4.4 Description of the ROIC

Refer to NCRIEO generic document [REF 3] for the general description of the MINI640 ROIC functions.

4.5 Description of the cooler

Refer to NCRIEO generic document [REF 5] for the general description of the XD-2 Stirling-cycle cooler.

5. MECHANICAL INTERFACES

The general mechanical description and the outline dimensions of the IDDCA are presented in the IDDCA drawings given at the end of this document.

All dimensions are given in millimeters and at room temperature.

The MINI640 IDDCA has to be attached in the system by the cooler mounting flange which shall be the only mechanical interface for the system.



It is forbidden to apply any mechanical stress on any part of the detector except on the attachment plane defined on the interface drawings.

The system electronics is attached to the IDDCA electrical interface. This attachment shall not, in any case, be used as a mechanical attachment of the IDDCA, because

- it would create a hyper static fixation with the cooler mounting flange;
- it is not mechanically designed for this purpose.

Note: The Stirling-cycle engine of the cooler generates some vibrations that induce audible noise. Depending on the camera design, this noise can be amplified and/or could be a constraint for the system. In that case, the use of a “**noise pad**” may be a useful option. Such a noise pad is implemented between the cooler attachment plane and the camera structure, and acts as a noise dumper as well as a thermal heat sink. It can be

manufactured by the cooler manufacturer. Please contact NCRIEO for more detail.

6. ELECTRICAL INTERFACES

6.1 Proxy board electrical interface

The IDDCA is delivered with the PROXYLITE proxy board and a connecting flex, enabling to power the IDDCA and to multiplex and digitize the IDDCA analog outputs.

Refer to NCRIEO generic documents [REF 6] for the detailed PROXYLITE electrical interface and the associated cautions for use, to drive the delivered IDDCA.

6.2 Cooler electrical interface

Refer to NCRIEO generic document [REF 5] and [REF 7] for the detailed description of the XD-2 and cooler driving electronics electrical interfaces, and the associated cautions for use.

6.3 ROIC electrical interface

Refer to NCRIEO generic document [REF 3] for the detailed description of the MINI640 ROIC electrical interface, and the associated cautions for use.

However, this document is for information only, since all electrical interfaces are defined through the delivered proxy board: see § 6.1.

6.4 Temperature sensor electrical interface

Refer to NCRIEO generic document [REF 3] for the detailed description of the temperature sensor electrical interface.

6.5 IDDCA electrical interfaces

The IDDCA electrical interface is the ring of pins at feedthru ceramics level.

The IDDCA drawing provided in annex detail the mechanical definition and the electrical functions distribution of this electrical interface.

One will refer to NCRIEO generic document [REF 3] for the detail of each FPA function.

7. OPTICAL INTERFACES

Refer to NCRIEO generic document [REF 2] for the detailed description of the MINI640-XD-2 optical interface, and the associated cautions for use.

The window thickness is 1 mm and the window coating is according to §4.2.2 of [REF 2] (Antireflective coated infrared window type 1).

8. THERMAL INTERFACES

Refer to NCRIEO generic document [REF 5] for the detailed description of the MINI640-XD-2 thermal interface, and the associated cautions for use.

9. STORAGE AND MAINTENANCE

9.1 Storage

Refer to NCRIEO generic document [REF 1] for the main facts concerning storage conditions of the MINI640-XD-2 IDDCA, and the associated cautions for use.

9.2 Cooler maintenance

Refer to NCRIEO generic document [REF 5] for the main facts concerning cooler maintenance of the MINI640 IDDCA, and the associated cautions for use.

9.3 Specific mission profile and consequences on maintenance

Paragraph is Not Applicable since there is no Customer-furnished mission profile.

10. CAUTIONS FOR USE

10.1 General

The delivered infrared (IR) Integrated Detector Dewar Cooler Assembly (IDDCA) is a semiconductor optronic device which needs to be manipulated with precautions, due to the associated risks to damage it.

IR detector interfaces are defined at four levels: mechanical, electrical, optical, and thermal levels.

- The standard cautions for use concerning the IDDCA (packaging, storage, handling,...) are detailed in [REF 1].
- The standard cautions for use concerning the IDDCA optical interface are detailed in [REF 2].
- The standard cautions for use concerning the IDDCA electrical interface are detailed in [REF 3] for the FPA and in [REF 5] for the cooler.

10.2 Handling of the IDDCA

Refer to NCRIEO generic document [REF 1] for the handling recommendations for the IDDCA.

10.3 Powering of the IDDCA

10.3.1 Powering of the cooler

Refer to NCRIEO generic document [REF 5] and [REF 7] for the detailed description of the XD-2 and cooler driving electronics electrical interfaces, and the recommendations for powering it.

Refer to the FPA electrical interface in [REF 3] for details on the temperature sensor information, to feed the cooler with the temperature control loop bias.

The cooler shall be turned ON before powering the IDDCA.

10.3.2 Powering of the proxy board

The proxy board shall not be powered when connecting it to the IDDCA.

The FPA shall not be ON without the cooling system ON.

When turning ON the system, it is usually required to bias the FPA only when it is at low operating temperature (below 120K for instance).

When turning OFF the system or in case of cooler failure that would lead to the stop of cooling while the general system is still ON, it is mandatory to stop biasing the FPA to avoid damage due to power dissipation inside the vacuumed dewar. It is therefore recommended to drive the ON/OFF of the IRFPA biases with respect to the operating temperature below 120K for instance.

The PROXYLITE proxy board includes this ON/OFF threshold security.

Refer to the FPA electrical interface in [REF 3] for details on the temperature sensor information, to monitor the FPA temperature.

10.4 Others / specificities

Not Applicable.

List of Annexes to the TDP

ANNEX 1: GENERAL HANDLING PROCEDURE OF NCRIEO ETECTORS.

ANNEX 2: OPTICAL INTERFACES OF MW IDDCAS.

ANNEX 3: ELECTRICAL INTERFACE OF THE FPA FOR MINI640.

ANNEX 4: XD-2 STIRLING-CYCLE MICROCOOLER.

ANNEX 5: MINI640 ELECTRICAL INTERFACE CRITICAL DOCUMENT.

**ANNEX 6: DIGITAL EXTERNAL CONTROLLER FOR XD-2 COOLER INTERFACES
CONTROL DOCUMENT.**

ANNEX 7: IDDCA DRAWINGS

Note: these annexes shall be found attached in the next pages.

Annex 1

Title: General Handling Procedure of

MINI640 IDDCA

(with 15um pixel pitch and XD-2 Stirling cooler)

Reference: GHP-MINI640-IDDCA-1801

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

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1. INTRODUCTION

This technical document describes the general handling procedure of a NCRIEO Infrared (IR) Integrated Detector Dewar Cooler Assembly (IDDCa) and the main cautions to be considered when using it.

The delivered IDDCa is a semi-conductor photonic device which needs to be manipulated with precautions, due to the associated risks to damage it.

IR detector interfaces are defined at three levels: mechanical, optical, and electrical levels. Specific care must be taken for the handling of the detector, regarding optical, mechanical and electrical interfaces. For instance, electrostatic discharge (ESD) protection, limitation for maximum voltages and currents to be applied, plugging on pins, mechanical stress as well as storage, have to be optimized for maximizing the detector life time.

The handling procedure does not directly depend on the Focal Plane Assembly (FPA) size or wavelength or of the cooler type. Therefore this document aims at being a generic document applicable to all standard IDDCAs, and as such, is an annex of the Technical Data Package (TDP) delivered to the customer when an IDDCa is ordered.

The contractual TDP delivered to the customer annexes this ICD in its latest update that will in any case supersede the ICD in lower issue that could be provided prior to the TDP.

It may also happen that the delivered IDDCa has specific recommendations with respect to some customization or to some particular requirements.

Therefore,

One shall always refer to the amendments pages of the contractual TDP (if any) to check for modifications that would amend and/or modify paragraphs of this ICD document.

2. REFERENCED DOCUMENTS

N/A

3. LIST OF ACRONYMS USED IN NCRIEO DOCUMENTS

ESD	Electro Static Discharge
FPA	Focal Plane Array
HgCdTe	Mercury Cadmium Telluride
I/O	Inputs / Outputs
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
ITR	Integration Then Read
IWR	Integration While Read
LW	Long Wave band = 8-12 μm
MC	Master Clock
MCT	Mercury Cadmium Telluride
MW	Mid Wave band = 3-5 μm
NUC	Non Uniformity Correction
PV	Photovoltaic
ROIC	Read Out Integrated Circuit
SW	Short Wave band = 0.8 – 2.5 μm
TDP	Technical Data Package

4. IDDCA GENERAL OVERVIEW

4.1 IDDCA main items

A standard IDDCA is composed of the following main items:

- Focal Plane Assembly (FPA), the IR sensitive array,
- sealed metal outer dewar,
- getter device designed for long vacuum life,
- window to allow infrared radiation to penetrate into the vacuum sealed dewar
- cold filter to precisely define the incident radiation flux on the IRFPA,
- high efficiency cold shield system to define the field of view of the IRFPA and limit parasitic fluxes,
- temperature sensors at the level of the IRFPA, for monitoring and feed back to the cooler,
- a cold plate and a cold finger which is the thermal interface between the cryocooler and the dewar,
- A cryogenic cooler.

A general overview of the main parts of the dewar is presented in Figure 4.1:

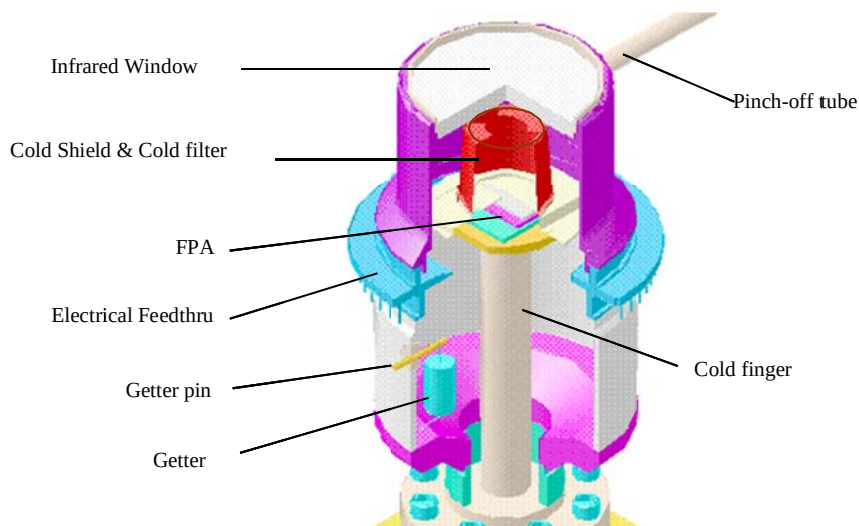


Figure 4.1: sketch of a dewar

4.2 IDDCA manufacturing process

The flow plan of Figure 4.2 points out the standard manufacturing steps of the main subassemblies of the IDDCA.

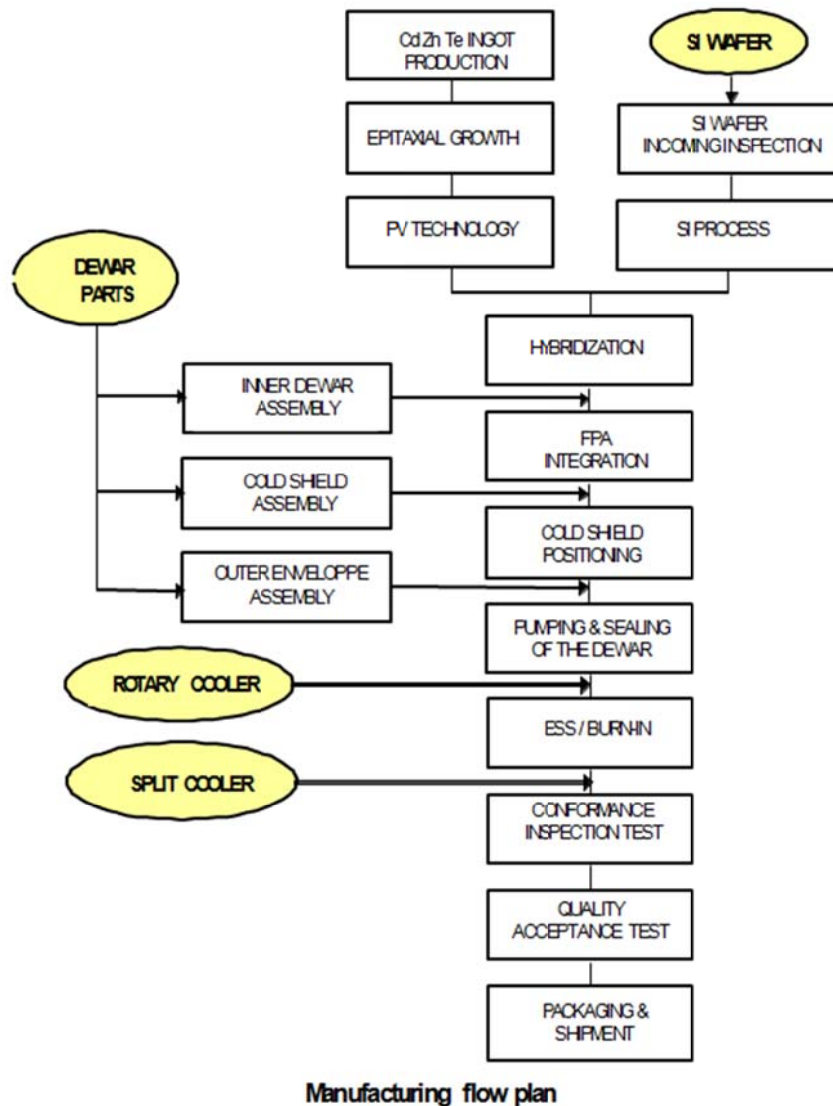


Figure 4.2: manufacturing flow plan

Main subassemblies are manufactured in house except dewar primary sub-parts, cooler and silicon wafers. All critical manufacturing steps are performed in house.

ESS (Environmental Stress Screening) is performed on each unit before NCRIEO quality acceptance test.

5. **STORAGE**

5.1 **Storage conditions**

For short term storage if waiting for system integration, the detector into its vacuumed anti-static plastic sheet can be left in its transportation box for a few weeks.

It is however recommended to unpack the detector from its collective transportation package after receipt, at least to ensure that the delivery is satisfactory.

For standard storage, the detector should not be left in the transportation box because the plastic foam is not designed for long duration storage nor for high temperature storage (>50°C).

Place the detector in storage area with the following conditions.

- Recommended ambient storage temperature: $23^{\circ}\text{C} \pm 5^{\circ}\text{C}$.
- Recommended Humidity : between 40% and 60%

It is not necessary to unpack the detector from its vacuumed-sealed anti-static plastic sheet (if applicable) but as soon as this bag is removed, it is recommended to handle the detector with gloves.

The short-circuit board shall not be removed for storage.

WARNING: if the storage temperature increases to “higher temperatures” (>35°C) for long duration, it will impact the detector life time since it will induce IRFPA ageing and acceleration of vacuum quality degradation.

6. HANDLING AND FIXATION OF THE DETECTOR

6.1 Handling / Manipulation

Gloves shall be worn when handling the detector in order to avoid finger prints which may favor corrosion.

Any manipulation shall be made with caution, by handling the detector at the level of the mechanical interface which is where the detector shall be attached.

It is the customer responsibility to estimate if a specific tooling needs to be developed to maintain the detector outside the system, by its mechanical interface.

6.2 Main fragile parts of the detector

Care must be taken not to touch the **window** during any detector handling. In the worst case a physical impact could irretrievably damage the window disc, which would result in loss of dewar vacuum associated with permanent unit failure. In order not to damage the window, it is recommended to maintain the window protection plastic cover in place during any manipulation.

Refer to the optical interface description in the contractual Technical Data Package of the delivered detector for recommendations concerning the window cleaning procedure.

The **pinch-off** of the detector and the **getter pins** are fragile parts which are unprotected against shocks. The user shall be extremely cautious when introducing the detector into the system, in order not to touch the pinch-off or the getter pins with other mechanical parts.

For Joule-Thomson cooled IDDCAs, the JT cooler may be delivered with a **gas pipe**. In that case, the user shall be extremely cautious when introducing the detector into the system, in order not to bend the inlet gas line.

The detector must not be sustained in the system by the electrical pins.

It is forbidden to apply any mechanical stress on any part of the detector except on the mechanical interface defined on the interface drawings.

6.3 Connecting the electrical interface of the cooler

Refer to the chapter “cooler” of the contractual Technical Data Package, or to the IDDCA mechanical drawings, for information concerning the connection of the cooler electrical interface.

6.4 Connecting the electrical interface of the FPA

The following paragraphs describe the main types of mechanical connections for the FPA electrical interface. If the delivered IDDCA is not described below, please refer to the custom pages of the contractual Technical Data Package.

Refer to the electrical interface description in the contractual Technical Data Package for description and cautions concerning the electrical interfaces.

The electrical interface connection has to be performed by a skilled person, in a convenient environment and suitable conditions such as required for a precise mounting and with ESD protection (see cautions in separate document on electrical interface).

The mechanical interface must be used to hold the detector.

Once connected, no stress shall be applied to the electrical interface, the detector must not be sustained in the system by the electrical connections.

6.4.1 Feedthru ceramics pins

Many standard detectors have their electrical interface defined by the ring of pins of the feedthru ceramics.

In that case, the electrical connection is insured by a direct clamp of a connector ring board or a proximity electronic board or a flex, onto the pins of the feedthru ceramics.

There are different feedthru ceramics diameters, therefore different quantity of pins, depending on the purchased detector:

- 64 pins
- 27 pins An + 27 pins Bn
- 26 pins An + 25 pins Bn
- 21 pins An + 21 pins Bn (as illustrated in Figure 6.1)
- 19 pins An + 19 pins Bn + 1 ground

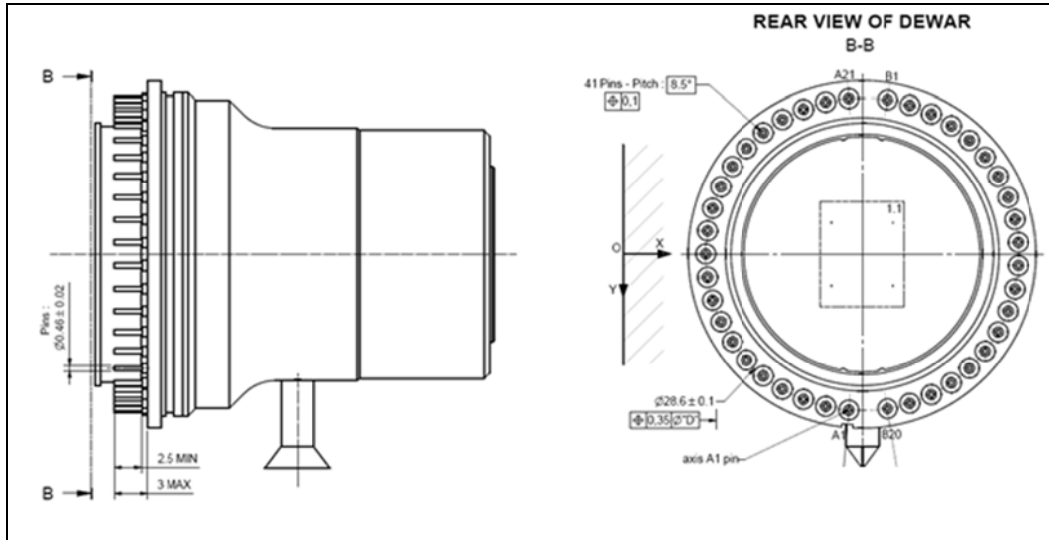


Figure 6.1: example of electrical interface with feedthru ceramics pins

Refer to the IDDCA mechanical interface for the exact definition of the delivered detector.

Carefully pull short circuit board out of the detector pins.

Pins are fragile and the force put on them must be minimized. A bent or twisted pin cannot be inserted in the electronic board / flex and can easily break if insertion is not performed with care. A strong effort on the pins could damage the ceramic feedthru on which the pins are fixed.



Extreme caution shall be considered when plugging the electrical interface, in order to avoid misplugging (rotation of flex with respect to pins, or inversion of pins An and pins Bn between flex and dewar) and ensure perfect correspondence between flex and pins functions.

In order to facilitate the maintenance operations and to minimize the risks of detector degradation, it is highly recommended to connect the detector thanks to a female connector on the male pins, instead of performing a direct soldering on the golden pins.

If absolutely requested, it is possible to solder on the dewar pins. It is mandatory to contact NCRIO for specific soldering procedure in order to avoid damage risks and drawbacks for future dismounting.

It is forbidden to use a soldering iron on the electronic board when the detector is connected.

(End of the document)

Annex 2

Title: Optical Interfaces of

MINI640 IDDCA

(with 15um pixel pitch and XD-2 Stirling cooler)

Reference: OI-MINI640-IDDCA-1801

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

OPTICAL INTERFACES OF NCRIEO DETECTORS		
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1. INTRODUCTION

This technical document describes the general optical interfaces of NCRIEO standard Mid Wave (MW) Infrared (IR) Integrated Detector Dewar Cooler Assemblies (IDDCAs).

The optical interface of an IDDCA consist of

- the Dewar window,
- the cold filter (in some specific cases, there is no cold filter; in that case, it is specified in the contractual Technical Data Package),
- the cold shield,
- the infrared focal plane array.

The optical interface does not depend on the cooling engine used to cool down the MW Focal Plane Assembly.

Therefore this document aims at being a generic document applicable to all standard MW IDDCAs, and as such, is an annex of the Technical Data Package (TDP) delivered to the customer when a MW IDDCA is ordered.

The contractual TDP delivered to the customer annexes this ICD in its latest update that will in any case supersede the ICD in lower issue that could be provided prior to the TDP.

However, it may happen that the delivered IDDCA has a specific or custom adaptation of the optical interface (window thickness, window material, specific spectral transmission of the window or of the cold filter, specific AR coating on the FPA, etc...).

Therefore, one shall always refer to the amendments pages of the contractual TDP (if any) to check for modifications that would amend and/or modify paragraphs of this ICD document.

2. **REFERENCED DOCUMENTS**

N/A

3. **LIST OF ACRONYMS USED IN THIS DOCUMENT**

FPA	Focal Plane Array
HgCdTe	Mercury Cadmium Telluride
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
MCT	Mercury Cadmium Telluride
MW	Mid Wave band = 3-5 μm
PV	Photovoltaic
ROIC	Read Out Integrated Circuit
TDP	Technical Data Package

4. OPTICAL INTERFACES OF THE IDDCA

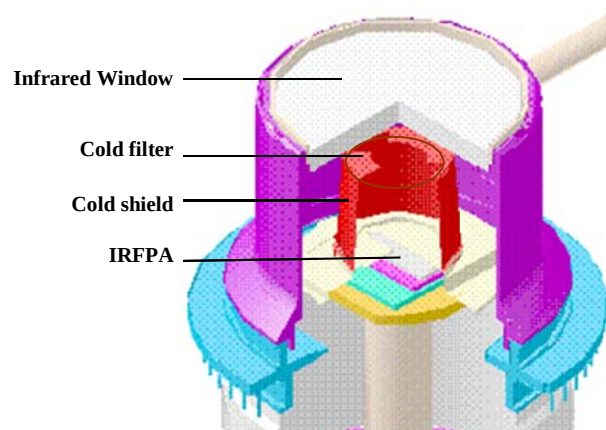
4.1 General

The optical interfaces for all incident referenced radiation provide well defined spectral bandwidth, focus location and field of view for any position within the focal plane.

The optical interface of an IDDCA consist of

- the Dewar window,
- the cold filter (in some specific cases, there is no cold filter; in that case, it is specified in the contractual Technical Data Package),
- the cold shield (in some specific cases, there may be no cold shield; in that case, it is specified in the contractual Technical Data Package),
- the infrared focal plane array.

A general overview of the IDDCA optical interfaces is presented in Figure 2.1:



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4.2 Antireflective coated infrared window

The function of the infrared window is to allow infrared radiation of the Mid Wave (3 to 5 μ m) bandwidth to penetrate into the dewar.

Because infrared radiation out of the required bandwidth has to be rejected while infrared radiation within the specified bandwidth should have maximum transmission, the infrared window of the standard MW IDDCA (except if otherwise specified in the contractual Technical Data Package) is made of Silicon (refractive index = 3.4) with a high efficiency antireflective coating.

When its diameter is ≤ 16 mm, the standard thickness of the window is 1 mm with a parallelism lower than 1 minute. The precise window thickness is stated in the delivered detector mechanical interface (2D drawings).

4.3 Infrared detector (FPA)

The central component of the detector dewar assembly is the infrared detector. Besides logical circuits of the ROIC, it consists of an array of infrared sensitive PV detector elements arranged within a rectangular plane. This active detector plane has to coincide with the actual focal plane position of the incident infrared radiation to provide proper detector performance.

Because the detector elements are backside illuminated, a slight shift of the focal plane due to the detector material refraction index is also considered.

The tolerances values for FPA location with respect to mechanical references (dewar/camera fixation plane) are specified in the IDDCA mechanical interface drawings delivered in the contractual TDP.

The FPA includes an anti-reflection coating in order to maximize the IR transmission in the HgCdTe material.

5. USER'S CAUTIONS REGARDING OPTICAL INTERFACES

5.1 General

The IDDCA is a vacuum sealed dewar, therefore the infrared window is the only optical interface that the user can contact and could damage.

5.2 IDDCA Handling

Such as any optical part, care must be taken not to touch the window during any detector handling. In the worst case a physical impact could irretrievably damage the window disc, which would result in loss of dewar vacuum associated with permanent unit failure.

In order not to damage the window, it is recommended to maintain the window protection plastic cover in place during any manipulation.

To avoid any finger prints on unprotected window, gloves shall be worn when handling the IDDCA.

5.3 Window Cleaning

Like most optical components the infrared window surface must not be treated chemically or mechanically to maintain the quality of optical parameters.

Cleaning of the window should only be considered in case of absolute necessity.



Follow classical rules for optics cleaning (only translation motion without return, from the center to the edge, never use the paper twice).

Compressed clean air or Nitrogen can be used to blow and dry the window after cleaning with isopropyl alcohol.



Do NOT use acetone to clean the window, as it could dissolve the edges of the window coatings.

(End of the document)

Annex 3

Title: Electrical Interfaces of the FPA

In MINI640 IDDCA

(with 15um pixel pitch and XD-2 Stirling cooler)

Reference: EIFPA-MINI640-IDDCA-1801

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

ELECTRICAL INTERFACES OF NCRIEO FPA in MINI640		
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1. INTRODUCTION

This technical document describes the electrical interfaces of the HgCdTe n/p Focal Plane Assembly (FPA) which is implemented in the 640×512 15µm pitch Mid Wave (MW) Infrared (IR) detectors - or Integrated Detector Dewar Cooler Assemblies (IDDCAs) - commercially referenced as MINI640 detectors.

This FPA is constituted of

- A Mercury Cadmium Telluride (HgCdTe) Photovoltaic (PV) IR sensitive array,
- A Read Out Integrated Circuit (ROIC) ,
- One or two discrete temperature sensors located on the cold table next to the ROIC.

This document details the following information for the FPA:

- General description
- Functions of the ROIC
- Electrical interfaces
 - o Input clocks
 - o Input biases
 - o Output signals
 - o Timing diagrams
- Cautions for use

This document aims at being a generic document applicable to all IDDCAs implementing this ROIC, and as such, is an annex of the Technical Data Package (TDP) delivered to the customer when a MINI640 IDDCA is ordered.

The contractual TDP delivered to the customer annexes this ICD in its latest update that will in any case supersede the ICD in lower issue that could be provided prior to the TDP.

It may also happen that the delivered IDDCA has a specific or custom adaptation of the ROIC electrical interface (restriction of number of inputs/outputs imposed by the dewar or by the system, windowing forced to a fixed format, master clock frequency forced to a fixed value, etc...).

Therefore, one shall always refer to the amendments pages of the contractual TDP (if any) to check for modifications that would amend and/or modify paragraphs of this ICD document.

If the MINI640 IDDCA is delivered with a NCRIO proxy board (Command & Control Electronics), one shall refer to the TDP of the proxy board for the detailed electrical interface of the IDDCA+C&CE set.

2. REFERENCED DOCUMENTS

N/A

3. LIST OF ACRONYMS USED IN THIS DOCUMENT

ATR	Acceptance Test Report
ESD	Electro Static Discharge
FPA	Focal Plane Array
HgCdTe	Mercury Cadmium Telluride
I/O	Inputs / Outputs
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
ITR	Integration Then Read
IWR	Integration While Read
LW	Long Wave band = 8-12 μm
MC	Master Clock
MCT	Mercury Cadmium Telluride
MW	Mid Wave band = 3-5 μm
NUC	Non Uniformity Correction
PV	Photovoltaic
ROIC	Read Out Integrated Circuit
TDP	Technical Data Package

4. GENERAL DESCRIPTION OF THE FPA

4.1 Principle of detection

The absorption of an IR photon by the MCT (Mercury Cadmium Telluride) sensitive material generates an electron-hole pair which is separated by the internal electrical field of the photodiode.

The photodiodes are electrically coupled to the readout circuit by direct injection and the charges generated by the absorption of IR photons during the integration time are stored in the capacitance of the ROIC. At the end of the integration time, the charges stored in the capacitance are transferred to a charge-voltage conversion amplifier.

Four multiplexers (160 to 1) operating simultaneously allow the transfer of the signals from the amplifiers to the four outputs of the ROIC (327680 diodes multiplexed in 512 rows and 640 columns).

Figure 4.1 presents the principle of the IR detection.

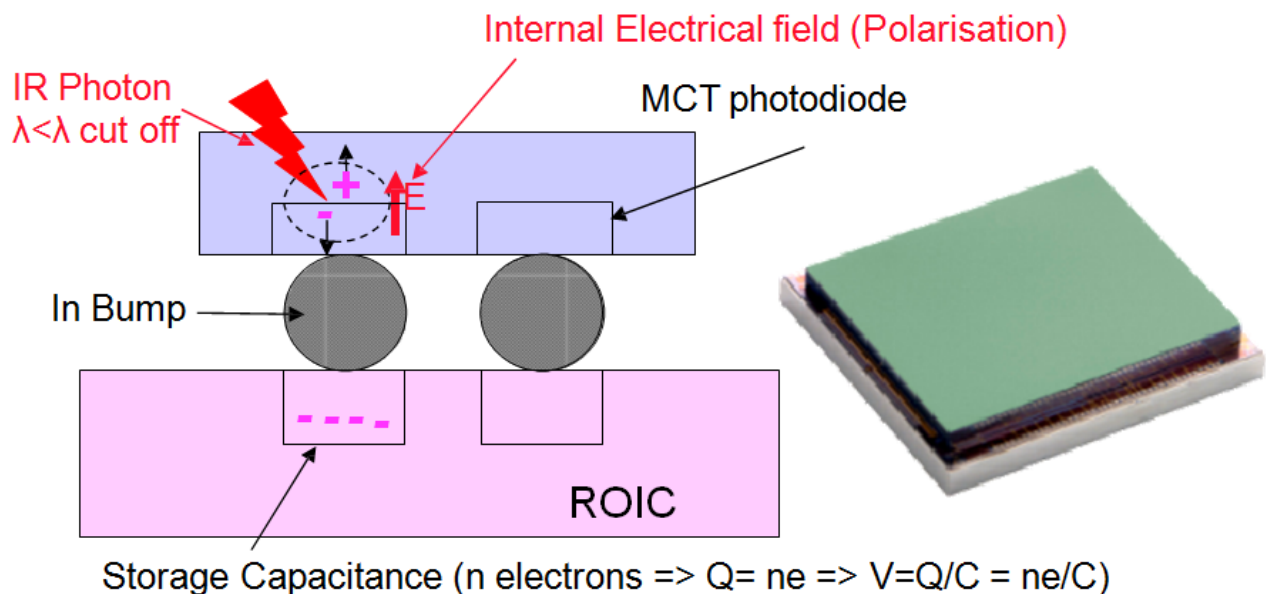


Figure 4.1: Principle of the IREPA

The output signal is proportional to the number of charges stored in the input capacitance, this number of charges being proportional to the number of electrons generated by the incident IR flux during the Integration time.

Depending on the flux on the detector, the integration time must be adjusted in such a way that the output signal is in the linear part of the dynamic range (generally about 30-50% but strongly dependant on the application).

As a general rule, the signal to noise ratio increases with the integration time, so it will be improved using the largest input capacitance (lowest gain) compatible with the other characteristics and requirements of the system (master clock frequency, frame rate, optical design, ...).

4.2 Temperature sensor

The FPA is operational at cryogenics temperatures ($< 120\text{K}$ (-153°C)) and this is made possible thanks to the use of electrically powered Stirling-cycle coolers, or of cryogenic gases using Joule-Thomson effect coolers.

Two temperature sensor diodes, 2N2222 diode type or equivalent, are integrated in the IDDCA at the level of the focal plane array.

- One must be used for ensuring the control loop of the Stirling-cycle cooler during regulation (in order to ensure a stabilized FPA temperature);
- The second one may be used for monitoring the focal plane array temperature (and measuring cool down time for instance).

For each temperature sensor, the electrical interfaces are the sensor anode (pin DTA) and the sensor cathode (pin DTK).

Note: in some IDDCA configuration (when I/O number is limited, or when there is no temperature control loop), there may be only one temperature sensor.

The electrical interface of the temperature sensor is described in § 7.

4.3 ROIC general description

The functions of the ROIC are detailed in § 5.

The electrical interfaces of the ROIC are detailed in § 6.

4.3.1 Architecture

The ROIC design is based on a Silicon CMOS technology. The ROIC is composed of 327680 diodes multiplexed in 512 rows and 640 columns.

Its organization permits the FPA

- to integrate simultaneously (snap-shot) the information from the PV diodes (coupled by direct injection),
- to store this information,
- to multiplex this information per row,
- to multiplex the rows information
- and to sample and hold the information.

4.3.2 Operating modes

The ROIC operates in the **snap shot mode** (all pixels of the array are integrated in the same time).

The ROIC design allows the user to select different operating modes depending on the application:

- Integration mode and gain adaptation
 - o **Integration Then Read (ITR)** : the signal of frame N integrated in the input stage is read out after the end of the integration time of frame N and the integration of the signal of frame N+1 starts after all the signals of frame N have been read out.
In ITR mode, the gain (storable charges) is **6.36 Me-**.
 - o **Integration While Read (IWR)**: the signal of frame N integrated in the input stage is read out during the integration time of frame N+1.
In IWR mode, the gain (storable charges) is **4.99 Me-**.
- Programmable integration time
 - o Minimum charge integration duration : 1.5 MC period (with min INT = 3077 MC)
 - o Minimum step for integration time change: 1 MC period
 - o Maximum integration time duration : no limitation
- Windowing mode
 - o 640×512 full array (default mode, at power ON)
 - o 640×480
 - o 512×512
 - o Random windowing mode within 640×512
- Horizontal and vertical Image flip mode
 - o Colum invert
 - o Line revert
 - o Image rotation (= invert + revert).

4.3.3 Overview of the electrical Interfaces

In order to give a global view of the electrical interfaces of the ROIC, all the Inputs and Outputs of the ROIC are listed in the two following tables. Refer to the corresponding paragraphs for a detailed description and requirements.

For optional Inputs the Default Mode gives the status of the function when the corresponding input is not connected (remained floating).

INPUTS			
Name	Type	Description	Comment
GPOL	Tunable (0 to 1.5 V)	PV diode bias. May have to be adjusted for each detector, the optimum value is given in the Acceptance Tests Report	This bias directly related to the PV diodes signal must be low noise
VDDA	Supply	Supply (3.3V) for the Analog Part of the ROIC	This bias directly related to the PV diodes signal must be low noise
VDDO	Supply	Supply (3.3 V) for the Analog Part of the ROIC	This bias directly related to the PV diodes signal must be low noise
VDD	Supply	Supply (3.3V) for the Digital Part of the ROIC	This bias must be low noise
VR	Supply	Supply (3V) used for capacitances reset	
GND	Ground	Ground for the Analog Part of the ROIC	
GNDD	Ground	Ground for the Digital Part of the ROIC	
SUBPV	Ground	Ground for PV diode	
IBP	Internal	Output stage current source: Internally generated, must be connected to the ground by a 42 kΩ resistance.	
MASTER CLOCK (MC)	Pulsed signal	This reference clock with a duty cycle of 50% is the base for the synchronism of the whole circuit	Maximum Frequency=10 MHz
INT	Pulsed signal	Integration time adjustment, allows the integration of charges in the pixel input capacitance. The charges are stored during the high level of INT signal	
SIZE-A	Digital Signal	Used with SIZE-B for format selection: 640x512, 640x480, 512x512 or windowing mode	Not compulsory. Default mode is SIZE-A = 1 and SIZE-B = 1 (640x512 format)
SIZE-B	Digital Signal	Used with SIZE-A for format selection: 640x512, 640x480, 512x512 or windowing mode	Not compulsory. Default mode is SIZE-A = 1 and SIZE-B = 1 (640x512 format)
SERCLR	Digital Signal	Reset of the counter which controls the number of input bits for the window definition.	Not compulsory. Used only in windowing mode
SERDAT	Digital Signal	Serial interface data, allows the input of data for the coordinates of the considered window	Not compulsory. Used only in windowing mode.
UPCOL	Digital Signal	Selection of columns readout direction	Not compulsory.
UPROW	Digital Signal	Selection of rows readout direction	Not compulsory
ITR	Digital Signal	Selection of the operating mode ITR=0 ⇒ Integration While Read (IWR) mode ITR=1 ⇒ Integration Then Read (ITR) mode	Not compulsory. Default Mode is ITR= 1 (Integration Then Read mode)

OUTPUTS			
Name	Type	Description	Comment
OUT0, OUT1, OUT2, OUT3,	Analog output	Output signals	
DATAVALID	Digital output	Indicates the presence of valid data on the analog outputs (OUT0 to OUT3) when on its high level	
ERROR	Digital output	Indicates the presence of valid data for the windowing mode when on its low level	Used only in windowing mode

In addition to these ROIC Inputs and Outputs and to the temperature sensors electrical interface, the IDDCA generally has additional electrical interfaces that can vary from detector to detector:

- one cold shield ground
- one dewar mechanical ground

Refer to the IDDCA interface drawings for the full detail of the I/O electrical interface at IDDCA level.

5. DETAILED DESCRIPTION OF THE ROIC FUNCTIONS

This chapter precisely describes the functions of the ROIC and the way to make it work.

If the customer develops its own proximity electronics, this chapter shall be carefully considered for understanding how to drive the FPA.

If the IDDCA is delivered with NCRIEO proximity electronics, this chapter should only be considered for information, and one will refer to the C&CE Technical Data Package for the description of the required electrical interface.

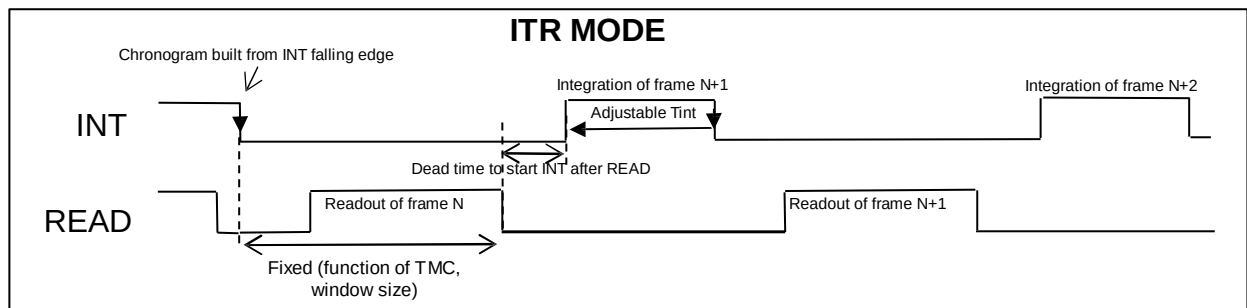
5.1 Input stage: integration mode (ITR function)

The input stage design is based on the direct injection principle.

During integration time, charges are stored in the input capacitance.

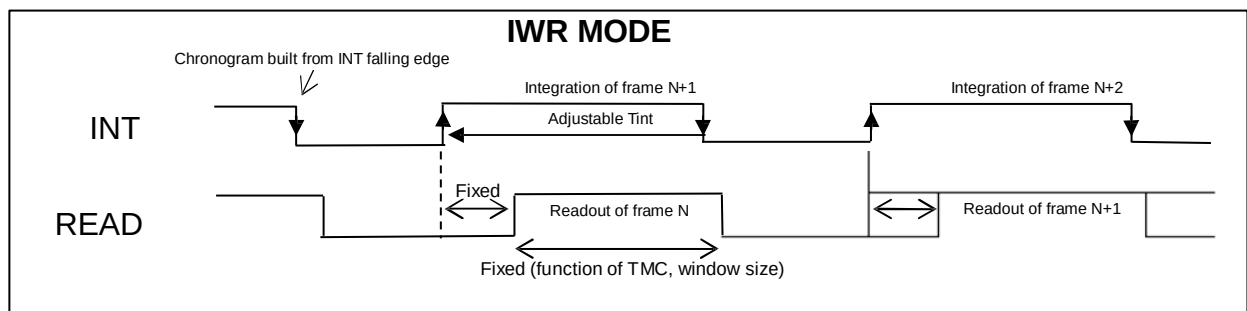
The input stage can be operated according to two different modes:

- **Integration Then Read (ITR)**
 - o the signal of frame N integrated in the input stage is read out after the end of the integration time of frame N and the integration of the signal of frame N+1 starts after all the signals of frame N have been read out.



- **Integration While Read (IWR)**

- o the signal of frame N integrated in the input stage is read out during the integration time of frame N+1.



5.2 Storable charges

ITR function allows choosing the operational mode according to the following table, with consequence on the storable charges:

ITR	Operation mode	Storable charges (gain)
1 (Default mode)	ITR	0.485pF over 2.10V = 1.02 pC ($6.36 \cdot 10^6 e^-$)
0	IWR	0.485pF over 1.65V = 0.8 pC ($4.99 \cdot 10^6 e^-$)

Figure 5.1: selection of operational mode with ITR

See § 6.3.1.2 for the Dynamic Range for each of the two modes/gains.

When input stage capacitance is full, charges which can't be stored are flowed away thanks to an antiblooming gate integrated in each input stage.

5.3 Charge-voltage conversion

At the end of the integration time, the charges are converted into voltages. The charge-voltage conversion is performed by amplifiers.

Each charge-voltage conversion stage is controlled by internally generated biases.

The readout capacitances are sequentially addressed line by line by 640 amplifiers located at each column end.

Four multiplexers (160 to 1) operating simultaneously allow the transfer of the signals from the amplifiers to the four outputs.

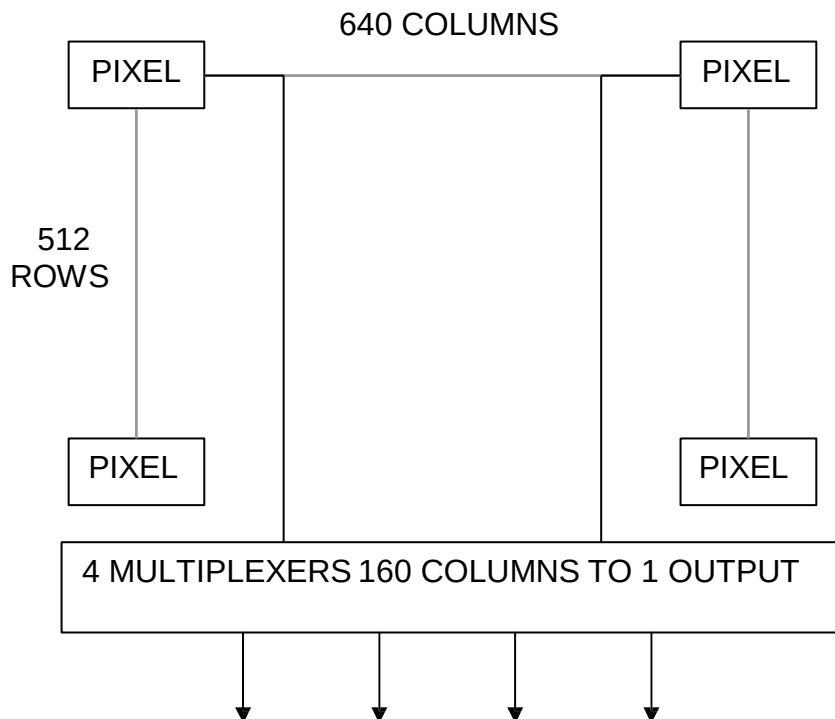


Figure 5.2: Synoptic of the circuit

5.4 Output stage and methodical arrangement

There are four output stages labeled **OUT0** to **OUT3**, one output stage for each multiplexer stage.

The pixel information is read out at a rate of one per master clock period.

The maximum output rate is given in § 5.5.1.

The output signal form and its settling time are detailed in § 6.3.1.1.

The output methodical arrangement is detailed in Figure 5.3 for the default mode and in Figure 5.4 for the windowing mode.

OUT0	OUT1	OUT2	OUT3
* C1.R1	* C2.R1	* C3.R1	* C4.R1
C5.R1	C6.R1	C7.R1	C8.R1
...	...	...	...
C637.R1	C638.R1	C639.R1	C640.R1
C1.R2	C2.R2	C3.R2	C4.R2
C5.R2	C6.R2	C7.R2	C8.R2
...	...	...	...
...	...	...	...
C637.R2	C638.R2	C639.R2	C640.R2
C1.R3	C2.R3	C3.R3	C4.R3
C5.R3	C6.R3	C7.R3	C8.R3
...	...	...	...
...	...	...	...
C637.R512	C638.R512	C639.R512	C640.R512

* first readout information for each output ; C = Column ; R = Row

Figure 5.3: Output methodical arrangement for the 640×512 mode (in default mode UPCOL=1, UPROW=1)

<table><tr><th>OUT0</th><th>OUT1</th><th>OUT2</th><th>OUT3</th></tr><tr><td>* (R_A;C_A)</td><td>* (R_A;C_A+1)</td><td>* (R_A;C_A+2)</td><td>* (R_A;C_A+3)</td></tr><tr><td>(R_A;C_A+4)</td><td>(R_A;C_A+5)</td><td>(R_A;C_A+6)</td><td>(R_A;C_A+7)</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>(R_A;C_B-3)</td><td>(R_A;C_B-2)</td><td>(R_A;C_B-1)</td><td>(R_A;C_B)</td></tr><tr><td>(R_A+1,C_A)</td><td>(R_A+1,C_A+1)</td><td>(R_A+1,C_A+2)</td><td>(R_A+1,C_A+3)</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>(R_B;C_B-7)</td><td>(R_B;C_B-6)</td><td>(R_B;C_B-5)</td><td>(R_B;C_B-4)</td></tr><tr><td>(R_B;C_B-3)</td><td>(R_B;C_B-2)</td><td>(R_B;C_B-1)</td><td>(R_B;C_B)</td></tr></table> * first readout information for each output (UPCOL=1, UPROW=1 : default mode)	OUT0	OUT1	OUT2	OUT3	* (R _A ;C _A)	* (R _A ;C _A +1)	* (R _A ;C _A +2)	* (R _A ;C _A +3)	(R _A ;C _A +4)	(R _A ;C _A +5)	(R _A ;C _A +6)	(R _A ;C _A +7)	...	...	...	...	(R _A ;C _B -3)	(R _A ;C _B -2)	(R _A ;C _B -1)	(R _A ;C _B)	(R _A +1,C _A)	(R _A +1,C _A +1)	(R _A +1,C _A +2)	(R _A +1,C _A +3)	...	...	...	...	(R _B ;C _B -7)	(R _B ;C _B -6)	(R _B ;C _B -5)	(R _B ;C _B -4)	(R _B ;C _B -3)	(R _B ;C _B -2)	(R _B ;C _B -1)	(R _B ;C _B)	<table><tr><th>OUT0</th><th>OUT1</th><th>OUT2</th><th>OUT3</th></tr><tr><td>* (R_B;C_B-3)</td><td>* (R_B;C_B-2)</td><td>* (R_B;C_B-1)</td><td>* (R_B;C_B)</td></tr><tr><td>(R_B;C_B-7)</td><td>(R_B;C_B-6)</td><td>(R_B;C_B-5)</td><td>(R_B;C_B-4)</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>(R_B;C_A)</td><td>(R_B;C_A+1)</td><td>(R_B;C_A+2)</td><td>(R_B;C_A+3)</td></tr><tr><td>(R_B-1,C_B-3)</td><td>(R_B-1,C_B-2)</td><td>(R_B-1,C_B-1)</td><td>(R_B-1,C_B)</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>(R_A;C_A+4)</td><td>(R_A;C_A+5)</td><td>(R_A;C_A+6)</td><td>(R_A;C_A+7)</td></tr><tr><td>(R_A;C_A)</td><td>(R_A;C_A+1)</td><td>(R_A;C_A+2)</td><td>(R_A;C_A+3)</td></tr></table> * first readout information for each output (UPCOL=0, UPROW=0; Image rotation)	OUT0	OUT1	OUT2	OUT3	* (R _B ;C _B -3)	* (R _B ;C _B -2)	* (R _B ;C _B -1)	* (R _B ;C _B)	(R _B ;C _B -7)	(R _B ;C _B -6)	(R _B ;C _B -5)	(R _B ;C _B -4)	...	...	...	...	(R _B ;C _A)	(R _B ;C _A +1)	(R _B ;C _A +2)	(R _B ;C _A +3)	(R _B -1,C _B -3)	(R _B -1,C _B -2)	(R _B -1,C _B -1)	(R _B -1,C _B)	...	...	...	...	(R _A ;C _A +4)	(R _A ;C _A +5)	(R _A ;C _A +6)	(R _A ;C _A +7)	(R _A ;C _A)	(R _A ;C _A +1)	(R _A ;C _A +2)	(R _A ;C _A +3)
OUT0	OUT1	OUT2	OUT3																																																																						
* (R _A ;C _A)	* (R _A ;C _A +1)	* (R _A ;C _A +2)	* (R _A ;C _A +3)																																																																						
(R _A ;C _A +4)	(R _A ;C _A +5)	(R _A ;C _A +6)	(R _A ;C _A +7)																																																																						
...	...	...	...																																																																						
(R _A ;C _B -3)	(R _A ;C _B -2)	(R _A ;C _B -1)	(R _A ;C _B)																																																																						
(R _A +1,C _A)	(R _A +1,C _A +1)	(R _A +1,C _A +2)	(R _A +1,C _A +3)																																																																						
...	...	...	...																																																																						
(R _B ;C _B -7)	(R _B ;C _B -6)	(R _B ;C _B -5)	(R _B ;C _B -4)																																																																						
(R _B ;C _B -3)	(R _B ;C _B -2)	(R _B ;C _B -1)	(R _B ;C _B)																																																																						
OUT0	OUT1	OUT2	OUT3																																																																						
* (R _B ;C _B -3)	* (R _B ;C _B -2)	* (R _B ;C _B -1)	* (R _B ;C _B)																																																																						
(R _B ;C _B -7)	(R _B ;C _B -6)	(R _B ;C _B -5)	(R _B ;C _B -4)																																																																						
...	...	...	...																																																																						
(R _B ;C _A)	(R _B ;C _A +1)	(R _B ;C _A +2)	(R _B ;C _A +3)																																																																						
(R _B -1,C _B -3)	(R _B -1,C _B -2)	(R _B -1,C _B -1)	(R _B -1,C _B)																																																																						
...	...	...	...																																																																						
(R _A ;C _A +4)	(R _A ;C _A +5)	(R _A ;C _A +6)	(R _A ;C _A +7)																																																																						
(R _A ;C _A)	(R _A ;C _A +1)	(R _A ;C _A +2)	(R _A ;C _A +3)																																																																						
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OUT0	OUT1	OUT2	OUT3																																																																						
* (R _B ;C _A)	* (R _B ;C _A +1)	* (R _B ;C _A +2)	* (R _B ;C _A +3)																																																																						
(R _B ;C _A +4)	(R _B ;C _A +5)	(R _B ;C _A +6)	(R _B ;C _A +7)																																																																						
...	...	...	...																																																																						
(R _B ;C _B -3)	(R _B ;C _B -2)	(R _B ;C _B -1)	(R _B ;C _B)																																																																						
(R _B -1,C _A)	(R _B -1,C _A +1)	(R _B -1,C _A +2)	(R _B -1,C _A +3)																																																																						
...	...	...	...																																																																						
(R _A ;C _B -7)	(R _A ;C _B -6)	(R _A ;C _B -5)	(R _A ;C _B -4)																																																																						
(R _A ;C _B -3)	(R _A ;C _B -2)	(R _A ;C _B -1)	(R _A ;C _B)																																																																						
OUT0	OUT1	OUT2	OUT3																																																																						
* (R _A ;C _B -3)	* (R _A ;C _B -2)	* (R _A ;C _B -1)	* (R _A ;C _B)																																																																						
(R _A ;C _B -7)	(R _A ;C _B -6)	(R _A ;C _B -5)	(R _A ;C _B -4)																																																																						
...	...	...	...																																																																						
(R _A ;C _A)	(R _A ;C _A +1)	(R _A ;C _A +2)	(R _A ;C _A +3)																																																																						
(R _A +1,C _B -3)	(R _A +1,C _B -2)	(R _A +1,C _B -1)	(R _A +1,C _B)																																																																						
...	...	...	...																																																																						
(R _B ;C _A +4)	(R _B ;C _A +5)	(R _B ;C _A +6)	(R _B ;C _A +7)																																																																						
(R _B ;C _A)	(R _B ;C _A +1)	(R _B ;C _A +2)	(R _B ;C _A +3)																																																																						

Figure 5.4: Output methodical arrangement for the windowing mode



Important Note: when UPCOL=0 (inverting or rotating the image), the columns are still output from the FPA by block of 4 columns, in a non-inverted order. Therefore the proxy board has to read out the OUTn signals in reverse order in order to correctly display the inverted or rotated image.

5.5 Sequencer / Frame rate / Clock diagram

The sequencer generates all internal signals necessary for the readout integrated circuit operation from the different external clocks and bias voltages. The readout integrated circuit operation is totally synchronous. All the pulses generated internally are multiple integers of the period of the master clock. A change in the master clock frequency induces a change in these pulses.

The clocks necessary to operate the sequencer are the master clock and the integration phase.

5.5.1 Clock diagram

Figure 5.5 illustrates the pulsed clock diagram for IWR mode and for ITR mode.

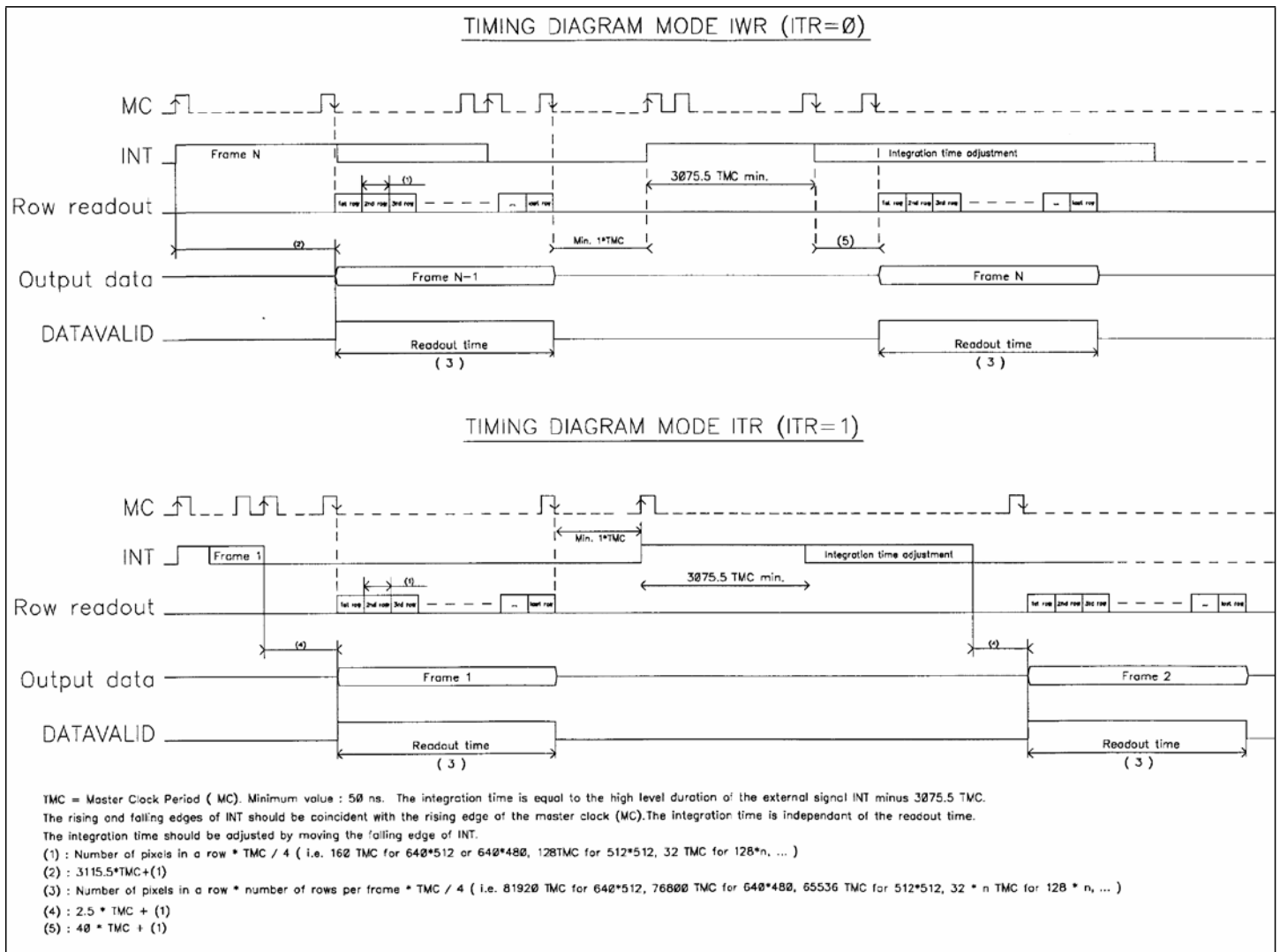


Figure 5.5: Clock diagram

5.5.2 Master Clock (MC)

The Master Clock (MC) has a maximum frequency of 10 MHz (100 ns MC period) and a duty cycle of 50 %;

It is the base for the synchronism of the operation of the whole circuit.

The pixel addressing is performed from static synchronous counters controlled by the master clock.

The pixel information is read out at a rate of one per master clock period.

5.5.3 Integration phase (INT)

The integration phase (INT) allows the integration of the charges in the pixel input capacitance (the PV diodes are biased during the high level of INT signal).

The INT rising edge (corresponding to MC rising edge) starts the integration capacitances reset during 3075.5 MC periods. **Only after these 3075.5MC starts the real integration of charges.**

Therefore, the minimum **INT** duration is 3077MC for an effective charge integration time of 1.5 MC.

Integration time is continuously programmable by 1MC step, by adjusting the start of INT rising edge.

The falling edge of INT corresponding to a rising edge of MC stops the integration phase by clamping the input MOS transistor.

A change in the integration time is applied at the very next snapshot frame.

Note : A change in INT duration must be followed by a new calibration (Non Uniformity Correction) to ensure an optimized image.

5.5.4 Transfer time

Note: in the formulas,

- MC is the master clock period
- SIZEcol is the number of pixels per column (≤ 512) of the chosen window
- SIZEline is the number of pixels per line (≤ 640) of the chosen window.

In function of the operation mode of the readout circuit (ITR or IWR), the transfer of the first window line to the charge-voltage conversion amplifier is performed during the following duration :

- for ITR mode : $2.5MC + (\frac{1}{4} \times SIZEline) \times MC$ period after the falling edge of INT.
(ex: 162.5MC for $640 \times n$, 130.5MC for $512 \times n$, 35.5MC for $132 \times n$,...)
- for IWR mode, the transfer of the first window line of frame N-1 starts 3115.5MC + $\frac{1}{4} \times SIZEline$ after the rising edge of frame N INT signal leading to the following duration for the transfer of the first window line :
 $3115.5MC + (\frac{1}{4} \times SIZEline) \times MC$ period (ex : 3275.5MC for $640 \times n$, 3243.5MC for $512 \times n$, 3148.5MC for $132 \times n$,...)

Then, the DATAVALID output is set automatically to its high level, indicating the availability of the first line data simultaneously on the four outputs OUT0 to OUT3.

5.5.5 Readout time

Since there are four outputs, each window line is output one after an other during the following duration : $(\frac{1}{4} \times SIZEline) \times MC$ period.

The window readout time is therefore $\frac{1}{4} \times SIZEline \times SIZEcol \times MC$ period.

The DATAVALID signal goes down to its low level when all the pixels of the array or of the selected window are read out.

Before starting a new integration phase of frame N+1, the minimum condition to be respected is to wait for 1MC minimum after the end of the readout phase of frame N for ITR mode or frame N-1 for IWR mode.

5.5.6 Frame time/maximum frame rate

The maximum frame rate (the minimum frame time) is depending on the output frequency that depends on the Master Clock frequency.

The maximum Master Clock frequency is stated in § 5.5.1.

- **Integrate Then Read Mode**

In the ITR mode, the minimum frame time is : $(3075.5 \times MC + \text{effective integration charge duration}) + (2.5 \times MC + \frac{1}{4} \times \text{SIZEline} \times MC) + (\frac{1}{4} \times \text{SIZEline} \times \text{SIZEcol} \times MC) + (1 \times MC)$

- **Integrate While Read Mode**

In the integrate while read mode, the minimum frame time is the maximum value between $(\text{INT duration} + 1 \times MC)$, and $(3115.5 + \frac{1}{4} \times \text{SIZEline} + \frac{1}{4} \times \text{SIZEline} \times \text{SIZEcol}) \times MC$.

The maximum frame rate for a given window is obtained with no dead time implemented before integration time phase

If the frame time needs to be lowered down without modifying the integration time, it can be done either by lowering down the Master Clock Frequency, or by implementing a dead time in the clock diagram at the end of the readout before the beginning of a new integration time.

5.6 Windowing mode (SIZE-A & SIZE-B)

5.6.1 Format selection

Three fixed formats and a windowing mode can be chosen thanks to **SIZE-A** and **SIZE-B** inputs.

These SIZE-A and SIZE-B inputs allow to choose at each frame between the full window size (640×512), two reduced fixed windows (640×480 or 512×512) and a random window defined through the serial interface.

SIZE-A	SIZE-B	FORMATS
1	1	640×512 (default mode, at power ON)
1	0	640×480
0	1	512×512
0	0	Random windowing mode within 640×512

Figure 5.6: Windowing programming

As SIZE-A and SIZE-B have an immediate impact on the pixels to be integrated or on the digital part for the selection of pixels to be read, **these inputs must only be changed between the falling edge of DATAVALID and the rising edge of INT.**

SIZE-A and SIZE-B must be set before the calibrations (NUC). A change in the following formats should be followed by a new calibration.

5.6.2 Windowing control and serial interface

The ROIC allows the choice of a reduced window for subframe imaging. As indicated in the previous paragraph, the windowing mode is obtained by the configuration SIZE-A=0 and SIZE-B=0. The coordinates of the opposite corners of the reduced window must be sent to the ROIC through the serial interface.

In case of windowing mode, the size of the selected window is determined by the C (column) and R (row) references of the corner points.

The following rules apply on the coordinates to be entered in the serial data interface :

- The (C;R) pixel coordinates can swing from (0;0) for the first array pixel to (639;511) for the last one (C = column, R = Row (or line)).
- The minimum size of the window (SIZEcol×SIZEline) is 132 columns × 1 line (within 640 columns × 512 lines) and the possible window width SIZEline (length of the line) is $132+4 \times n$ ($0 \leq n \leq 127$).

The position of pixel (0;0)=(column 1;line 1) relative to the (X;Y) dewar referential is given in Figure 5.7.

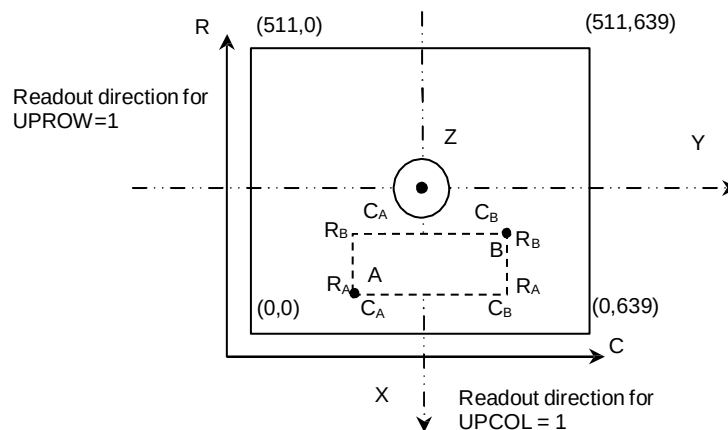


Figure 5.7: Referential axes and pixel coordinates

Note: except as otherwise specified in the IDDCA TDP following a specific customer request, pixel (0;0) is always located in the (+X;-Y) quadrant of the IDDCA referential.

(C_A ; R_A) are the coordinates of first corner of the window (point A), and (C_B ; R_B) the coordinates of the opposite corner (point B) and are addressed through the values R_{MIN} , C_{MIN} , R_{MAX} , C_{MAX} .



At system/screen level, the pixel coordinates usually consider the convention “1st pixel = (1;1)”, (A & B swing between (1 ;1) and (640 ;512)), instead of (0;0) considered in the ROIC (where A & B swing between (0 ;0) and (639 ;511)).

Therefore, the following tables refer to the two conventions.

The window coordinates data (R_{MIN} , C_{MIN} , R_{MAX} , C_{MAX}) are provided to the IRFPA through a serial interface consisting in the following inputs : **SERCLR** and **SERDAT**. The clock used for the windowing logic is Master Clock (MC), the same clock as for the sequencer.

- **SERCLR** : SERCLR falling edge indicates the beginning of SERDAT data.
SERCLR must vary on MC rising edge, and is considered on MC falling edge.
- **SERDAT** : allows the input of serial interface data. These data must vary on MC rising edge, and are considered on MC falling edge. These data are entered in the following specific order : R_{MIN} (9 bits), R_{MAX} (9 bits), C_{MIN} (8 bits), C_{MAX} (8 bits), with each of them beginning with their high weight.

SERDAT is therefore a 34 bit sequence : [$R_{MIN}(8) \dots C_{MAX}(0)$].

Since the ROIC has four outputs, the pixels are addressed in the serial interface by blocks of 4 columns (one for each output) among 160 and then C_A and C_B shall be [modulo 4] and the columns coordinates C_{MIN} , and C_{MAX} shall be 4 times less than C_A and C_B .

Screen image	with convention « 1st pixel = (0 ;0) » (A & B can swing between (0 ;0) and (639 ;511))	with convention « 1st pixel = (1 ;1) » (A & B can swing between (1 ;1) and (640 ;512))
Authorized window columns on screen	• $C_A = 4.n$, with $0 \leq n \leq 127$ • $C_B = 4.m-1$, with $n+33 \leq m \leq 160$	• $C_A = 1+4.n$, with $0 \leq n \leq 127$ • $C_B = 4.m$, with $n+33 \leq m \leq 160$
Window column programming in SERDAT	• $C_{MIN} = C_A / 4$ • $C_{MAX} = (C_B - 3)/4$	• $C_{MIN} = (C_A - 1)/ 4$ • $C_{MAX} = (C_B - 4)/4$

C_{MIN} and C_{MAX} are then expressed over 8 bits.

Screen image	with convention « 1st pixel = (0 ;0) » (A & B can swing between (0 ;0) and (639 ;511))	with convention « 1st pixel = (1 ;1) » (A & B can swing between (1 ;1) and (640 ;512))
Authorized window rows on screen	• $R_A = n$, with $0 \leq n \leq 511$ • $R_B = m$, with $n \leq m \leq 511$	• $R_A = n$, with $1 \leq n \leq 512$ • $R_B = m$, with $n \leq m \leq 512$
Window row programming in SERDAT	• $R_{MIN} = R_A$ • $R_{MAX} = R_B$	• $R_{MIN} = R_A - 1$ • $R_{MAX} = R_B - 1$

R_{MIN} and R_{MAX} are then expressed over 9 bits.

- $C_{MIN} = C_A / 4$ with $0 \leq C_{MIN} \leq C_{MAX} - 32$
- $C_{MAX} = (C_B - 3)/4$ with $C_{MIN} + 32 \leq C_{MAX} \leq 159$
- $R_{MIN} = R_A$ with $0 \leq R_{MIN} \leq 511$
- $R_{MAX} = R_B$ with $R_{MIN} \leq R_{MAX} \leq 511$

ERROR bit

When these rules are respected, the error bit (output **ERROR**), set to its high level by SERCLR rising edge, is set back to its low level at the end of the 34 bit sequence, indicating that the window definition is correct. If ERROR remains at high level, there is incoherence in the window definition and SERDAT shall be verified and sent again.

Example: Selection of the central window 384×288 within the 640×512 array.

This 384 columns and 288 rows central window starts at pixel A ($C_A = 128$; $R_A = 112$) and ends at pixel B ($C_B = 511$; $R_B = 399$).

window coordinates for SERDAT	decimal	binary	length
$R_{min} = R_A =$	112	001110000	over 9 bits
$R_{max} = R_B =$	399	110001111	over 9 bits
$C_{min} = C_A/4 =$	32	001000000	over 8 bits
$C_{max} = (C_B-3)/4 =$	127	01111111	over 8 bits
$SERDAT = <R_{min}><R_{max}><C_{min}><C_{max}>$		0011100001100011110010000001111111	over 34 bits
SERDAT (in hexadecimal from right to left)		0E18F207F	

5.6.3 Windowing function operation

The data necessary for the window definition (R_{MIN} , R_{MAX} , C_{MIN} and C_{MAX}) are put into a first register through the serial interface. If there is no error (ERROR at low level), these data are then put in a second register using parallel loading, between a falling edge of DATAVALID (thus at the end of an integration - readout cycle) and the next rising edge of INT.

When the readout circuit is turned on, this second register using parallel loading is forced for being in the full 640×512 format. This avoids any undetermined state and enables to immediately operate the circuit in a windowing mode when turned on.

For a change of the window definition, the windowing data can be sent to the serial interface at any moment. Their transfer to the second register is performed (after error-free verification) only after complete readout of the current frame.

Windowing sequence :

- the data of the first chosen window are sent through the serial interface ;

- when all the data are entered, the ERROR signal must be tested at the low level. If there has been an error during the data input (ERROR at high level after all bits input), the data must be sent again ;
- after this verification, the data are sent automatically to the parallel loading register after the falling edge of DATAVALID. Then, this register contains the valid data ;
- the data defining the next window can then be sent at any time. They will be automatically applied, if there was no error in the transmission, at the end of a readout time, and will be valid for the next integration and readout cycles.

Figure 5.8 describes the clock chronogram for the windowing mode

Serial Input Diagram

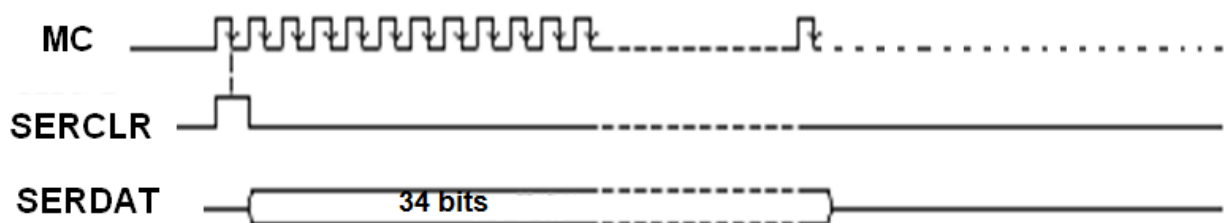


Figure 5.8: Serial input clock diagram

Note: A change in the window size should be followed by a new calibration (NUC).

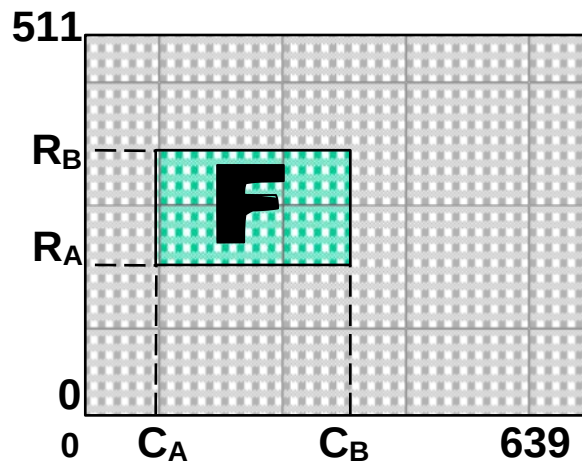
5.7 Image flip function (UPCOL & UPROW)

UPROW and UPCOL are the logic inputs that determine the readout direction of the addressed window (it can be the full frame if the windowing function is not used).

- **UPROW** drives the readout direction for the lines for an image vertical flip (along axes indicated X and R in the drawing of paragraph 5.6.2)
- **UPCOL** drives the readout direction for the columns for an image horizontal flip (axes indicated Y and C in the drawing of paragraph 5.6.2).

In case of use of the image flip function, the signals coming from the pixels of the addressed window are readout line after line.

The correspondence between UPCOL / UPROW and the effect on the image is illustrated in the following table:



UPCOL	UPROW	Readout direction	
		ROIC Side	Screen Side
1 (Default mode)	1 (Default mode)		
0	0		
1	0		
0	1		

Figure 5.9: Rotation Function Operation

6. DETAILED DESCRIPTION OF THE ROIC ELECTRICAL INTERFACES

One will refer to the IDDCA TDP for the definition of the electrical interface pinout (see the IDDCA drawings) identifying the available inputs/outputs at dewar level. Indeed, it may happen that the delivered IDDCA has a specific or custom adaptation of the ROIC electrical interface (restriction of number of inputs/outputs imposed by the dewar or by the system, windowing forced to a fixed format).

6.1 Silicon processor schematic

Figure 6.1 shows the silicon processor schematic with the inputs/outputs. All other signals are generated internally.

Note: in this schematic, the temperature sensors are identified; they are not part of the ROIC but are additional discrete diodes located on the FPA cold table.

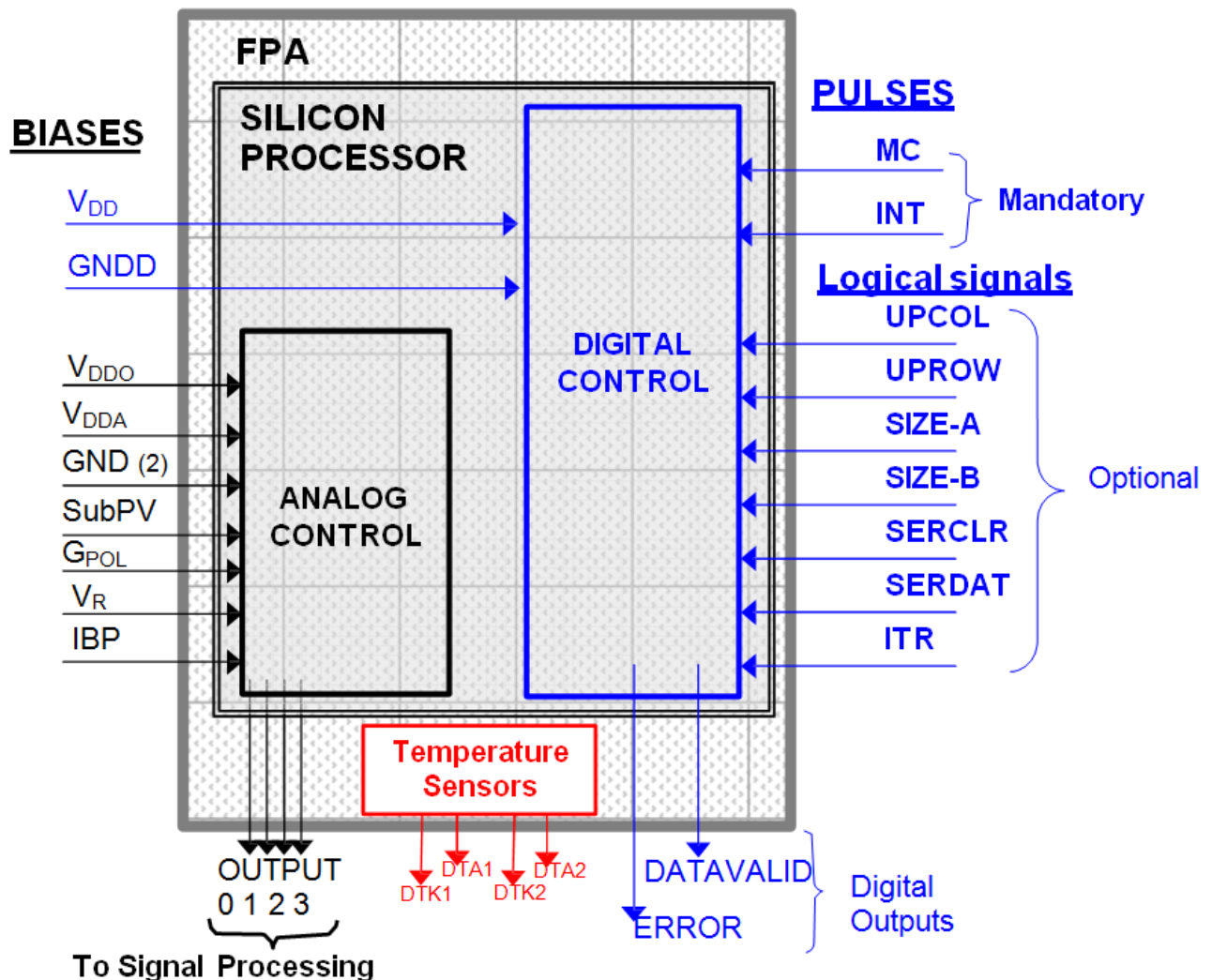


Figure 6.1: ROIC: Silicon processor schematic

6.2 Inputs interfaces

The adjustments of all inputs are defined by the value of the bias noise, the bias rejection and the maximum current, and their tolerances (i.e. the possible variation range without performance degradation).

The adjustments given in this document are valid for 3.3 V supplies.

6.2.1 Bias voltages

The bias voltages necessary to operate the readout integrated circuit are :

- **GPOL** : one tuned external bias voltage which may have to be adjusted from detector to detector to optimize the overall photodiode bias voltages. **This bias voltage is indicated in the Acceptance Test Report (ATR) of the detector.**
- **V_{DDA} and V_{DDO} and V_{DD}** : three fixed external power supplies;
- **V_R** : one fixed external bias voltage for integration and readout capacitances reset;
- **GNDD, SUBPV and two GND** : four grounds.

These nine biases are all mandatory.

In addition, **IBP** is an internal current source which must be connected to the ground by a resistance fixed between 15 kΩ to 100 kΩ (typical value 42 kΩ).

Electrical function name	Bias type	Range value	Typical value	Tolerance	Maximum current ⁽¹⁾	Bias rejection
V_{DDA} (analog supply)	fixed	-	3.3 V	± 0.05 V	< 20 mA	- 80 dB
V_{DDO} (analog supply)	fixed	3.3 – 3.6 V	3.3 V	± 0.05 V	< 50 mA	-55 dB
V_{DD} (digital supply)	fixed	-	3.3 V	± 0.3 V	< 30 mA	- 80 dB
GPOL (PV diode bias)	tunable	0 – 1.5 V	0.68V	± 0.05 V	< 2 mA	-13 dB
V_R (capacitances reset)	fixed	-	3.0 V	± 0.05 V	< 30 mA	0 dB
GND (analog electrical ground)	fixed	Ground	0 V	-	-	-
GNDD (digital electrical ground)	fixed	Ground	0 V	-	-	-
GND (output stage electrical ground)	fixed	Ground	0 V	-	-	-
SUBPV (PV detector substrate)	fixed	Ground	0 V	-	-	-
IBP ⁽²⁾ (Output stage current source)	fixed	15-100 kΩ	42 kΩ			

(1) The maximum current corresponds to the peak current in fixed window mode if all diodes were bloomed to 100 times the current saturating one input stage

(2) IBP is internally generated but must be connected to the ground by a resistance.

Figure 6.2: input biases

Noise requirements for V_{DDA}, V_{DDO}, V_{DD} and G_{POL} are shown in Figure 6.3 and Figure 6.4.

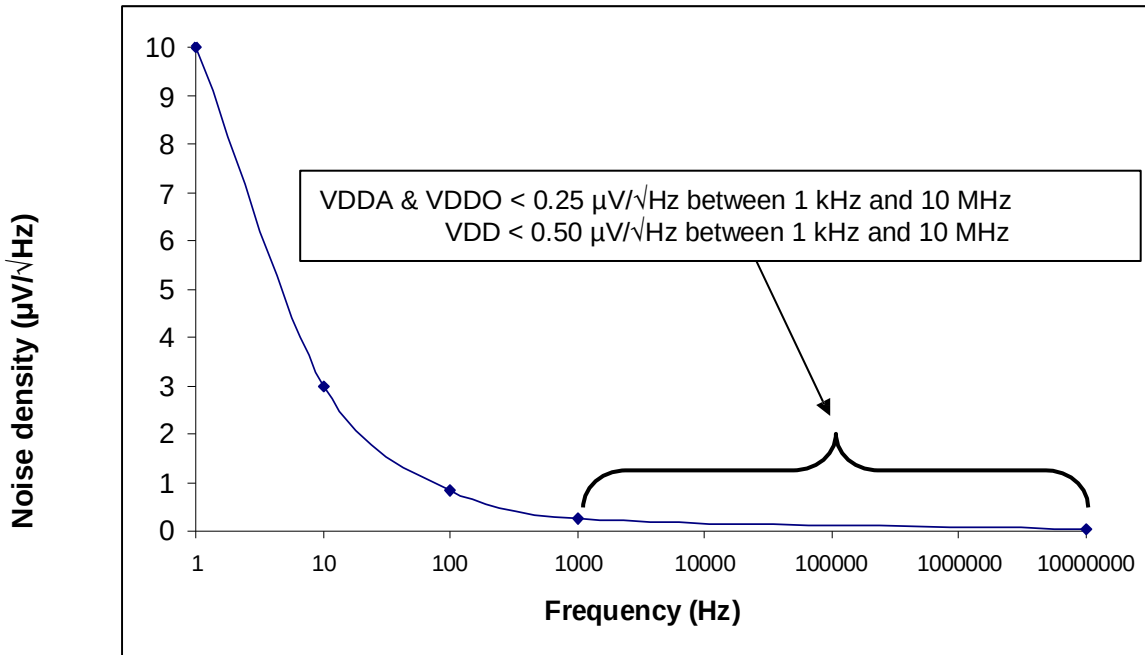


Figure 6.3: V_{DDA}, V_{DDO} and V_{DD} noise spectral density

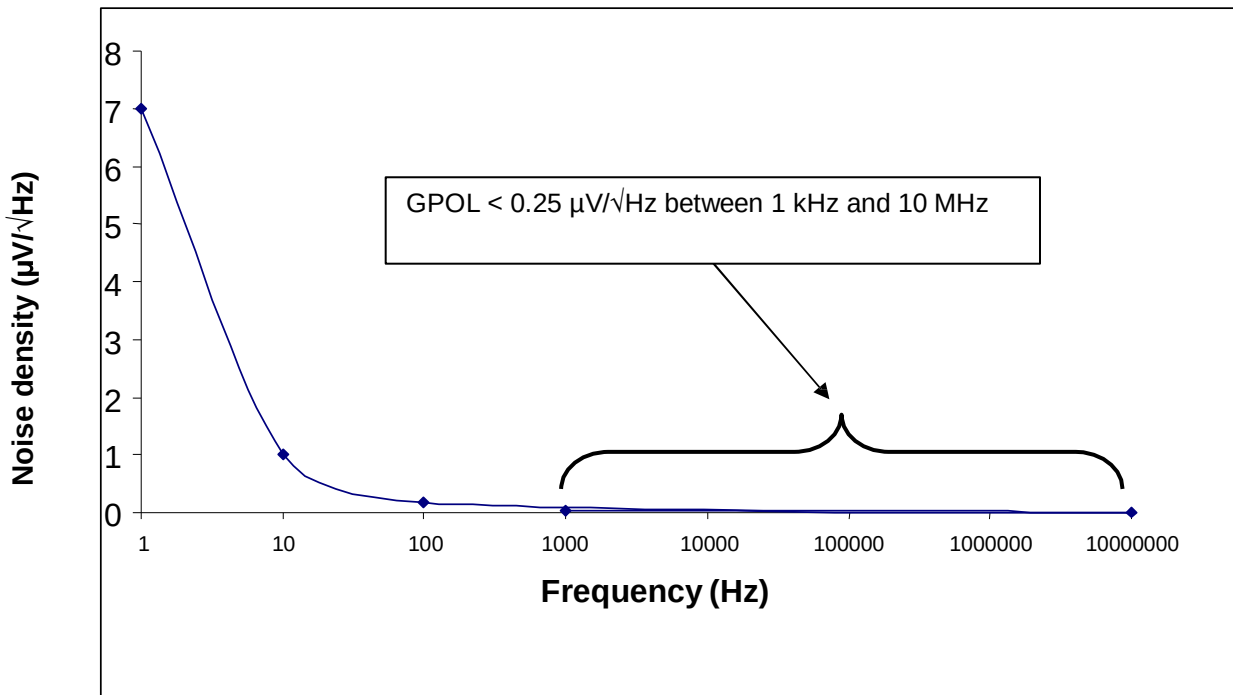


Figure 6.4: GPOL noise spectral density

Note: As G_{POL} is directly related to the polarization of the detector, it's obvious that noise on this bias voltage will induce a perturbation of the signal of the detector and consequently of the output signal. In the same way the output signals are sensitive to a perturbation on the analog supply (V_{DDA}, V_{DDO}) or on the digital supply (V_{DD}).

6.2.2 Pulsed voltages

Only the two pulsed voltages **MC** and **INT** are necessary to operate the sequencer of the CMOS processor.

MC step up and step down times shall be less than 20 ns with a 10 MHz MC.

In order not to generate any jitter on integration time, INT must change its state during the time frame [MC step up $\pm \frac{1}{4}$ MC period] (± 25 ns at 10 MHz) and must be stable outside this interval.

Besides these two pulsed voltages, seven other pulsed voltages can be used for optional modes (input stage operational mode, windowing mode capabilities, rotation mode capabilities):

- **ITR** is used to choose between Integration While Read (IWR) mode or Integration Then Read (ITR) mode, with the associated storable charges (see § 5.1).
- **SIZE-A** and **SIZE-B** are used for the selection of the format (three fixed formats and windowing mode). (See § 5.6)
- **SERCLR** and **SERDAT** are used to send the window coordinates in the windowing mode. (See § 5.6.2)
- **UPROW** and **UPCOL** are used to determine the readout direction of the addressed window. (See § 5.7)

Figure 6.5 summarizes the characteristics of these pulsed voltages.

Electrical function name	Low level (V)		High level (V)	
	Typical	Required	Typical	Required
MC (master clock)	0	< 0.6	V_{DD}	$> V_{DD} - 0.6$
INT (integration time adjustment)				
SIZE-A * (format adjustment 640×512, 640×480, 512×512 or window mode)				
SIZE-B * (format adjustment 640×512, 640×480, 512×512 or window mode)				
SERCLR * (reset of serial interface clock)				
SERDAT * (serial interface data)				
UPCOL * (selection of columns readout direction)				
UPROW * (selection of rows readout direction)				
ITR * (selection of input stage operating mode)				

(*) options not compulsory

Figure 6.5: Pulsed voltages

6.3 Outputs signal characteristics

Two types of outputs are used for the CMOS processor:

- four analog outputs : **OUT0**, **OUT1**, **OUT2** and **OUT3**
- two digital outputs: **DATAVALID** and **ERROR**.

6.3.1 Analog outputs

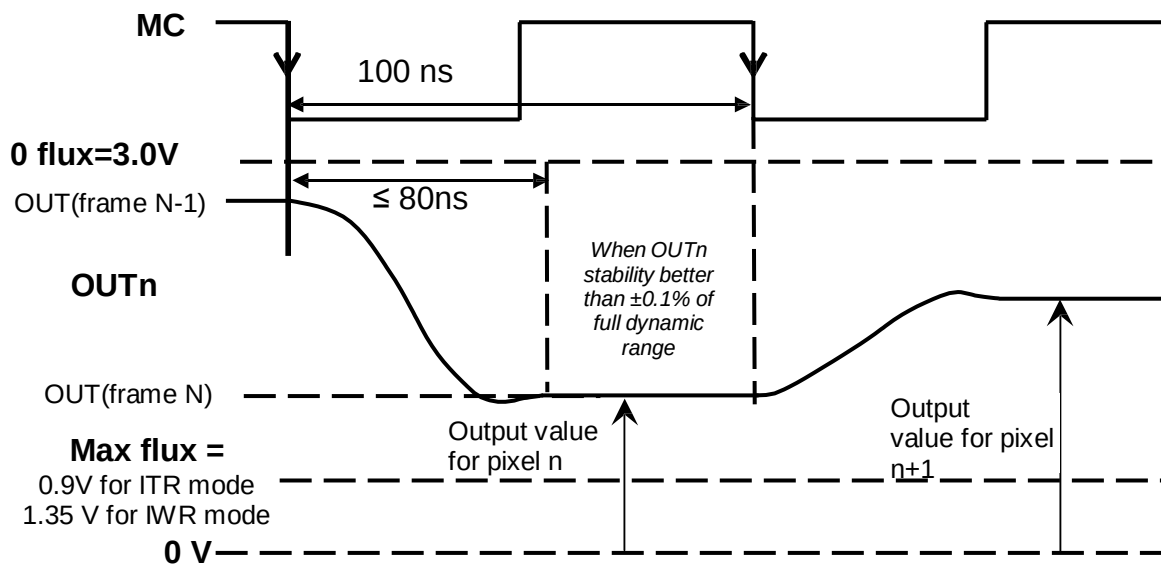
6.3.1.1 Signal waveform, settling time

The information output period is the master clock period.

Output signal waveform is described in Figure 6.6 with a 10 MHz MC.

The settling time of the output signal is defined as the time for the output signal to reach a stability better than 0.1% of the maximum output dynamic range as specified in § 6.3.1.2

The settling time is ≤ 80 ns ($\Rightarrow$ stability range for signal sampling at 10 MHz is about 20 ns).



Example: OUTn settling time with 10 MHz MC $\Rightarrow$ 100 ns MC period; and with $R \geq 100\text{k}\Omega$ and $C \leq 15\text{pF}$

Figure 6.6: Output signal waveform

6.3.1.2 Analog outputs dynamic range

With the external load of § 6.3.3, the OUTn output signal typical levels are :

Mode	Low level (saturation)	High level (zero flux)	maximum output dynamic range	usefull output dynamic range
ITR	0.9 V	3 V (zero flux)	2.1 V	1.5V (between 2.9V and 1.4V)
IWR	1.35 V	3 V (zero flux)	1.65 V	1.5V (between 2.9V and 1.4V)

Figure 6.7: Analog outputs table

The useful dynamic range considers a linearity criteria $> 99.8\%$ (non linearity $< \pm 0.2\%$).

Note: when considering average DC levels over the full array, the cold shield dome effect of the IDCCA may limit the dynamic range over the whole array to a lower value than the above one, in order to avoid saturation of central diodes.

In addition in the ITR mode, the signal varies from 3 V for the zero-flux level down to 0.4-0.5V for couple fluxes-integration time greater than saturation level indicated above.

6.3.2 Digital Outputs

The digital output **DATAVALID** on its high level indicates the presence of valid data on the analog outputs (see § 5.5.1).

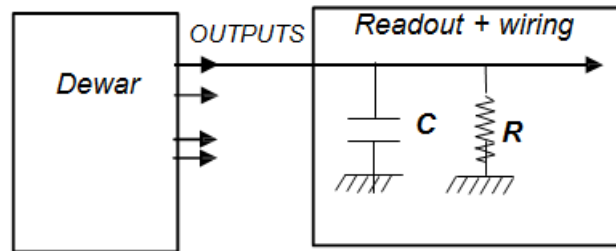
The digital output **ERROR** on its low level indicates the presence of valid data in SERDAT for the windowing mode and is only used for this mode (see § 5.6.2).

Electrical function name	Low level	High level
DATAVALID (digital output)	0 V	V_{DD}
ERROR (serial interface digital output)	0 V	V_{DD}

Figure 6.8: Digital outputs table

6.3.3 Load impedances

The analog outputs **OUTn** can be loaded by a resistance $R \geq 100 \text{ k}\Omega$ in parallel with a capacitance $C \leq 15 \text{ pF}$ external to the dewar.



DATAVALID and ERROR digital outputs can be loaded by a resistance $R \geq 10 \text{ k}\Omega$ in parallel with a capacitance $C \leq 80 \text{ pF}$ external to the dewar.

6.4 ROIC electrical power dissipation

The typical electrical power dissipation of the ROIC is less than 50 mW in the default mode (SIZE-A = 1, SIZE-B = 1, MC = 10MHz, Ribp = 42k Ω).

The maximum electrical power dissipation of the ROIC is less than 70 mW in any other cases.

6.5 Default mode and minimum electrical interface

The ROIC default operation mode is the following:

- Array Format : 640×512 (SIZE-A =1 and SIZE-B = 1)
- Integration Mode : Integration Then Read (ITR = 1) $\Rightarrow$ 6.36Me- storable charges
- Image flip mode : No flip (UPCOL = 1 and UPROW = 1)

The minimum dewar electrical interface to operate the ROIC in the default mode is 18 I/O as follows :

- the 9 biases listed in § 6.2.1, including 4 grounds
- The IBP internal current source which must be connected to the ground by a resistance fixed between 15 k Ω to 100 k Ω (typical value 42 k Ω).
- the 2 pulsed voltages MC and INT
- the 4 outputs OUT0 to OUT3
- the interface DTA and DTK of one temperature sensor for cooler control loop

7. USER'S CAUTIONS REGARDING ELECTRICAL INTERFACES

The electrical interfaces of the FPA are defined at IDDCA level by the pinout described in the mechanical interfaces of the delivered detector (refer to the contractual TDP).

If a command and control electronic for the IRFPA (proxy board) is part of the delivery, the user shall refer to the proxy board TDP for the electrical interfaces of the set (IDDCA + FLEX + PROXY board).

The user shall take all the cautions when handling the detector, when connecting the electrical interface of the detector, and when powering the detector, to avoid any electrical misuse or electrostatic discharges.

7.1 Cautions against electrostatic discharges



Such as any electronic component, infrared detectors are highly sensitive to electrostatic discharges (ESD), when the short circuit board is taken off the IDDCA electrical pins or connector.

To avoid any electrostatic discharge on the FPA, the following shall be respected:

- Do not remove the short-circuit board of the IDDCA, unless when required to plug the Flex/electronics board during integration of the IDDCA in the system
- Manipulate IDDCA in clean room with humidity level between 40% and 60%;
- Handle the IDDCA on a static free work station (grounded workbenches through a 1MOhm resistor) and equipped with grounded carpet;
- Handle the IDDCA while grounding wrist-strap is properly worn and grounded (through a 1MOhm resistor);
- Clean work surfaces and wrist-straps to ensure maximum efficiency;
- Wear protective clothing containing conductive material;
- Ground the electronic equipment / board before plugging it onto the IDDCA;
- Short-circuit the IDDCA electrical interface with the specific short-circuit board delivered with the IDDCA, when IDDCA is

disconnected from the Flex/electronics board.

7.2 Cautions against electrical damage

To avoid any electrical damage on the FPA, the following shall be respected:

- Ensure that pulsed voltages values in Figure 6.5 are respected,
- Ensure that input biases values in Figure 6.2 are respected,
- Never apply voltage $> 3\text{ V}$ on GPOL,
- Never apply voltage $> 2\text{ V}$ on SUBPV (PV substrate),
- Never apply voltage $> 3.3\text{ V}$ on other bias voltages,
- Never apply negative voltage (reverse bias) with respect to Silicon substrate,
- In any case, the mean direct current must be lower than 50 mA.
- Never turn ON the FPA or let the FPA turned ON, when it is not cooled down in a cryogenic temperature range. Indeed, the FPA being in a vacuumed sealed dewar, its own electrical power dissipation could damage it by self heating.

7.3 Electrical cautions for obtaining correct performances

To ensure the best performance out of the IDDCA, the following shall be respected:

- Ensure that pulsed voltages values in Figure 6.5 are respected,
- Ensure that input biases values in Figure 6.2 are respected,
- Follow the recommendations on the signal noise in Figure 6.3 and Figure 6.4,
§ 6.2.1
- Follow the recommendations on the outputs load impedances § 6.3.3
- Some IDDCA pins shall be grounded. Ensure to ground them as indicated on the lead distribution of the mechanical interface drawings provided in the TDP.

Ground connections are dependant on the system design.

For general cases, we recommend to connect the cold shield ground to the ROIC ground (GND or GNDD) and to connect the dewar mechanical ground to the ground of the proximity electronic board.

Otherwise, as a minimum, all different grounds should be short-circuited together on the same ground plane.

7.4 Cautions when powering the proxy board

The proxy board developed by the customer shall respect the specified maximum authorized voltages and currents values of this document.

The proxy board shall not be powered when connecting it to the IDDCA.

The FPA shall not be ON without the cooling system ON. Therefore when turning ON the proxy board, it is required to bias the FPA only when it is in cryogenic temperature range (below 120K for instance).

When turning OFF the system or in case of cooler failure that would lead to the stop of cooling while the general system is still ON, it is mandatory to stop biasing the FPA to avoid damage due to power dissipation inside the vacuumed dewar. It is therefore recommended to drive the ON/OFF of the IRFPA biases with respect to the operating temperature below 120K for instance.

(End of the document)

Annex 4

Title: XD-2 Stirling Cooler Interface

Reference: XD-2-SC-1801

Issue 1 Revision 0

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

[illegible]

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1. INTRODUCTION

This technical document describes the interfaces of the XD-2 coolers, delivered as an integrated microcooler in NCRIEO Integrated Detector Dewar Cooler Assemblies (IDDCAs).

The cooler electrical and thermal interfaces do not depend on the Focal Plane Assembly (FPA) size or wavelength (MW, LW) implemented in the dewar cooled by this cooler.

Therefore this document aims at being a generic document applicable to all standard IDDCAs delivered with a XD-2 cooler, and as such, is an annex of the Technical Data Package (TDP) delivered to the customer when such an IDDCA is ordered.

The contractual TDP delivered to the customer annexes this ICD in its latest update that will in any case supersede the ICD in lower issue that could be provided prior to the TDP.

It may also happen that the delivered IDDCA is required with a cooler with modified mechanical and/or electrical interfaces and/or thermal management requirements.

Therefore, one shall always refer to the amendments pages of the contractual TDP (if any) to check for modifications that would amend and/or modify paragraphs of this document.

2. REFERENCED DOCUMENTS

N/A

3. LIST OF ACRONYMS USED IN THIS DOCUMENT

ATR	Acceptance Test Report
FPA	Focal Plane Array
I/O	Inputs / Outputs
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
MW	Mid Wave band = 3-5 μm
TDP	Technical Data Package

Note:

The yellow warning sign  highlights an important note to be respected.

The red warning sign  highlights an important note that could lead to a permanent failure if not respected.

4. COOLER GENERAL DESCRIPTION

4.1 XD-2 cooler

The XD-2 cryogenic cooler is a miniature closed-cycle Stirling.

This machine consists of a pressure wave generator and a cold finger attached together to the same crankcase in a monobloc (integral) configuration.

The monobloc cooler includes the electrical motor and the rotating arrangement driving the compressor piston as well as the displacer. The displacer contains the heat exchanger of the system.

The displacer is fitted in a dewar in an IDDCA concept in order to reduce thermal losses and therefore enhanced the specific power of the cooler.

4.2 Cooler controller

The XD-2 microcooler controller is not included in the cooler stator. This COOLER DRIVE cooler controller is described in document [REF 1].

If the COOLER DRIVE cooler controller is part of the XD-2 cooled IDDCA delivery, on shall refer to document [REF 1] for the applicable electrical interfaces to drive the cooler, and the following cooler electrical interface should be considered only for information.

If the customer develops its own cooler controller, it is mandatory that the following paragraphs shall be respected.

The cooler controller must follow the principle below:

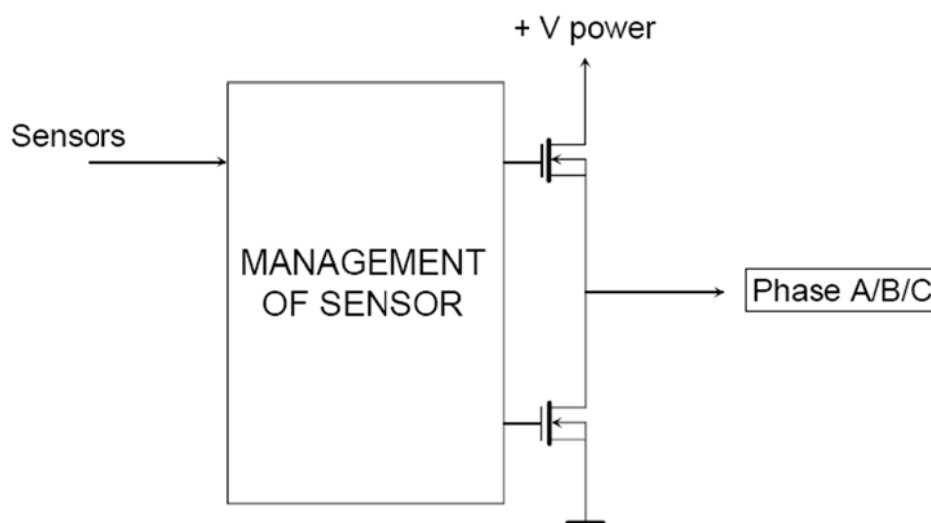


Figure 4.1: cooler controller logic

5. COOLER MECHANICAL INTERFACES

5.1 Cooler size

The cooler being an integral part of the IDDCA, its mechanical interface is described in the IDDCA mechanical interface (drawings) delivered in the contractual Technical Data Package.

5.2 Electrical connection

The XD-2 cooler does not have a connector. Its electrical interfaces are ensured by eight pins described in Figure 6.4.

The pins have the following characteristics:

- Diameter : $\varnothing 1\text{mm} \pm 0.03\text{mm}$,
- length = $4\text{mm} \pm 0.2\text{mm}$,
- angular position: at 45° on a $\varnothing 13.2\text{mm}$ diameter.

A circular connecting board shall be developed at the end of the controller cable for easy plugging on these pins. The electrical connection of the motor must be achieved with such a specific connector to avoid soldering damage.



It is forbidden to solder the controller electrical interfaces on the cooler pins.

5.3 Attachments in the system

5.3.1 Cooler mounting Surface

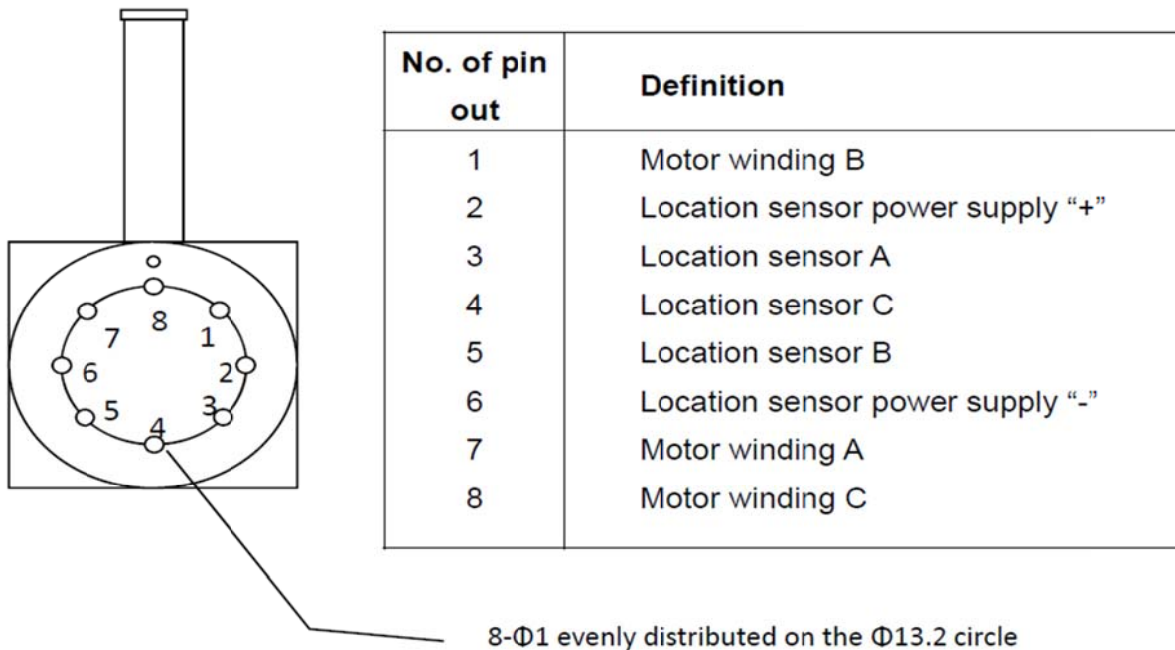
The cooler's mounting surface with the system is the IDDCA attachment plane and is defined in the drawing in the IDDCA Technical Data Package.

5.3.2 Cooler Clamping Method

For the XD-2, the cooler mounting surface shall be clamped to the system by four M2.5 screws, using the index hole and groove in order to achieve the desired accuracy with respect to the optical line of sight.

6. COOLER ELECTRICAL INTERFACES

6.1 Definition of the pin out of the electrical interface of the cooler.



6.2 Definition of the electrical interface of the cooler drive control PCB.

Red wire	power supply + (9V)
White wire	power supply –
Yellow wire	Diode (+)
Blue wire	Diode (–)

6.3 Connection of the cooler

- 1) Plug the pins on the cooler motor into the corresponding pin caps on the ring interface of the cooler drive control PCB.
- 2) Connect the DC power supply to the cooler drive control PCB and adjust the power supply voltage to 9V. Connect red wire to power supply (+). Connect white wire to power supply (-).
- 3) Connect the diode feedback wires to the corresponding diode anode and cathode leads (check the tape on the lead) on the A/D conversion PCB to form a feedback loop. Connect yellow wire to the diode (+) and blue wire to the diode (-).
- 4) Power on and the cooler should operate properly.

7. COOLER THERMAL INTERFACES

7.1 General

An infrared system (FLIR camera, gimbal, seeker,...) is a complex and compact system whose electrical and electro-optical performances have to be maintained during the full operational hours, as well as during the whole system life.

The thermal interface shall be precisely known and addressed in order to design a system with a correct thermal management avoiding global performance degradation.

The cooler body dissipates heat. It is requested to ensure that inside the system, this heat is correctly evacuated from the skin of the running cooler. Take care to design the mechanical interface of your system from a thermal point of view.

7.2 Heat sink

Caution: Care must be taken to reject heat generated by the compressor/motor assembly.



The surface temperature of the whole cooler body shall not exceed the ambient temperature by more than +15°C.

(refer to the IDDCA specification for the maximum power consumptions at different ambient temperatures during cool down and during regulated phase).

Therefore
:

- **Adequate heat sink and/or convective cooling shall be provided by the user** at least through the cooler attachment plane
- and/or adequate air flow of thermal link shall be provided by the user around the cooler stator.

8. TEMPERATURE REGULATION LOOP CONTROL

8.1 FPA regulation temperature settling point

One temperature sensor of the IDDCA is provided to feed the cooler controller for the temperature regulation.

- The sensor anode and cathode have to be connected to the cooler controller electrical interface.

One other temperature sensor of the IDDCA is provided to monitor the detector temperature.

- Refer to the FPA electrical interface document provided in the TDP, for the full

description of the temperature sensor interface (current and $T=f(V)$ function).

8.2 Standard temperature set point

The regulation temperature is adjusted at NCRIEO during Acceptance test.

Refer to the Acceptance Test Report (ATR) of the delivered IDDCA for the FPA temperature set during acceptance test and for which the electro-optical performances have been recorded.

8.3 Modification of the FPA regulation temperature settling point

The customer shall set the cooler controller regulation point in order to obtain the same FPA temperature as the one recommended by NCRIEO in the ATR.



If the customer wishes to change the FPA temperature set by NCRIEO during acceptance test, the Electro-Optical and cryogenic performances may not be guaranteed anymore.

9. USER'S CAUTIONS

9.1 Cooler assembly and dismantling

The cooler is an integral part of the detector since they share the same cold finger.

As a consequence, the integrated cooler must not be dismantled and reassembled without following a specific procedure.

This operation is only performed by NCRIO or maintenance.

Contact NCRIO for possibility of cooler maintenance at customer site.

9.2 Cautions on electrical interface

Refer to § 4 for details on electrical interface to be respected.

The Hall Effect sensors of the cooler electrical engine can be damaged by an inappropriate electrical connection. It is therefore highly recommended to connect the cooler with great care in order to ensure a connection conform to the electrical specifications.



The cooler pins location being symmetrical by 45° step, great care shall be taken when plugging the controller connector in order to avoid an incorrect plug following a 45° angle error. Pin n°8 is the one closest to cold finger, identified by a reference feature as shown in Figure 6.4.

The cooler pins are fragile. The mounting and dismantling of the controller connection shall be performed with great care, and shall be minimized.

(End of the document)

Annex 5

Title: Electrical Interface of

MINI640 IDDCA

(with 15um pixel pitch and XD-2 Stirling cooler)

Reference: EI-MINI640-IDDCA-1801

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

ELECTRICAL INTERFACE OF NCRIEO DETECTORS		
ISSUES	EVENTS	DATES
1	Creation of the document	01/2018
2	Modification of Gpol description and Annex	12/2018
3		
4		

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1. INTRODUCTION

The purpose of this document is to describe the main features and electrical interfaces of the 640x512 MWIRCMOS Integrated Detector Dewar Cooler Assemblies (IDDCAs) for MINI640.

These IDDCAs are equipped with a non dismountable Control and Command Electronics (C&CE) in order to provide a multiplexed and digitized video output to the system at 60Hz maximum frame rate. As such, the IDDCA + C&CE constitute the MINI640 Engine.

In the whole document,

- “*board*” or “*C&CE*” will stand for “*MINI640 board*”.
- “*MINI640 engine*” will stand for “*the MINI640 IDDCA associated with its C&CE*”.

Note: the coolers electrical and mechanical ICDs are not described in this document but in documents [REF 3], [REF 4], and [REF 5].

The C&CE enables the user of a MINI640 engine to easily interface its system electronics (image processing board) with the detector, by driving the IRFPA with the specific input biases and clocks, and by providing the system electronics with a digitized detector output, after ADC conversion and multiplexing of the analog outputs of the infrared detector.



We strongly recommend that the Users read this document before operating the MINI640 engine.

All the people who have to handle this product should know the precautions to be used for handling it.

This document aims at being a generic document applicable to MINI640 engines, and as such, is an annex of the Technical Data Package (TDP) delivered to the customer when a MINI640 engine is ordered.

The contractual TDP delivered to the customer annexes this ICD in its latest update that will in any case supersede the ICD in lower issue that could be provided prior to the TDP.

2. REFERENCED DOCUMENTS

[REF 1] AIA Camera Link version 1.2: Specifications of the Camera Link Interface

Standard for Digital Cameras and Frame Grabbers

[REF 2] MODBUS over serial line specification and implementation guide V1.02: Specifications of Cyclical Redundancy Checking generation.

- [REF 3] NTG029: K563 Cooler Interfaces Control Document
- [REF 4] NTG008: XD-2 Cooler Interfaces Control Document.
- [REF 5] NTG015: Digital external controller for XD-2 cooler, Interfaces Control Document.

3. **LIST OF ACRONYMS USED IN THIS DOCUMENT**

ADC	Analog to Digital Converter
FPA	Focal Plane Array
FPGA	Field-Programmable Gate Array
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
IRFPA	Infra Red Focal Plane Assembly
MC	Master Clock
ROIC	Read Out Integrated Circuit
TDP	Technical Data Package

Nota:

The yellow warning sign  highlights an important note to be respected.

The red warning sign  highlights an important note that could lead to a permanent failure if not respected.

4. **GENERAL DESCRIPTION**

4.1 General description of MINI640 Engine

The product is an infrared opto-electronic device sensitive to radiation in the 3.7 to 4.8 micrometer spectral region. It includes a high-sensitivity focal plane array (FPA) comprised of a 640x512 element two dimensional detector array made of photovoltaic (PV) Mercury Cadmium Telluride (HgCdTe / MCT) connected to a Silicon readout integrated circuit (ROIC) by indium bump technology, and a 1/4 Watt Stirling cycle rotary microcooler integrated into a metal dewar.

A C&CE is mounted on the dewar housing and is in charge of focal plane management and Analog to Digital conversion of the video signal.

As a summary, the sub-assemblies of the product concerned by this technical specification are:

- IDDCA
 - Dewar (including cold shield, cold filter and window)
 - Detector or IRFPA, composed of :
 - PV array (640x512 format, MW , 15μm pitch)
 - ROIC.
 - Cooler
- Cooler Controller board.
- C&CE.

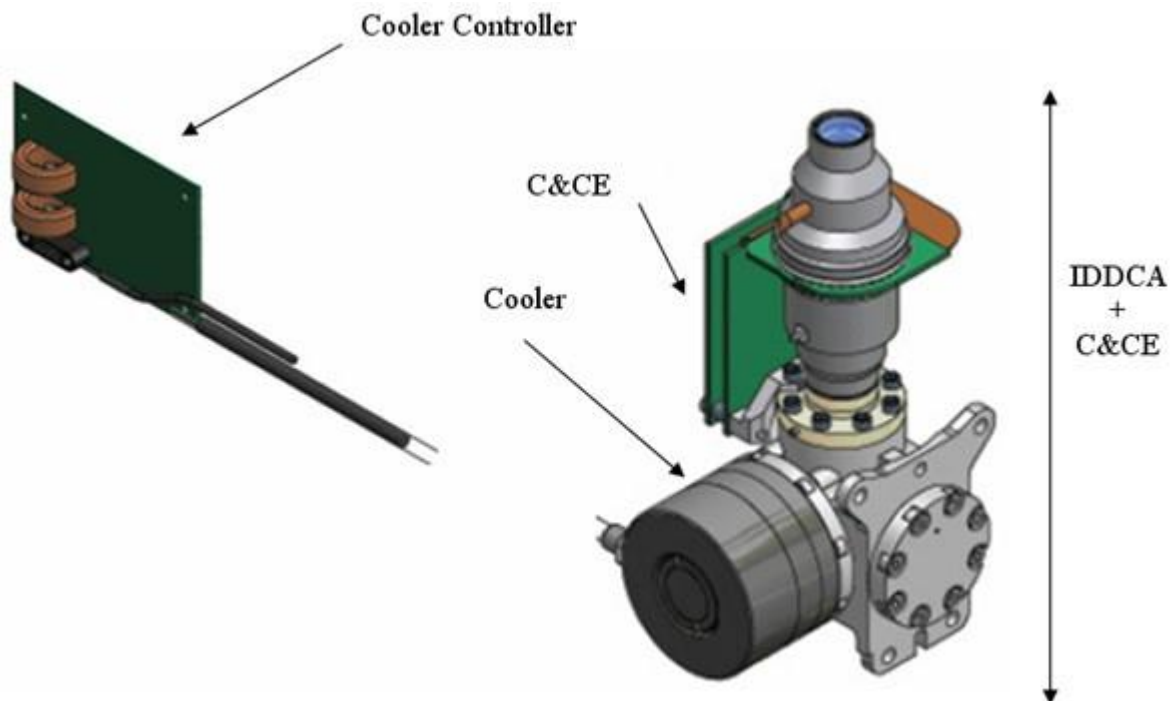


Figure 1: MINI640-XD-2

The block diagram of Figure 2 illustrates how the MINI640 engine interfaces with a host system in order to acquire the IR detector signal, to process it and to visualize it.

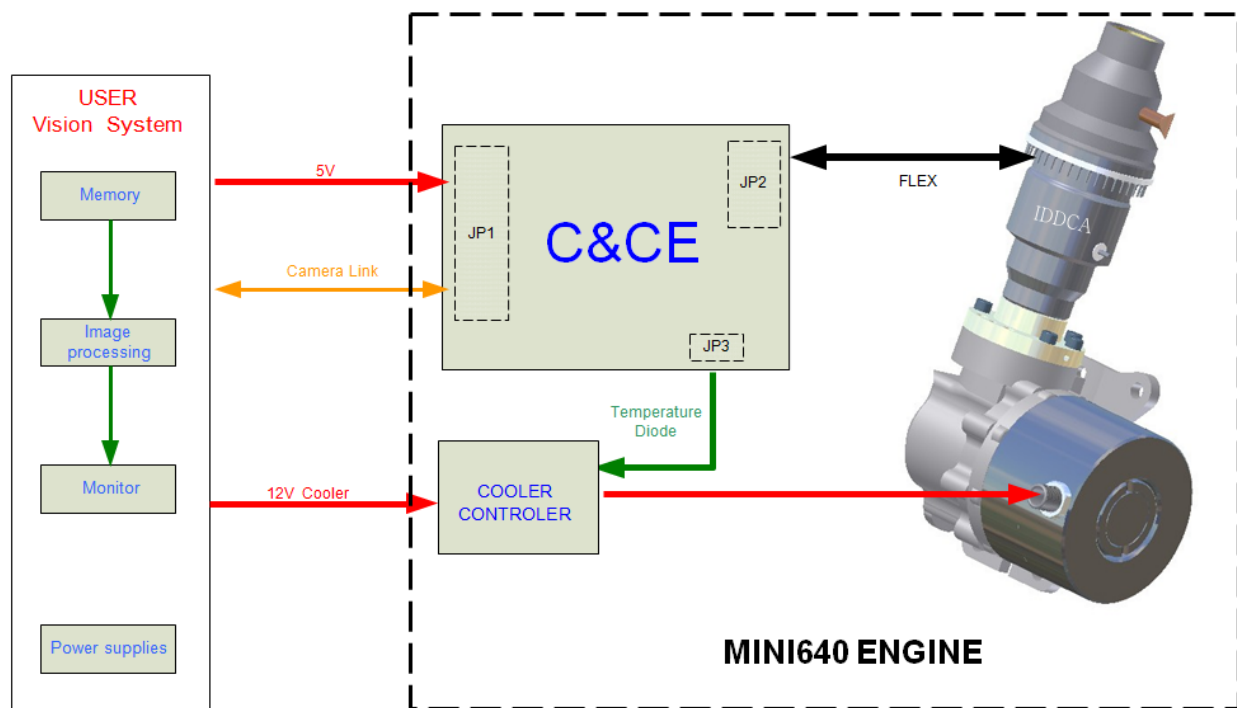


Figure 2: Synoptics of MINI640 engine

The user has to complete the MINI640 engine with the followings in order to have a fully operational system:

- Power supplies,
- Customer designed or commercially of the shelf frames grabber,
- Image processing system (hardware and software) for bad pixel replacement, Non Uniformity Correction, Automatic Gain Control and other signal analysis,
- Personal Computer and/or video monitor.

4.2 Focal plane array description

4.2.1 Photovoltaic array topology

The Mercury Cadmium Telluride (MCT) Photovoltaic (PV) array is composed of 640 columns $\times$ 512 rows of pixel.

The pitch is 15 μm in both directions.

4.2.2 ROIC general description

4.2.2.1 ROIC Architecture

The ROIC design is based on a Silicon CMOS technology. The ROIC is composed of 327680 diodes multiplexed in 512 rows and 640 columns.

Its organization permits the FPA

- to integrate simultaneously (snap-shot) the information from the PV diodes (coupled by direct injection),
- to store this information,
- to multiplex this information per row,
- to multiplex the rows information
- and to sample and hold the information.

4.2.2.2 Operating modes

The ROIC operates in the **snap shot mode** (all pixels of the array are integrated in the same time).

The ROIC design allows the user to select different operating modes depending on the application:

- Integration mode and gain adaptation
 - **Integration Then Read (ITR)** : the signal of frame N integrated in the input stage is read out after the end of the integration time of frame N and the integration of the signal of frame N+1 starts after all the signals of frame N have been read out.
In ITR mode, the gain (storable charges) is **6.36 Me-**.
 - **Integration While Read (IWR)**: the signal of frame N integrated in the input stage is read out during the integration time of frame N+1.
In IWR mode, the gain (storable charges) is **4.99 Me-**.
- Programmable integration time
 - Minimum charge integration duration : 1.5 MC period (with min INT = 3077 MC)
 - Minimum step for integration time change: 1 MC period
 - Maximum integration time duration : no limitation
- Windowing mode
 - 640×512 full array (default mode, at power ON)
 - 640×480
 - 512×512
 - Random windowing mode within 640×512
- Horizontal and vertical Image flip mode
 - Column invert
 - Line revert
 - Image rotation (= invert + revert).

4.3 **C&CE description**

The board is made of a single folded set (see Figure 3) connected to the IDDCA electrical interface and screwed onto the IDDCA body by three screws.

The board shall not be disconnected from the dewar without NCRIEO authorization.

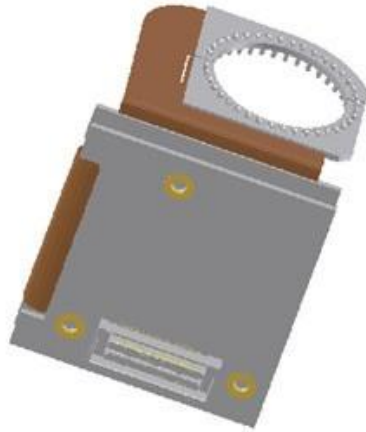


Figure 3: MINI640 C&CE

The board main features are:

- Providing all low noise power supplies to the ROIC
- Generating all internal power supplies out of one input voltage
- Containing a clock generator for all necessary timing signals to be sent to ROIC
- Providing an input interface for external synchronization of the ROIC
- Providing an input interface for control and commands of the ROIC
- Including the analog signal processing between ROIC output stages and ADC Input stages
- Including four 14 bit ADCs for each video channel
- Providing a user interface to the IDDCA temperature sensors
- Providing a built in test (BIT) including test Pattern (output of a grey level wedge (Intensity ramp) test image).

The C&CE converts the analog video signals of the detector into a digital video channel at up to 60 Hz frame rate.

The internal Master Clock value is dependent on the MINI640 configuration:

- The programming of internal Master clock at maximum 5.55MHz (50/9 MHz) enables a frame rate at 60Hz.
- The programming of internal Master clock at 5.0MHz (50/10 MHz) limits the frame rate at 58Hz).

Note: This configuration is not set dynamically, this setting is done in factory.

The analog data from each pixel are converted to digital form. The ADC digitizes the entire dynamic range to 14-bit resolution. The digital code is unsigned binary code and an increasing code represents an increasing scene temperature (it can be inverted by C&CE register).



The C&CE has followed a ROHS process.

4.3.1 Board synoptic and functional diagram

The global architecture is presented by the following figure:

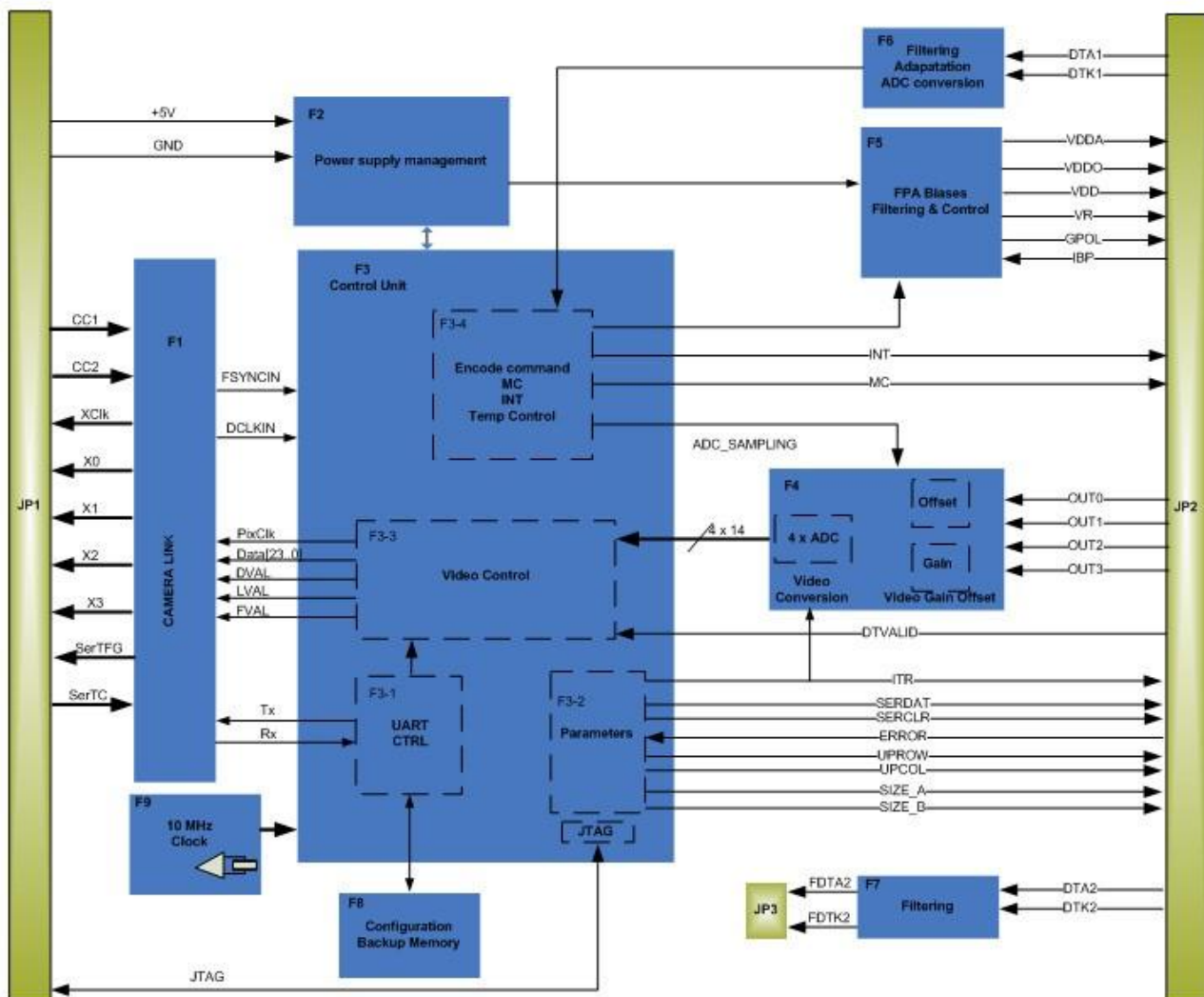


Figure 4: C&CE block diagram overview

All the functionalities of the board are managed by a **FPGA** with external flash memory. Communication with system is done through **CAMERALINK UART** protocol allowing to access to the flash configuration memory and to vital board status information.

The board allows the user to select with communication interface different operating modes depending on:

- Definition of the integration time duration.
- Selection of the operating mode (ITR or IWR), the gain of the input stage (storage capacitance) depends on the selected mode.
- Selection of the format: 3 fixed formats (640x512, 640x480 and 512x512) and a random windowing mode for sub frame imaging are available.
- Selection of rotation functions: choice of the readout direction for lines and/or columns.

4.3.2 Board physical interfaces (connectors)

Referring to Figure 4, the C&CE has two accessible connectors **JP1** and **JP3**:

Connector	JP1: to system	JP3: to cooler controller
		
Manufacturer	SAMTEC	NICOMATIC
Manufacturer reference	QSH-30-01-C-D-A	101R03LS Male Connector 1x3 pins Single row 90° SMT
Mates with	QTH-30-01-C-D-A: High Speed Terminal Strip. or HQCD High Speed Cable (Remark: holes of this cable are not compliant with holes on this board for locking)	NICOMATIC – 102.S.03 (version crimp assembly for cable AWG=#24 - #28). or Others versions possible (PCB/ SMT, 90°, Straight..). See Nicomatic website for more information
Latch	This connector doesn't include latching system. If necessary, latch must be done with the two holes located on both sides of the connector. See mechanical drawings for more details.	latch on connector

For more details, refer to [SAMTEC Website](#) or NICOMATIC website.

The **JP2** connection between the board and the IDDCA is made by plugging the board onto the ring of pins of the IDDCA. The board is delivered mounted on the IDDCA and shall not be dismounted.

4.3.3 **Board functions**

Referring to Figure 4, the main functions of the C&CE are defined in the table below:

C&CE internal functions	
F1	Video stream serialization with a Cameralink protocol
F2	Power supply management: in charge of the filtering and generation of all the internal voltages.
F3	Control unit: generates the synchronization signals, the integration time and the master clock used by the detector; configures the detector through the FPA serial link, through the system serial link. • F3-1: UART control • F3-2: serial interface Parameters • F3-3: video control • F3-4: Encode command, MC, INT, temperature control.
F4	FPA video Gain Offset adjustment and analog to digital conversion
F5	FPA Biases
F6	Temperature sensor #1 filtering (provides the temperature feedback for user interface).
F7	Temperature sensor #2 filtering (provides the temperature feedback to the cooler).
F8	FPA Configuration backup memory.
F9	Main clock used for internal mode and board management.

4.4 **Mechanical interfaces**

4.4.1 **Weight**

The weight of MINI640-XD-2 is:

IDDCA + C&CE: < 355g.

External cooler controller board: < 30g for the board and < 30g for the cables.

4.4.2 **Size**

The overview drawings of the MINI640 products are given in annex for information. All dimensions are given in millimeters.

However the full detailed drawings of the delivered IDDCA will be provided in the contractual Technical Data Package

5. DETAILED DESCRIPTION OF THE ROIC FUNCTIONS

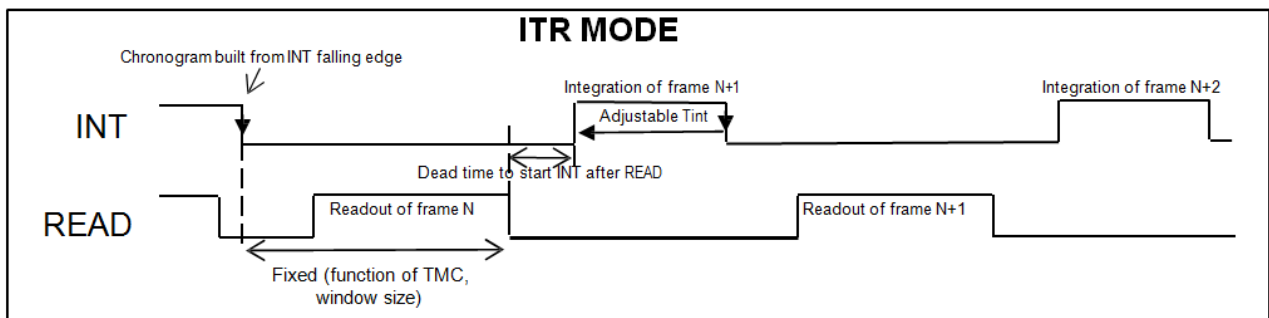
5.1 Input stage: integration mode (ITR function)

The input stage design is based on the direct injection principle.

During integration time, charges are stored in the input capacitance.

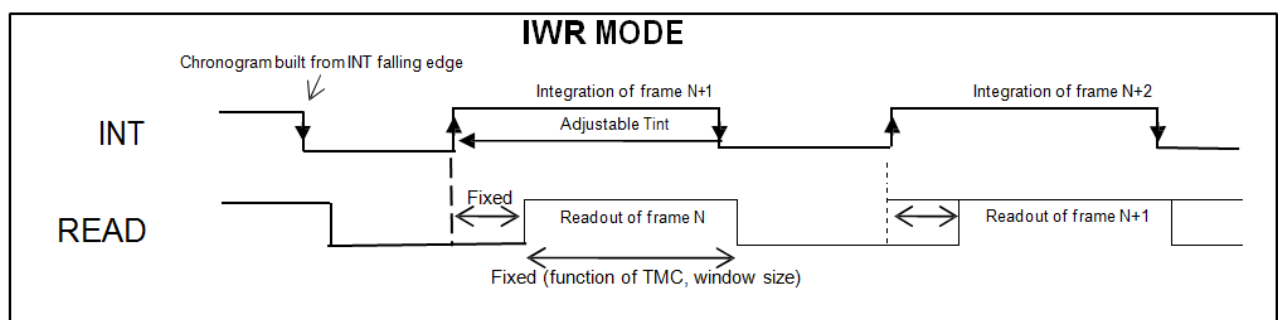
The input stage can be operated according to two different modes:

- **Integration Then Read (ITR)**
 - the signal of frame N integrated in the input stage is read out after the end of the integration time of frame N and the integration of the signal of frame N+1 starts after all the signals of frame N have been read out.



- **Integration While Read (IWR)**

- the signal of frame N integrated in the input stage is read out during the integration time of frame N+1.



5.2 Storable charges and dynamic range

ITR function allows choosing the operational mode according to the following table, with consequence on the storable charges:

ITR	Operation mode	Storable charges under	
		maximum output dynamic range	linear dynamic range
1 (Default)	ITR	0.485pF over 2.10V = 1.02 pC ($6.36 \cdot 10^6 e^-$)	0.485pF over 1.5V = 0.78 pC ($4.84 \cdot 10^6 e^-$)
0	IWR	0.485pF over 1.65V = 0.8 pC ($4.99 \cdot 10^6 e^-$)	0.485pF over 1.5V = 0.78 pC ($4.84 \cdot 10^6 e^-$)

Figure 5.1: selection of operational mode with ITR

Dynamic output is automatically managed by C&CE (with ITR register) for Signal to Noise Ration optimization on the analog to digital converter chain.

When input stage capacitance is full, charges which can't be stored are flowed away thanks to an antiblooming gate integrated in each input stage.

5.3 Charge-voltage conversion

At the end of the integration time, the charges are converted into voltages. The charge-voltage conversion is performed by amplifiers.

Each charge-voltage conversion stage is controlled by internally generated biases.

The readout capacitances are sequentially addressed line by line by 640 amplifiers located at each column end.

Four multiplexers (160 to 1) operating simultaneously allow the transfer of the signals from the amplifiers to the four outputs.

5.4 Output stages

There are four output stages (one for each multiplexer stage) labeled OUT0 to OUT3 which are digitized by the C&CE at a maximum output rate of 5.55MHz per output for a maximum frame of 60Hz.

5.5 Sequencer

The sequencer generates all internal signals necessary for the readout integrated circuit operation from the different external clocks and bias voltages. The readout integrated circuit operation is totally synchronous. All the pulses generated internally are multiple integers of the period of the master clock. A change in the master clock frequency induces a change in these pulses.

The clocks necessary to operate the sequencer are the master clock and the integration phase managed by C&CE :

5.5.1 Master Clock (MC)

The Master Clock (**MC**) has a maximum frequency of 5.55 MHz (180 ns MC period) generated by C&CE and a duty cycle of 50 %;

It is the base for the synchronism of the operation of the whole circuit.

The pixel addressing is performed from static synchronous counters controlled by the master clock.

The pixel information is read out at a rate of one per master clock period.

5.5.2 Integration phase (INT)

The integration phase (**INT**) allows the integration of the charges in the pixel input capacitance (the PV diodes are biased during the high level of INT signal).

Integration time can be tuned through C&CE registers.

- If the INT signal is externally forced by the system, then INT must respect the requirement from the ROIC:

The INT rising edge (corresponding to MC rising edge) starts the integration capacitances reset during 3075.5 MC periods. **Only after these 3075.5MC starts the real integration of charges.**

Therefore, the minimum **INT** duration is 3077MC for an effective charge integration time of 1.5 MC.

- If the system simply sends the integration duration to the C&CE through the registers, then the C&CE takes care of all the above specificities of INT signal and then:

The minimum integration time duration is 1 MC

In any case :

Integration time is continuously programmable by 1MC step, by adjusting the start of INT rising edge.

A change in the integration time is applied at the next snapshot frame.

Note: A change in INT duration must be followed by a new calibration (Non Uniformity Correction) to ensure an optimized image.

5.6 Operational mode selection

The readout circuit can be operated either in the integration then read mode (ITR) or in the integration while read mode (IWR). This operational mode is selected thanks to the C&CE ITR register according to the following table.

ITR	Operational mode
0	Integration while read (IWR)
1 (Default mode)	Integration then read (ITR)

Read-Out modes

5.7 Format selection and optional windowing mode

Three fixed formats and a windowing mode can be chosen thanks to **SIZE-A** and **SIZE-B** inputs.

These SIZE-A and SIZE-B inputs allow to choose at each frame between the full window size (640×512), two reduced fixed windows (640×480 or 512×512) and a random window defined through the serial interface.

SIZE-A	SIZE-B	FORMATS
1	1	640×512 (default mode, at power ON)
1	0	Central 640×480
0	1	Central 512×512
0	0	Random windowing mode within 640×512 down to 132x1 (see C&CE description part)

Figure 5.2: Windowing programming

As SIZE-A, SIZE-B have an immediate impact on the pixels to be integrated or on the digital part for the selection of pixels to be read, these inputs are only changed by C&CE between the falling edge of DATAVALID and the rising edge of INT.

SIZE-A, SIZE-B must be set before the calibrations (NUC). A change in the following formats must be followed by a new calibration.

5.8 Rotation function operation

UPROW and UPCOL are the logic inputs that determine the readout direction of the addressed window (it can be the full frame if the windowing function is not used). UPROW drives the readout direction for the lines and UPCOL drives the readout direction for the columns.

In case of use of the rotation function, the signals coming from the pixels of the addressed window are readout line after line according to the following table.

UPCOL	UPROW	Mode	Readout direction	
			ROIC Side	Screen Side
1 (Default mode)	1 (Default mode)	Normal		
0	1 (Default mode)	Invert (horizontal flip)		
1 (Default mode)	0	Revert (vertical flip)		
0	0	Rotation		

Figure 5.3: Flip Function Operation in Full Array Size

6. ELECTRICAL INTERFACES

6.1 Power supply

The power supply on JP1 shall comply with the following characteristics:

Signal Type	Input/output	Nominal voltage	Max Voltage	Min Voltage	Ripple
POWER	Input	+5.0 Volts	+5.25 Volts	4.75 Volts	50 mVpp Max.

6.2 Currents and power

Max peak current at power-on: 1 A during 1 ms

The total electrical power consumption of the proximity electronics (without the detector) is therefore ≤ 2 W (at 20°C ambient temperature and with IR_MC = 5.55 MHz).

6.3 Electrical Characteristics for CAM IN and CAM OUT Pins

Signal Type	Type (1)	Vid (mV) (2)		Vicm (V) (3)		
		Min	Max	Min	Max	Conditions
CAM IN	Input DE	100	--	0.05	1.80	Dmax < 500Mbps
				0.55	1.80	500Mbps < Dmax < 700Mbps

Signal Type	Type (1)	Vod (mV) (4)(6)		Vos (V) (5)(6)		
		Min	Max	Min	Typ	Max
CAM OUT	Output DE	247	600	1.125	1.250	1.375

- (1): SE : Single ended / DE : Differential ended
(2): Vid : Input differential Voltage Swing. The difference in voltage between the positive and complementary conductors of a differential transmission at the receiver.
(3): Vicm : Input Common Mode Voltage. The common mode of the differential signal at the receiver.
(4): Vod : Output differential Voltage Swing—The difference in voltage between the positive and complementary conductors of a differential transmission at the transmitter. $V_{OD} = V_{OH} - V_{OL}$.
(5): Vos : Output offset voltage— $V_{OS} = (V_{OH} + V_{OL}) / 2$.
(6): RL range: 90 Ohms < RL ≤ 110 Ohms.

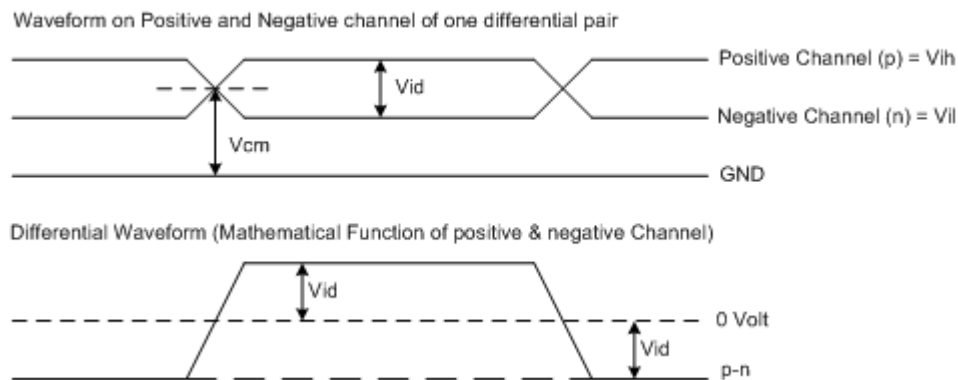


Figure 5 : Waveform for CAM IN Signal Type

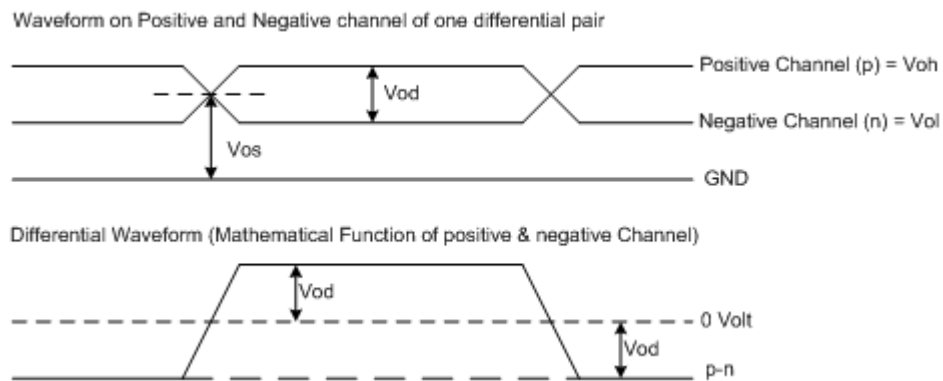


Figure 6 : Waveform for CAM OUT Signal Type

6.4 **Connectors electrical interface**

6.4.1 **Temperature sensor connector pinout (JP3)**

The JP3 connector outputs the analog signal of the FPA temperature sensor n°2 to feed the Stirling cooler driving electronics for temperature regulation loop.

Pin	Name	Range	Function
1	DTA2	0 to 5V	Anode of FPA Diode temperature sensor N°2
2	GND	0 V	GND reference and shield
3	DTK2	0 to 5V	Cathode of FPA diode temperature sensor N°2

Figure 7 : JP3 Cooler Connector Pinout

6.4.2 CameraLink connector pinout (JP1)

QSH-30-01-C-D-A							
Function	Type	Name	Pin	Pin	Function	Type	Name
Power Supply 1	POWER	P5V	1	2	Power Supply 3	POWER	P5V
Power Supply 2	POWER	P5V	3	4	Power Supply 4	POWER	P5V
GROUND SIGNAL REFERENCE	POWER	GND	5	6	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential X0 data pair	OUT	X0_M	7	8	Not Connected	-	NC
Positive signal of differential X0 data pair	OUT	X0_P	9	10	Not Connected	-	NC
GROUND SIGNAL REFERENCE	POWER	GND	11	12	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential X1 data pair	OUT	X1_M	13	14	Not Connected	-	NC
Positive signal of differential X1 data pair	OUT	X1_P	15	16	Not Connected	-	NC
GROUND SIGNAL REFERENCE	OUT	GND	17	18	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential X2 data pair	OUT	X2_M	19	20	Not Connected	-	NC
Positive signal of differential X2 data pair	OUT	X2_P	21	22	Not Connected	-	NC
GROUND SIGNAL REFERENCE	POWER	GND	23	24	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential X3 data pair	OUT	X3_M	25	26	Not Connected	-	NC
Positive signal of differential X3 data pair	OUT	X3_P	27	28	Not Connected	-	NC
GROUND SIGNAL REFERENCE	POWER	GND	29	30	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential PIX CLK pair (channel X)	OUT	XCLK_M	31	32	Not Connected	-	NC
Positive signal of differential PIX CLK pair (channel X)	OUT	XCLK_P	33	34	Not Connected	-	NC
GROUND SIGNAL REFERENCE	POWER	GND	35	36	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential CC1 pair	IN	CC1M	37	38	Negative signal of differential TC pair	IN	TC_M
Positive signal of differential CC1 pair	IN	CC1P	39	40	Positive signal of differential TC pair	IN	TC_P
GROUND SIGNAL REFERENCE	POWER	GND	41	42	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential CC2 pair	IN	CC2_M	43	44	Negative signal of differential TFG pair	OUT	TFG_M
Positive signal of differential CC2 pair	IN	CC2_P	45	46	Positive signal of differential TFG pair	OUT	TFG_P
GROUND SIGNAL REFERENCE	POWER	GND	47	48	GROUND SIGNAL REFERENCE	POWER	GND
Negative signal of differential CC3 pair	IN	CC3_M	49	50	Negative signal of differential Spare Data pair	IN	SP
Positive signal of differential CC3 pair	IN	CC3_P	51	52	Positive signal of differential Spare Data pair	IN	SP
GROUND SIGNAL REFERENCE	POWER	GND	53	54	GROUND SIGNAL REFERENCE	POWER	GND
Reserved Factory	*	Fact	55	56	Reserved Factory	*	Fact
Reserved Factory	*	Fact	57	58	Reserved Factory	*	Fact
GROUND SIGNAL REFERENCE	POWER	GND	59	60	GROUND SIGNAL REFERENCE	POWER	GND
GROUND SIGNAL REFERENCE	POWER	GND	Metal Plane		GROUND SIGNAL REFERENCE	POWER	GND

* : Must not be connected

Figure 8 : JP1 System Connector Pinout

7. FPA TEMPERATURE FILTERING, MONITORING AND MANAGEMENT

The board biases the IDDCA 2N2222 temperature sensor n°1 with an internal 25µA current source, to monitor the FPA temperature and to drive the FPA ON/OFF with respect to a temperature threshold.

The temperature sensor analog bias V_{diode} is adapted and converted to digital value.

The system can read The FPA temperature (in Kelvin) in registers #160 and #161 through CAMERALINK UART protocol, at a 0.25 Hz conversion rate (1 sample every 4 seconds).

The measurement precision of the FPA temperature is

- ± 0.2 K in the range [70K – 120K]
- ± 0.5 K in the range [120K – 200K].

The board automatically ensures that after being powered ON,

- the FPA is only biased when it is at low operating temperature (below a given threshold).
- The FPA bias is stopped when the operating temperature is higher than the given threshold (in case of cooler failure that would lead to the stop of FPA cooling).

For this purpose, the V_{diode} digital value is compared to a threshold value programmed by NCRIO for MINI640 application (below 150K). This threshold defines the cold state of the FPA.

- When the FPA temperature is below the programmed threshold, the FPA is powered within less than 400 ms after threshold crossing.
- If FPA temperature increases, the FPA is powered OFF considering a hysteresis of 2K above the temperature threshold.

This power-up sequence consists in switching-on step by step :

- Mains power supplies: VDD, VDDA, VDDO.
- Bias voltage: GPOL, VR.
- Digital signals: MC, INT, SERDAT, SERCLR, UPROW, UPCOL, SIZE_A, SIZE_B, ITR.

Although the temperature threshold is defined in registers, a hysteresis cycle of ± 2 Kelvin is applied.

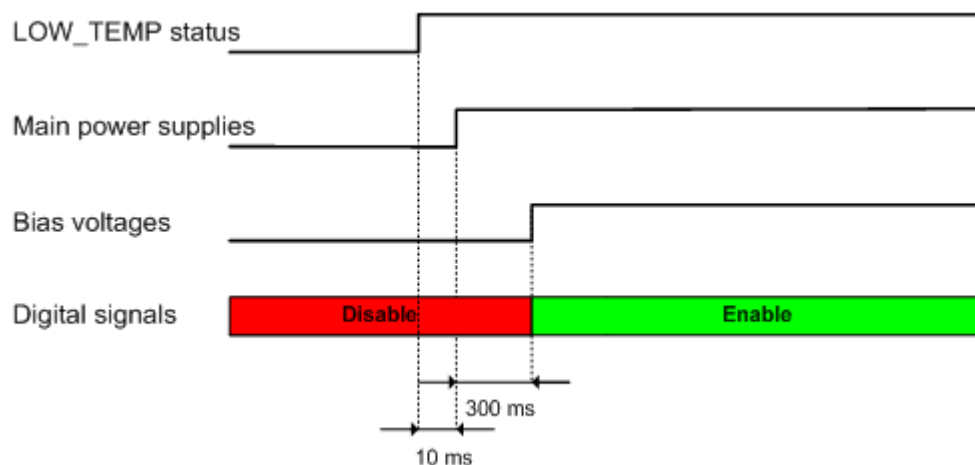


Figure 9: Power-Up Sequence

For FPA power-down procedure: all analog and digital signals are switched-off simultaneously.

8. THERMAL INTERFACES

8.1 Thermal interface of the board

It is the customer responsibility to ensure that the board power dissipation is correctly evacuated from the system, when operational.



In order to ensure that the board power dissipation is correctly evacuated, a sufficient air volume around the board shall be maintained, in order that the air flow ensures a sufficient convective cooling, or a forced air flux or a thermal drain should be considered.

8.2 Thermal interface of the FPA

The FPA shall not be ON without the cooling system ON, in order to avoid damage due to power dissipation inside the vacuum Dewar. The 2N2222 temperature sensor n°1 is used by the board to monitor the temperature, and to power ON/OFF the IDDCA with respect to a programmed temperature threshold (see § 7).

The cooler controller is connected to JP3 for the focal plane temperature regulation loop. The current source provided by the cooler controller will enable to feed the controller with the 2N2222 temperature sensor n°2 V_{diode} bias, to perform the control loop (see § 6.4.1).

8.3 Thermal interface of the IDDCA

Refer to the cooler handling procedure for requirements and/or recommendations regarding the thermal interface of the IDDCA (heat dissipation of the Stirling engine).

9. CAMERALINK COMMUNICATION PORT

Communication between System and the board is built on Cameralink UART link.

Two LVDS pairs are allocated for asynchronous serial communication to and from the camera and frame grabber. These signals are:

- TC - Differential pair with serial communications to the camera
- TFG - Differential pair with serial communications to the frame grabber.

The serial interface is used for controlling the board with the parameters below:

- 9600 bauds,
- 8 data bits,
- 1 start bit,
- 1 stop bit without parity
- No handshaking.
- 16 CRC bits. (Critical Redundancy Control). For more details, see [REF 2].

9.1 Communication Protocol

9.1.1 Description

		Write Access	Read Access	Copy registers data to flash user data	Copy flash factory data to flash customer data
COMMAND					
Start of frame		"@" (0x40)			
Command		"W" (0x57)	"R" (0x52)	">" (0x3E)	"<" (0x3C)
Address	ADDR[11:8]	ADDR	ADDR	none	none
	ADDR[7:4]				
	ADDR[3:0]				
Data	D[7:4]	WR_ DATA	none	none	none
	D[0:3]				
CRC	CRC[15:12]	CRC4	CRC4	CRC4	CRC4
	CRC[11:8]	CRC3	CRC3	CRC3	CRC3
	CRC[7:4]	CRC2	CRC2	CRC2	CRC2
	CRC[3:0]	CRC1	CRC1	CRC1	CRC1
Number of byte		11	9	6	6
RESPONSE					
Start of frame		"@" (0x40)			
Response or acknowledge		Y or N	Y or N and RD_DATA	Y or N	Y or N
		none			
		none		none	none
CRC	CRC[15:12]	CRC4	CRC4	CRC4	CRC4
	CRC[11:8]	CRC3	CRC3	CRC3	CRC3
	CRC[7:4]	CRC2	CRC2	CRC2	CRC2
	CRC[3:0]	CRC1	CRC1	CRC1	CRC1

	Write Access	Read Access	Copy registers data to flash user data	Copy flash factory data to flash customer data
Number of byte	6	8	6	6

Notes:

- “N” Acknowledge is sent by the C&CE board in the following cases:
 - Command was not understood
 - Number of bytes is not correct
 - C&CE board is busy
 - Address area access is protected (Read or Read/Write)
- ADDR, WR_DATA and RD_DATA are ASCII encoded.
- CRC is ASCII encoded.
- CRC is computed according to [REF 2] MODBUS CRC16 Specification, chapter 6.2.2 **EXCEPT the order to send CRC value: MSB first for C&CE** (instead of LSB first on MODBUS).
- “@E” response means CRC ERROR detected by the board.

9.1.2 CRC Calculation examples

Here find 3 examples of frame on the serial link:

Read address 0x100 / Answer of C&CE : 0x03

System									C&CE							
@	R	1	0	0	6	D	F	1	@	Y	0	3	E	2	9	1
Start of frame	Command	Address				CRC			Start of frame	Ack	Data			CRC		

2/ Write 0x03 at 0x100 / Acknowledgement of C&CE Board

System										C&CE						
@	W	1	0	0	0	3	F	D	7	1	@	Y	4	A	F	0
Start of frame	Command	Address			Data		CRC				Start of frame	Ack	CRC			

3/ Copy registers datas to flash user data

System						C&CE					
@	>	A	0	B	1	@	Y	4	A	F	0
Start of frame	Command	CRC				Start of frame	Ack	CRC			

9.2 CameraLink control port

For external mode, the system must provide to the board the clock frequency and the Frame synchronization signal as necessary.

System sends these two signals by means of differential lines:

- Camera Control 1 (**CC1**) : **FSYNCIN** (Frame synchronization)
- Camera Control 2 (**CC2**) : **DCLKIN** (External clock)

9.2.1 Synchronization on C&CE Board:

Synchronization on C&CE can be done as

- Internal mode (default mode) :

C&CE generates its own MC and INT derived from on board oscillator. Default MC frequency is 5 MHz or 5.555MHz, depending on MINI640 configuration for the required frame rate. Period and high level duration of INT signal are user programmable with MC pitch.

- MC External mode :

User must send the DCLKIN clock to C&CE by the mean of Camera Control 2 (CC2) from Camera Link. The frequency of MC will be DCLKIN frequency and must be equal to or lower than 5.555MHz.

- INT External mode :

In this mode, INT signal is synchronized with rising edge of Camera Control 1 (CC1) signal from Camera Link.

High Level duration of Integration Time can be performed in two manners:

Locally generated as in internal mode with the same registers parameters. In this case, C&CE will take into account INT offset to generate really integration time of flux.

Externally sent by user by the mean of the high level length of CC1. In this mode, C&CE will NOT take into account the INT offset that must be generated to the ROIC. The INT rising edge (corresponding to MC rising edge) starts the integration capacitances reset during 3075.5 MC periods. **Only after these 3075.5 MC the**

actual integration of charges starts. It's the reason why the system has to add

this offset to the nominal width of CC1.

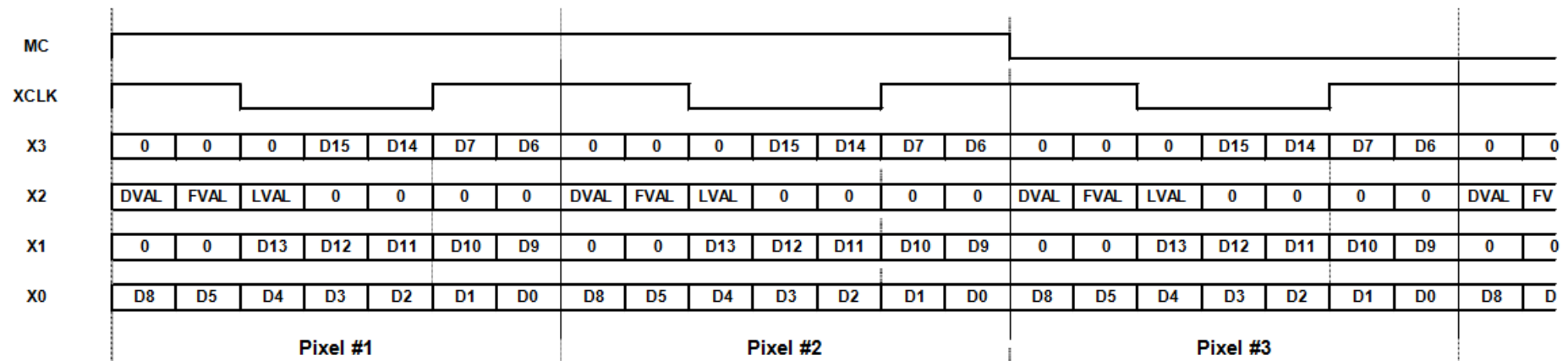
Internal/external modes for INT and MC signals can be combined.

CC1 : FSYNC	CC2 DCKLIN
External	Internal
External	External
Internal	External

9.3 **CameraLink data port**

The CameraLink data port uses four LVDS pairs, these signals are:

- X0 – video data,
- X1 – video data,
- X2 – video data and control (line valid, frame valid, and data valid),
- X3 – video data,
- XCLK – pixel clock



Base Mode - Bit assignement for 4 Video outputs

FPA video Outputs could be assigned to any pixel (Register configuration).

XCLK frequency = 4 x MC frequency.

Bit Rate on X differential pairs = 4 x 7 x MC frequency.

Video data are transmitted with D[2..15] bits. D0 and D1 values are always 0 for this board (14 bits Conversion)

Figure 10: Bit assignment

10. MANAGEMENT OF CONFIGURATION

The C&CE configuration is stored in a non volatile memory divided in two parts:

- First part is used for Current Configuration (Customer part): Read and Write access.
- Second is the Default value (Factory Part): Read Access.

This memory is accessible through the Camera Link UART interface.

The system interface is represented by the synoptic of Figure 11:

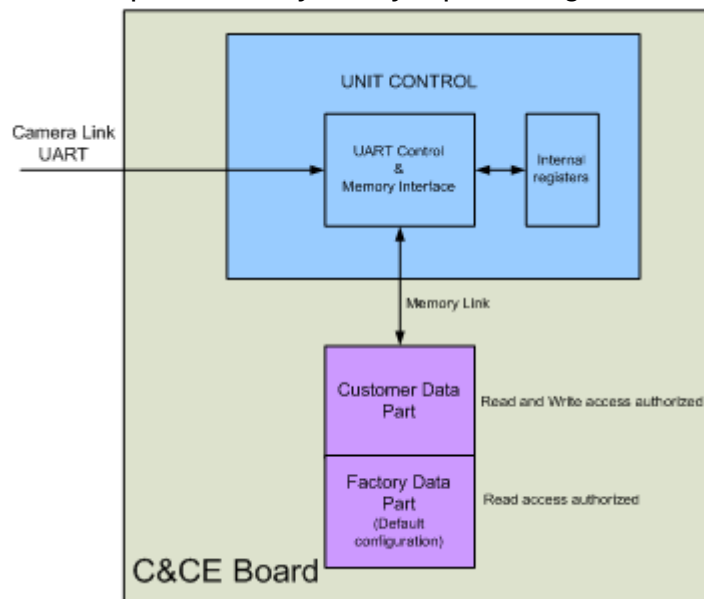


Figure 11: Memory Configuration

Data configurations are recorded in the registers area of the memory.

After Power-On, the C&CE stores current configuration from Customer Part into the dynamic internal registers (FPGA registers). Internal registers can be modified by Camera Link UART Interface.

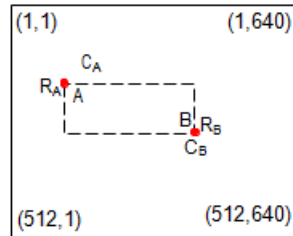
Customer data part is used to store current configuration of FPGA registers and it is initialized with defaults values at Factory. FPGA registers can be read out or modified by the system. This is done by Read or Write Command. Values in the FPGA register's and can be recorded in memory with the specific command ">".

Factory data Part is used to store Factory configuration and cannot be modified by the user. At any time the system can transfer these values into the Customer Data Part by sending a UART Command "<", to recover the factory configuration. **This recovering process will be taken into account after next Power-On.**

10.1.1.1 Windowing Mode

In case of windowing mode (Video size selection register = "00"), the selected window is determined by the R (row) and C (column) references of the corner points A and B.

Screen side :



Whatever the UPCOL and UPROW registers configuration, coordinates of A and B corners are those that can be displayed on a screen.

Window's coordinates are set by the system in the corresponding C&CE registers :

- (R_A;C_A) are the coordinates of first corner of the window (point A).
- (R_B;C_B) the coordinates of the opposite corner (point B).

The following rules apply on the coordinates to be entered in the serial data interface :

- The (R;C) pixel coordinates can swing from (1;1) for the first array pixel to (512;640) for the last one (R = line (or row), C = column).
- The minimum size of the window (SIZEcol×SIZEline) is 1 line × 132 columns (within 512 lines × 640 columns).
- $A < R_B$ and $C_A < C_B$.

C&CE board manage automatically the coordinates which must be sent to the ROIC depending on the UPCOL/UPROW configuration.

Note : Column coordinates must be a multiple of 4. If this rule is not applied, C&CE will traduce this column coordinate by the nearest inferior correct value.

The window coordinates data are provided to the IRFPA through a serial interface by C&CE consisting in the following inputs: SERCLR and SERDAT. The clock used for the windowing logic is Master Clock (MC), the same clock than for the sequencer.

- **SERCLR** : SERCLR falling edge indicates the beginning of SERDAT data.
- **SERDAT** : allows the input of serial interface data.

Note : a change in the window size shall be followed by a new calibration (NUC).

11. REGISTER DESCRIPTION

11.1 CCE Configuration

Address	Protected	Access	Name	Description
100	N	R/W	CCE_CONF	C&CE Board configuration

[3..0]: Unused

[4]: Video Inversion (black/white inversion)

0: Video data not inverted.

1: Video data inverted.

This bit allows inverting the video data on the video output stream (Black/Hot or White/Hot).

[7..5]: Unused

Address	Protected	Access	Name	Description
102	N	R/W	CCE_Input_CTRL	C&CE Board Synchronization

[0]: Master clock sources (MC)

0: MC master clock source = internal FPGA clock

1: MC master clock source is external CC2 signal (DCLKIN)

[2:1]: Master clock frequency (if MC is internal)

x0: MC is 5MHz when MC is local x1:

MC is 2.5MHz when MC is local

[3]: Frame sources (INT)

0: INT is generated inside the FPGA

1: INT source is external CC1 (FSYNC)

[4]: Frame sources mode

0: CC1 (FSYNC) is used directly to drive the sensor INT signal.

1: CC1 is used to start the onboard generation of INT.

[6:5]: Position of the sampling in the pixel

00: Position 1 (0°)

01: Position 2 (90°)

10: Position 3 (180°)

11: Position 4 (270°)

[7]: Not Used

11.2 CCE Status registers

Address	Protected	Access	Name	Description
110	N	R	CCE_PBIT	Power-Up Built-in-test result

[0]: Unused

[1]: Memory Checksum status

0: Memory checksum NOK.

1: Memory checksum OK.

[7:2]: Unused

Address	Protected	Access	Name	Description
111	N	R	CCE_CBIT	Continuous Built-in-test result

[0]: Power Supply status

0: Power supply NOK.

1: Power supply OK.

This bit indicates that all power supplies of the board are OK.

[1]: Unused

[2]: Serdat Error

0: No default detected on the serdat signal

1: Default detected on the serdat signal

This bit indicates if an error has occurred during a windowing configuration of the FPA.

[6:3]: Unused

[7]: Factory reserved

Address	Protected	Access	Name	Description
112	N	R	CCE_Temperature status	

[0]: Temperature Low

0: The FPA temperature is higher than the temperature threshold.

1: The FPA temperature is lower than the temperature threshold.

This bit indicate if the FPA temperature is lower or upper of the temperature threshold programmed in the register \$162 and \$163 by factory.

[7:1]: Unused

Address	Protected	Access	Name	Description
113	N	R	CCE_Power_status	Internal power supplies status

[0]: Ref 2V5 Status

0: Power supply Ref 2V5 OK
1: Power supply Ref 2V5 NOK

[1]: Power supply 3.3V ADC Status

0: Power supply 3.3V ADC OK
1: Power supply 3.3V ADC NOK

[2]: Power supply 5V Status

0: Power supply 5V OK
1: Power supply 5V NOK

[3]: Power supply +5V / -4V Status

0: Power supply of +5V and -4V OK
1: Power supply of +5V and -4V NOK

[7:4]: Unused

These values are refreshed every 4 seconds

Address	Protected	Access	Name	Description
114	N	R	MONITOR_REF2.5V LSB	LSB Power supply REF2.5V Value

[7:0]: LSB REF 2.5V value

LSB of the value of the REF 2.5V internal power supply.
The value of the voltage is coded in 16 bits format in mVolt (Hexadecimal coded)
This value is refreshed every 4 seconds

Address	Protected	Access	Name	Description
115	N	R	MONITOR_REF2.5V MSB	MSB Power supply REF2.5V Value

[7:0]: MSB REF 2.5V value

MSB of the value of the REF 2.5V internal power supply.
The value of the voltage is coded in 16 bits format in mVolt (Hexadecimal coded)
This value is refreshed every 4 seconds

Address	Protected	Access	Name	Description
116	N	R	MONITOR_3.3V_ADC LSB	LSB Power supply 3.3V ADC Value

[7:0]: LSB Power supply 3.3V

LSB of the value of the 3.3V_ADC internal power supply.

The value of the voltage is coded in 16 bits format in mVolt (Hexadecimal coded)
This value is refreshed every 4 seconds

Address	Protected	Access	Name	Description
117	N	R	MONITOR_3.3V_ADC MSB	MSB Power supply 3.3V ADC Value

[7:0]: MSB Power supply 3.3V value

MSB of the value of the 3.3V_ADC internal power supply.
The value of the voltage is coded in 16 bits format in mVolt (Hexadecimal coded)
This value is refreshed every 4 seconds

Address	Protected	Access	Name	Description
118	N	R	MONITOR_5V_N4V LSB	LSB Power supply 5V_N4V Value

[7:0]: LSB Power supply +5V and -4V value

LSB of the value of the 5V_N4V internal power supply.
The value of the voltage is coded in 16 bits format in mVolt (Hexadecimal coded)
This value is refreshed every 4 seconds

Address	Protected	Access	Name	Description
119	N	R	MONITOR_5V_N4V MSB	MSB Power supply 5V_N4V Value

[7:0]: LSB Power supply +5V and -4V value

MSB of the value of the 5V_N4V internal power supply.
The value of the voltage is coded in 16 bits format in mVolt (Hexadecimal coded)
This value is refreshed every 4 seconds

11.3 FPA Configuration

Address	Protected	Access	Name	Description
120	N	R/W	PATTERN_CTRL	Test Pattern activation

[1..0]: Test pattern activation

00: Normal Mode.
01: Test pattern forced.
10: Automatic mode.
11: Not used.

[2]: Power-On FPA

0: FPA powers supplies are shut down.
1: FPA power supplies are ON.

[7..3]: Not used

11.4 FPA Bias Configuration

Address	Protected	Access	Name	Description
130	N	R/W	GPOL_DAC LSB	LSB of the voltage Bias GPOL value x 43.6

[7..0]: LSB GPOL Voltage

The GPOL voltage value is programmed with 2 registers. LSB and MSB register.
The voltage value is coded on 16 bits and expressed in mVolts.

Address	Protected	Access	Name	Description
131	N	R/W	GPOL_DAC MSB	MSB of the voltage Bias GPOL value x 43.6

[7..0]: MSB GPOL Voltage

The bias voltage value is programmed with 2 registers. LSB and MSB register.
The voltage value is coded on 16 bits and expressed in mVolts.

Example:

To program a Gpol Value to 0.700 Volts, the LSB and MSB register must be programmed as follow:

$700 \times 43.6 = 30520$, which is 0x7738, therefore

- LSB Register (@130) : 0x38
- MSB Register (@131) : 0x77

Address	Protected	Access	Name	Description
150	Y	R	CCE_Ref_1	Reference of the board

[3..0]: 1st character of the serial number (Hexadecimal coded). [7..4]:
2nd character of the serial number (Hexadecimal coded).

Address	Protected	Access	Name	Description
157	Y	R	CCE_VersionFPGA_2	FPGA software version

[3..0]: 3rd character of the FPGA software version (Hexadecimal coded).
[7..4]: 4th character of the FPGA software version (Hexadecimal coded).

Example:

If the embedded software version is : 1.2.3.4 :

CCE_VersionFPGA_1 (@156) : 0x34

CCE_SN_2 (@153): 0x98

Address	Protected	Access	Name	Description
154	Y	R	CCE_VersionSft_1	Embedded software version

[3..0]: 1st character of the embedded software version (Hexadecimal coded).
[7..4]: 2nd character of the embedded software version (Hexadecimal coded).

Address	Protected	Access	Name	Description
155	Y	R	CCE_VersionSft_2	Embedded software version

[3..0]: 3rd character of the embedded software version (Hexadecimal coded).
[7..4]: 4th character of the embedded software version (Hexadecimal coded).

Example:

If the embedded software version is : 4.3.2.1 :

CCE_VersionSft_1 (@154) : 0x21

CCE_VersionSft_1 (@155) : 0x43

Address	Protected	Access	Name	Description
156	Y	R	CCE_VersionFPGA_1	FPGA software version

[3..0]: 1st character of the FPGA software version (Hexadecimal coded).
[7..4]: 2nd character of the FPGA software version (Hexadecimal coded).

Address	Protected	Access	Name	Description
157	Y	R	CCE_VersionFPGA_2	FPGA software version

[3..0]: 3rd character of the FPGA software version (Hexadecimal coded). [7..4]:
4th character of the FPGA software version (Hexadecimal coded).

Example:

If the embedded software version is : 1.2.3.4 :

CCE_VersionFPGA_1 (@156) : 0x34

CCE_VersionFPGA_2 (@157) : 0x12

11.5 Temperature parameters

Address	Protected	Access	Name	Description
160	Y	R	CCE_Temp_LSB	FPA Temperature LSB

[7..0]: FPA Temperature LSB

Indicates the FPA temperature expressed in 0.01 Kelvin (LSB part of a 16 bits unsigned value coded in hexadecimal). This value is refreshed every 4 seconds

Address	Protected	Access	Name	Description
161	Y	R	CCE_Temp_MSB	FPA Temperature MSB

[7..0]: FPA Temperature MSB

Indicates the FPA temperature expressed in 0.01 Kelvin (MSB part of a 16 bits unsigned value coded in hexadecimal). This value is refreshed every 4 seconds

Example:

If the temperature of the FPA is 90.14 K :

CCE_Temp_LSB (@160) : 0x36

CCE_Temp_MSB (@161) : 0x23

Address	Protected	Access	Name	Description
162	P	R/W	CCE_Temp_Thres_LSB	Temperature threshold LSB

[7..0]: FPA Temperature Threshold LSB

Indicates the FPA temperature threshold expressed in 0.01 Kelvin (MSB part of a 16 bits unsigned value coded in hexadecimal).

The temperature threshold is the temperature below the FPA can be switch-on.

Address	Protected	Access	Name	Description
163	P	R/W	CCE_Temp_Thres_MSB	Temperature threshold MSB

[7..0]: FPA Temperature Threshold MSB

Indicates the FPA temperature threshold expressed in 0.01 Kelvin (MSB part of a 16 bits unsigned value coded in hexadecimal).

The temperature threshold is the temperature below the FPA can be switch-on.

Example:

For the temperature threshold is 110.00 K:

CCE_Temp_Thres_LSB (@162): 0xF8

CCE_Temp_Thres_MSB (@163) : 0x2A

Address	Protected	Access	Name	Description
164	P	R	A1_Temp_Poly_LSB	Polymonial coefficient A1 (LSB)
165	P	R	A1_Temp_Poly_Mid	Polymonial coefficient A1 (Mid)
166	P	R	A1_Temp_Poly_MSB	Polymonial coefficient A1 (MSB)
167	P	R	A2_Temp_Poly_LSB	Polymonial coefficient A2 (LSB)
168	P	R	A2_Temp_Poly_Mid	Polymonial coefficient A2 (Mid)
169	P	R	A2_Temp_Poly_MSB	Polymonial coefficient A2 (MSB)
16A	P	R	A3_Temp_Poly_LSB	Polymonial coefficient A3 (LSB)
16B	P	R	A3_Temp_Poly_Mid	Polymonial coefficient A3 (Mid)
16C	P	R	A3_Temp_Poly_MSB	Polymonial coefficient A3 (MSB)
16D	P	R	A4_Temp_Poly_LSB	Polymonial coefficient A4 (LSB)
16E	P	R	A4_Temp_Poly_Mid	Polymonial coefficient A4 (Mid)
16F	P	R	A4_Temp_Poly_MSB	Polymonial coefficient A4 (MSB)
170	P	R	A5_Temp_Poly_LSB	Polymonial coefficient A5 (LSB)
171	P	R	A5_Temp_Poly_Mid	Polymonial coefficient A5 (Mid)
172	P	R	A5_Temp_Poly_MSB	Polymonial coefficient A5 (MSB)

These register are the coefficient of the polynomial function to convert the diode voltage to a temperature expressed in Kelvin. The value of one coefficient is coded on 24 bits signed (1 sign bit +23 magnitude bit).

The coefficient stored in the registers is multiplied by 1000.

$$T(Vdiode) = A_4.Vdiode^4 + A_3.Vdiode^3 + A_2.Vdiode^2 + A_1.Vdiode + A_0$$

T in Kelvin

Vdiode in Volts

The diode sensor is polarized by a current source of 25μA.

Example:

<i>Coefficient</i>	<i>Value</i>	<i>MSB</i>	<i>Middle</i>	<i>LSB</i>
A0	539.646	08	3B	FD
A1	-227.763	FC	86	4D
A2	-812.665	F3	99	86
A3	1052.813	10	10	8D
A4	-460.663	F8	F8	88

11.6 FPA Configuration

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1A0	N	R/W	FPA_CONF	FPA control

[1..0]: Readout mode selection (UPCOL, UPROW)

00: From Pixel RB,CB to RA,CA

01: From Pixel RA,CB to RB,CA

10: From Pixel RB,CA to RA,CB

11: From Pixel RA,CA to RB,CB0: Direction Inverted.

The column inversion with the bit Column inversion (UPCOL) inverts automatically the columns by package of 4 columns.

[3..2]: Video size selection (SIZE_A, SIZE_B)

00: 640x512 (windowing mode)
 01: 512x512
 10: 640x480
 11: 640x512

[7..4]: Unused

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1B0	N	R/W	INT_PERIOD_TIME_1	Integration period (Byte 1)

[7..0]: INT_PERIOD_TIME_1:

Bit 7 to Bit 0 of the integration period (coded in 24 bits / Expressed in number of master clock period).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1B1	N	R/W	INT_PERIOD_TIME_2	Integration period (Byte 2)

[7..0]: INT_PERIOD_TIME_2:

Bit 15 to Bit 8 of the integration period (coded in 24 bits / Expressed in number of master clock period).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1B2	N	R/W	INT_PERIOD_TIME_3	Integration period (Byte 3)

[7..0]: INT_PERIOD_TIME_3:

Bit 23 to Bit 16 of the integration period (coded in 24 bits / Expressed in number of master clock period).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1B4	N	R/W	INT_HIGH_LEVEL_1	Integration high level (Byte 1)

[7..0]: INT_HIGH_LEVEL_1:

Bit 7 to Bit 0 of the high level duration (coded in 24 bits / Expressed in number of master clock period).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1B5	N	R/W	INT_HIGH_LEVEL_2	Integration high level (Byte 2)

[7..0]: INT_HIGH_LEVEL_2:

Bit 15 to Bit 8 of the high level duration (coded in 24 bits / Expressed in number of master clock period).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1B6	N	R/W	INT_HIGH_LEVEL_3	Integration high level (Byte 3)

[7..0]: INT_HIGH_LEVEL_3:

Bit 23 to Bit 16 of the high level duration (coded in 24 bits / Expressed in number of master clock period).

11.7 FPA Windowing

The user should first write LSB register and finish access with the MSB.

For Y and X coordinates, the user should finish the access with the MSB of Xmax.

Even if the register Xmax is not modified the user should write it in order to launch SERDAT configuration.

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D0	N	R/W	X1MIN_LSB	LSB of X1min

[7:0] : LSB of X1min :

LSB of X1min (Lower Column of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D1	N	R/W	X1MIN_MSB	MSB of X1min

[7:0] : MSB of X1min :

MSB of X1min (Lower Column of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D2	N	R/W	Y1MIN_LSB	LSB of Y1min

[7:0] : LSB of Y1min :

LSB of Y1min (Lower Row of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D3	N	R/W	Y1MIN_MSB	MSB of Y1min

[7:0] : MSB of Y1min :

MSB of Y1min (Lower Row of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D4	N	R/W	X1MAX_LSB	LSB of X1max

[7:0] : LSB of X1max :

LSB of X1max (Upper Column of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D5	N	R/W	X1MAX_MSB	MSB of X1max

[7:0] : MSB of X1max :

MSB of X1max (Upper Column of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D6	N	R/W	Y1MAX_LSB	LSB of Y1max

[7:0] : LSB of Y1max :

LSB of Y1max (Upper Row of window) (coded in 16 bits).

<i>Address</i>	<i>Protected</i>	<i>Access</i>	<i>Name</i>	<i>Description</i>
1D7	N	R/W	Y1MAX_MSB	MSB of Y1max

[7:0] : MSB of Y1max :

MSB of Y1max (Upper Row of window) (coded in 16 bits).

11.8 Register default value

ADDR[11:0] (Hex)	R/W	Register description	Default Value (Hex)
000 - 0FF	-	RESERVED FACTORY	
CCE CONFIGURATION			
100	R/W	CCE_CONF : C&CE Board configuration [3..0]: Unused [4]: Video Inversion (black/white inversion). 0: Video data not inverted. 1: Video data inverted. [7..5]: Unused	00 0 : NA 0: Video data not inverted. 0 : NA
101	-	UNUSED	
102	R/W	CCE_Input_CTRL : C&CE Board Synchronization [0]: Master clock sources (IR_MC) 0: MC master clock source = internal FPGA clock 1: MC master clock source is external CC2 cameralink signal (DCLKIN) [2:1]: Master clock frequency (if MC is internal) x0: IR_MC is 5MHz (or 5.55MHz) when IR_MC is local x1: unused [3]: Frame sources (INT) 0: INT is generated inside the FPGA 1: INT source is external CC1 (FSYNC) [4]: Frame sources mode 0: CC1 (FSYNC) is used directly to drive the sensor INT signal. 1: CC1 is used to start the onboard generation of INT. [6:5]: Master clock offset phase 00:00 01: 90° 10: 180° 11: 270° [7]: Not Used	30 0 : Internal MC 00 : MC = 5MHz or 5.55MHZ ¹ 0 : Internal Frame Sync 1: CC1 is used to start the onboard generation of INT 01 : 90° 0 : NA
103 - 10F	-	UNUSED	
CCE STATUS REGISTERS			
110	R	CCE_PBIT : Power-Up Built-in-test result [0]: Unused [1]: Memory Checksum status 0: Memory checksum NOK. 1: Memory checksum OK.	

ADDR[11:0] (Hex)	R/W	Register description	Default Value (Hex)
		[7:2]: Unused	
111	R	CCE_CBIT : Continuous Built-in-test result [0]: Power Supply status 0: Power supply NOK. 1: Power supply OK. [1]: Unused [2]: Serdat Error 0: No default detected on the serdat signal 1: Default detected on the serdat signal [3]: Unused [6:4]: Unused [7]: Administrator Mode Active 0: Customer Mode 1: Administrator Mode Activated	
112	R	CCE_Temperature status [0]: Temperature Low 0: The FPA temperature is higher than the temperature threshold. 1: The FPA temperature is lower than the temperature threshold. [1]: Unused [7:2]: Unused	
113	R	CCE_Power_status : Internal power supplies status [0]: ref 2.5V status 0: Power supply ref 2.5V NOK 1: Power supply ref 2.5V OK [1]: 3.3V ADC status 0: Power supply ref 3.3V ADC NOK 1: Power supply ref 3.3V ADC OK [2]: 5V status 0: Power supply ref 5V NOK 1: Power supply ref 5V OK [3]: -4V status 0: Power supply ref -4V NOK 1: Power supply ref -4V OK [7:4]: Unused	
114	R	MONITOR_REF2.5V LSB [7:0]: LSB Power supply REF2.5V Value	
115	R	MONITOR_REF2.5V MSB [7:0]: MSB Power supply REF2.5V Value	
116	R	MONITOR_3.3V_ADC LSB [7:0]: LSB Power supply 3.3V ADC Value	
117	R	MONITOR_3.3V_ADC MSB [7:0]: MSB Power supply 3.3V ADC Value	

ADDR[11:0] (Hex)	R/W	Register description	Default Value (Hex)
118	R	MONITOR_5V_N4V_LSB [7:0]: LSB Power supply 5V_N4V Value	
119	R	MONITOR_5V_N4V_LSB [7:0]: MSB Power supply 5V_N4V Value	
11A - 11F	-	UNUSED	
FPA CONFIGURATION			
120	R/W	PATTERN_CTRL[7:0] : Pattern control register [1:0]: Test pattern activation 00: Normal mode 01: Test pattern forced 10: Automatic mode [2]: Detector Power mode 0: Detector is off 1: Detector is on [7:3] : Not used	06 10: Automatic mode 1: Detector is on 0 : NA
121 - 12F	-	UNUSED	
FPA BIAS CONFIGURATION			
130	R/W	GPOL_DAC_DATA[7:0] [7:0] : Bit 7 to Bit 0	0D
131	R/W	GPOL_DAC_DATA[15:8] [7:0] : Bit 15 to Bit 8	74
132 - 14F	-	UNUSED	
SERIAL NUMBER / VERSION			
150	R/W	CCE_Ref_1 : Indicates the reference number. (i.g: PEL012A) [3:0]: 1 st character of the reference board. (e.g: A) [7:4]: 2 nd character of the reference board. (e.g: 2)	
151	R/W	CCE_Ref_2 : Indicates the reference number. (e.g: PEL012A) [3:0]: 3 rd character of the reference board. (e.g: 1) [7:4]: 4 th character of the reference board. (e.g: 0)	
152	R/W	CCE_SN_#1 : Indicates the serial number. (e.g: 9812) [3:0]: #1 of the serial number of the board. (e.g : 2) [7:4]: #2 of the serial number of the board. (e.g : 1)	
153	R/W	CCE_SN_#2 : Indicates the serial number [3:0]: #3 of the serial number of the board. (e.g : 8) [7:4]: #4 of the serial number of the board. (e.g : 9)	
154	R	CCE_VersionSft_#1 : Indicates the software version embedded on board (e.g : V4.321) [3:0]: #1 of the version software. (e.g : 1) [7:4]: #2 of the version software. (e.g : 2).	
155	R	CCE_VersionSft_#2 : Indicates the software version embedded on board (e.g : V4.321) [3:0]: #3 of the version software. (e.g : 3)	

ADDR[11:0] (Hex)	R/W	Register description	Default Value (Hex)
		[7:4]: #4 of the version software. (e.g : 4)	
156	R	CCE_VersionFPGA#1 : Indicates the FPGA version embedded on board (e.g : V4.321) [3:0]: #1 of the version software. (e.g : 1) [7:4]: #2 of the version software. (e.g : 2).	
157	R	CCE_VersionFPGA#2 : Indicates the FPGA version embedded on board (e.g : V4.321) [3:0]: #3 of the version software. (e.g : 3) [7:4]: #4 of the version software. (e.g : 4)	
158 - 15F	R/W	FACTORY RESERVED (PROCESS)	
FPA TEMPERATURE			
160	R	FPA_Temperature #LSB : FPA temperature in 0.01Kelvin [7:0]: LSB FPA temperature	
161	R	FPA_Temperature #MSB : FPA temperature in 0.01Kelvin [7:0]: MSB FPA temperature	
162 - 1FF	R/W	FACTORY RESERVED (PROCESS)	
FPA CONFIGURATION			
1A0	R/W	FPA_CONF : FPA Control Register [1:0]: Readout mode selection (UPCOL, UPROW) 00: From Pixel RB,CB to RA,CA 01: From Pixel RA,CB to RB,CA 10: From Pixel RB,CA to RA,CB 11: From Pixel RA,CA to RB,CB [3:2]: Video size selection (SIZE_A, SIZE_B) 00: 640x512 (windowing mode) 01: 512x512 10: 640x480 11: 640x512 [4]: Unused [5]: Unused [6]: Unused [7] : ITR control 0: integrate while read 1:integrate then read	0F 11: From Pixel RA,CA to RB,CB 11: 640x512 0 : NA 0 : NA 0 : NA 0: integrate while read
1A1 - 1AF	-	UNUSED	
TIMING			
1B0	R/W	INT_PERIOD_TIME_1 : Integration Period duration - Byte 1 (LSB) [7:0] : Bit 15 to Bit 8 of the integration period (coded in 24 bits / Expressed in number of master clock period).	
1B1	R/W	INT_PERIOD_TIME_2 : Integration period duration - Byte 2	

ADDR[11:0] (Hex)	R/W	Register description	Default Value (Hex)	
		[7:0] : Bit 15 to Bit 8 of the integration period (coded in 24 bits / Expressed in number of master clock period).		
1B2	R/W	INT_PERIOD_TIME_3 : Integration period duration - Byte 3 (MSB) [7:0] : Bit 23 to Bit 16 of the integration period (coded in 24 bits / Expressed in number of master clock period).		
1B3		UNUSED		
1B4	R/W	INT_HIGH_LEVEL_TIME : Integration high level duration – Byte 1 (LSB) [7:0] : Bit 7 to Bit 0 of the integration period (coded in 24 bits / Expressed in number of master clock period).	LEO F/5.5 19.615ms 1B	LEO F/4 10ms 50
1B5	R/W	INT_HIGH_LEVEL_TIME : Integration high level duration – Byte 2 [7:0] : Bit 15 to Bit 8 of the integration period (coded in 24 bits / Expressed in number of master clock period).	7F	C3
1B6	R/W	INT_HIGH_LEVEL_TIME : Integration high level duration – Byte 3 (MSB) [7:0] : Bit 23 to Bit 16 of the integration period (coded in 24 bits / Expressed in number of master clock period).	01	00
1B7 - 1CF	-	UNUSED		
WINDOWING				
1D0	R/W	X1MIN_LSB : LSB of X1min (Lower Column of window) [7:0] : LSB of X1min	01	
1D1	R/W	X1MIN_MSB : MSB of X1min (Lower Column of window) [7:0] : MSB of X1min	00	
1D2	R/W	Y1MIN_LSB : LSB of Y1min (Lower Row of window) [7:0] : LSB of Y1min	01	
1D3	R/W	Y1MIN_MSB : MSB of Y1min (Lower Row of window) [7:0] : MSB of Y1min	00	
1D4	R/W	X1MAX_LSB : LSB of X1max (Upper Column of window) [7:0] : LSB of X1min	80	
1D5	R/W	X1MAX_MSB : MSB of X1max (Upper Column of window) [7:0] : MSB of X1max	02	
1D6	R/W	Y1MAX_LSB : LSB of Y1max (Upper Row of window) [7:0] : LSB of Y1max	00	
1D7	R/W	Y1MAX_MSB : MSB of Y1max (Upper Row of window) [7:0] : MSB of Y1max	02	

ADDR[11:0] (Hex)	R/W	Register description	Default Value (Hex)
1D8 - 1EF	-	UNUSED	
200 - 3FF	-	RESERVED FACTORY	
400 - 4FF	-	INTERNAL REGISTERS	
500 - FFF	-	RESERVED FACTORY	

12. TEST PATTERN

This video test pattern is activated by the bits “Test pattern activation” in the register Pattern_CTRL (@120).

This test pattern is useful to check the camera link.

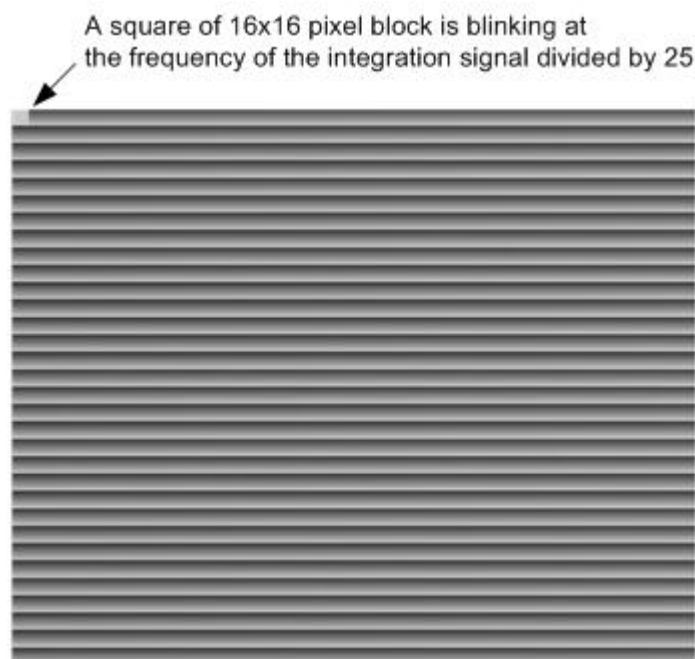


Figure 12: Test Pattern Definition

The rules of the pattern generator value are defined below:

- Video_pattern_pixel (line, column) $\leq$ Number_line[3:0] & Number_column / 4 [9:0].
- Pixel value is the same for 4 consecutive columns.
- A square of 16x16 pixel block is blinking at the frequency of the integration signal divided by 30.

13. CAUTIONS FOR USE

13.1 Cautions against electrostatic discharges

Such as any electronic component, the MINI640 engine is sensitive to electrostatic discharges. The following cautions are therefore specified:

- Handling of the MINI640 engine on a static free work station equipped with grounded carpet;
- not to handle the MINI640 engine unless grounding wrist strap is properly worn and grounded;
- Ground the electronic equipment before plugging it onto the MINI640 C&CE.

13.2 General electrical precautions

The MINI640 C&CE is taking care of generating all necessary biases to drive the detector; therefore the customer is not in charge of applying the electrical precautions described in the ICD document of the FPA.



The electrical interface presented in this document shall be respected.

If the recommended bias values are not respected and correctly applied on the I/O of the C&CE, the C&CE could be damaged or destroyed.

Make sure to follow the recommendations on the outputs impedance.

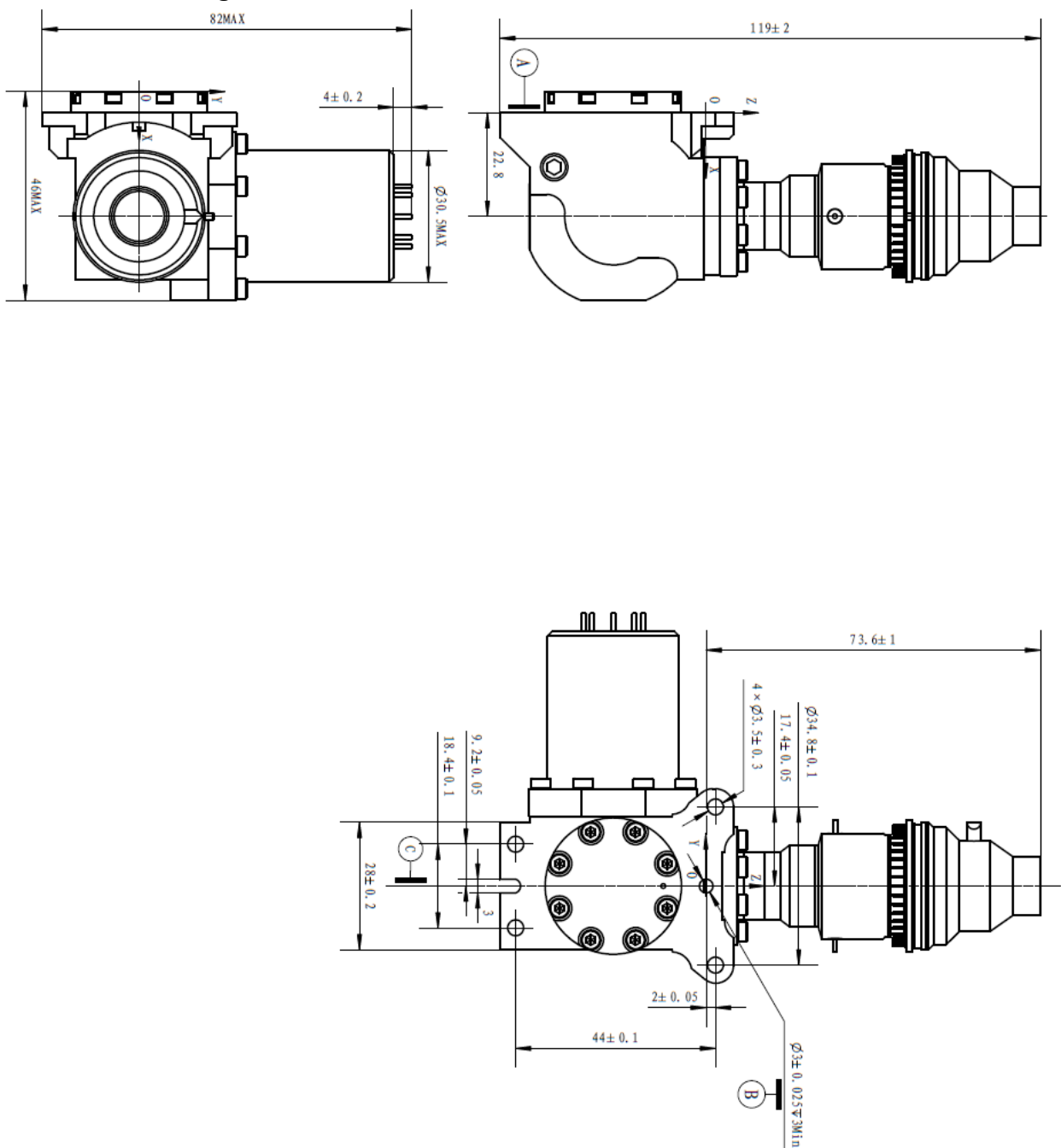
ANNEX

MINI640 MECHANICAL INTERFACE OVERVIEW

The following drawings give an overview of the MINI640 products:

- Drawing for MINI640-XD-2 IDDCA (without the cooler C&CE)

The full detailed drawings of the delivered IDDCA will be provided in the contractual Technical Data Package.



Annex 6

Title: CoolReg for XD-2 External Controller

Reference: CoolReg-XD-2-EC-1801

Issue 1 Revision 0

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

[illegible]

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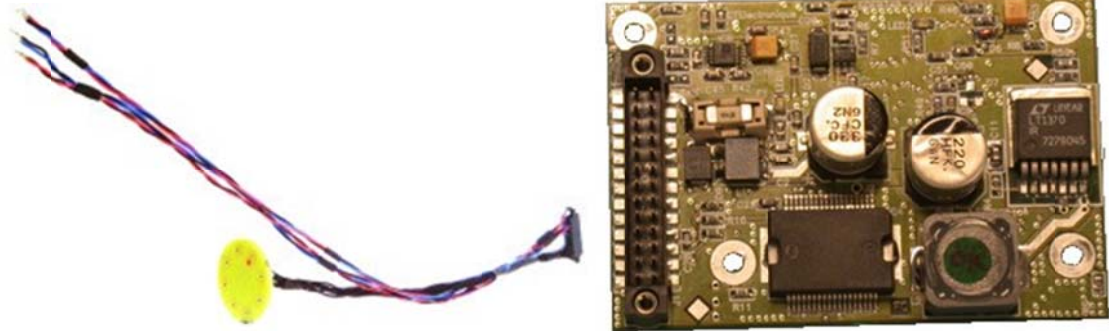
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Application

COOLER DRIVE is a digital external controller enabling to easily drive the XD-2 rotary microcooler and to ensure the Integrated Detector Dewar Cooler Assembly (IDDCA) temperature regulation.

Hardware

- One rectangle board
- One cable for the interface to the cooler and to the system



Features / typical performances

- Single Connector and supply voltage
- Input Voltage: 7.5V-12 V (nominal +9V)
- Output Power max: 30 W
- Efficiency: 70% to 82% (ambient temperature dependent)
- Temperature sensor, Bias Current: 100 μ A
- Setpoint Range: 0.300V to 1.600V
- Stability: 600mK
- Accuracy: Sigma max: 15mK
- Drift (-40°C to +71°C): 600mK
- Dimensions: 63.4 x 43 x 17 mm³
- Mass: < 30 g (board), < 30g (cable)
- Operating Temperature: -40°C to +71°C
- EMC: low emission

1. INTRODUCTION

This technical document describes the interfaces of the COOLER DRIVE digital external controller enabling to easily drive the XD-2 rotary microcooler and to ensure the Integrated Detector Dewar Cooler Assembly (IDDCA) temperature regulation.

The IDDCA temperature regulation is ensured by driving the microcooler motor as a function of the temperature sensor and the internal temperature setpoint through a specific algorithm.

The COOLER DRIVE programming and its interface to the IDDCA is different for each IDDCA configuration (type of cooler and temperature setpoint).

In the document, COOLER DRIVE will stand for “cooler controller”, or “electronic control board”, or “Demand Control Electronics” (DCE).

This document aims at being a generic document applicable to all standard COOLER DRIVE boards, and as such, is an annex of the Technical Data Package (TDP) delivered to the customer when such a cooler controller is ordered.

The contractual TDP delivered to the customer annexes this ICD in its latest update that will in any case supersede the ICD in lower issue that could be provided prior to the TDP.

It may also happen that the delivered IDDCA is required with a cooler controller with modified interfaces;

Therefore, one shall always refer to the amendments pages of the contractual TDP (if any) to check for modifications that would amend and/or modify paragraphs of this document.

2. LIST OF ACRONYMS USED IN THIS DOCUMENT

DC	Direct Current
FPA	Focal Plane Array
I/O	Inputs / Outputs
IDDCA	Integrated Detector Dewar Cooler Assembly
IR	Infra Red
TDP	Technical Data Package

Note:

The yellow warning sign  highlights an important note to be respected.

The red warning sign  highlights an important note that could lead to a permanent failure if not respected.

3. COOLER DRIVE GENERAL DESCRIPTION

As described in the synopsis of Figure 4.1, COOLER DRIVE has the 3 major following functions:

- DC brushless motor interface.
- Temperature sensor biasing and measurement.
- Temperature regulation control.

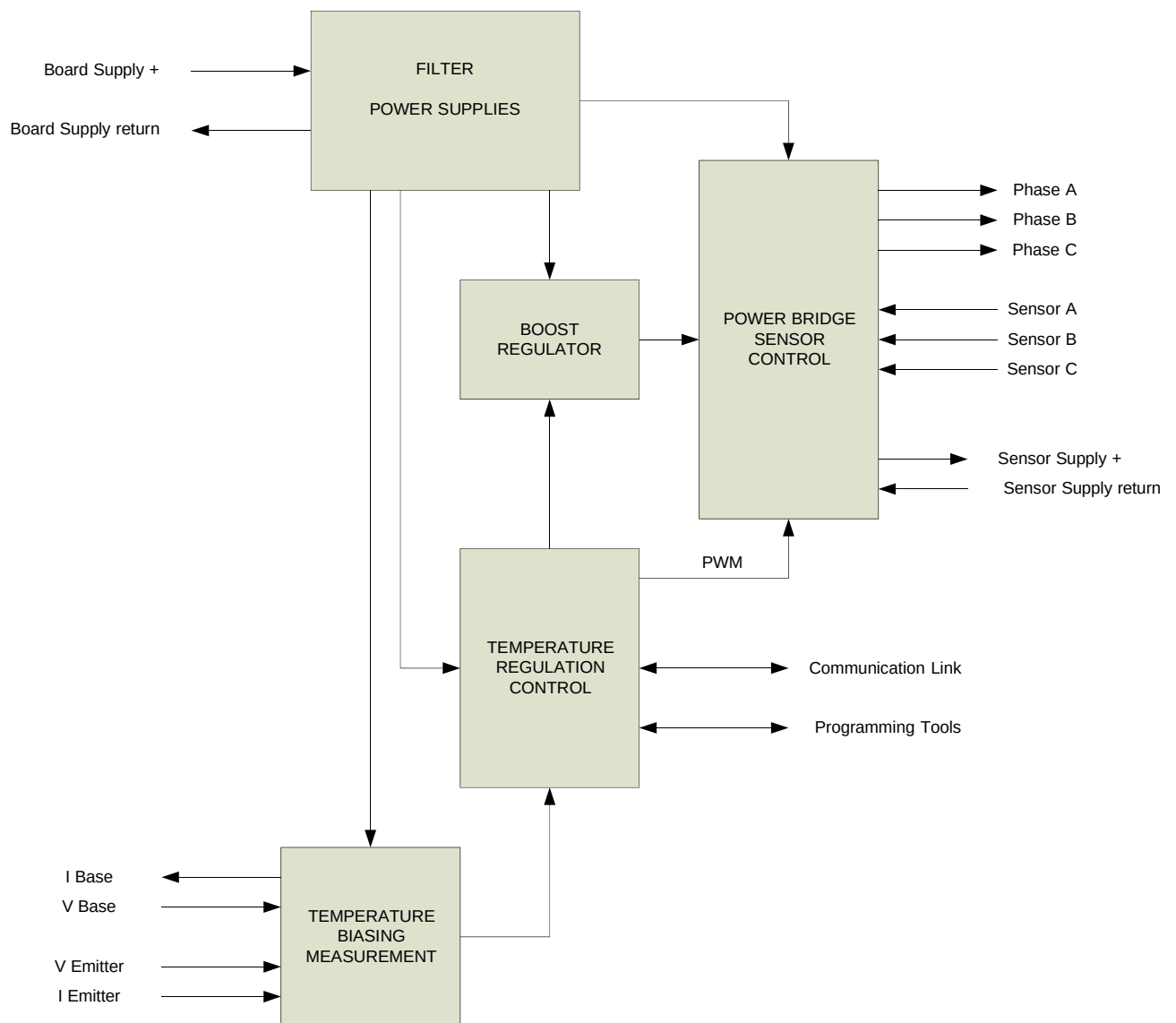


Figure 4.1: COOLER DRIVE Electronic Synopsis

4. TEMPERATURE REGULATION LOOP CONTROL

4.1 FPA regulation temperature setting point

One temperature sensor of the IDDCA is provided to feed the cooler controller for the temperature regulation.

- The sensor anode and cathode have to be connected to the cooler controller electrical interface as described in § 7.3.2.

One other temperature sensor of the IDDCA is provided to monitor the detector temperature.

- Refer to the FPA electrical interface document provided in the TDP, for the full description of the temperature sensor interface (current and $T=f(V)$ function).

4.2 Standard temperature set point

The regulation temperature of COOLER DRIVE is programmed at NCRIEO during Acceptance Test, for the associated delivered IDDCA.

Refer to the Acceptance Test Report (ATR) of the delivered IDDCA for the FPA temperature set during acceptance test and for which the electro-optical performances have been recorded.

4.3 Modification of the FPA regulation temperature setting point

The modification of the FPA regulation temperature setting point is realized by reprogramming COOLER DRIVE with serial communication between the COOLER DRIVE and the customer's system.



Changing the FPA regulation setting point will change the IDDCA performances, and the Electro-Optical and cryogenic performances may not be guaranteed anymore.

The standard cable delivered with COOLER DRIVE does not implement this serial communication lead, and therefore prevents modification of the FPA regulation temperature.



If the customer wishes to change the FPA temperature set by NCRIEO during acceptance test, please contact NCRIEO

See § 7.3.3 and §9 for more details on the serial communication with COOLER DRIVE.

5. COOLER DRIVE MECHANICAL INTERFACE

5.1 Size

The size of the controller board is $63.4 \times 43 \times 17\text{mm}$.

The outline dimensions of the COOLER DRIVE controller are given in Figure 5.1:

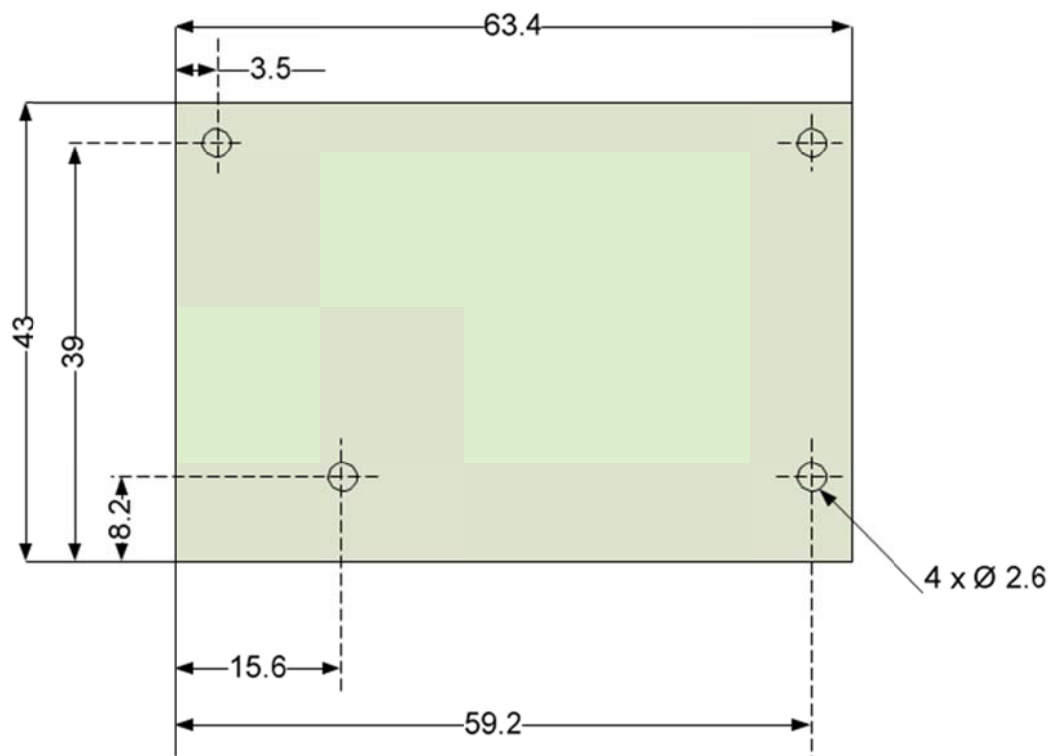


Figure 5.1: COOLER DRIVE dimensions

5.2 Weight

The weight of the controller is

- less than 30g for the board alone (typical $27\text{g} \pm 2\text{g}$).
- less than 30g for the cable and PCB connector (typical $26\text{g} \pm 2\text{g}$).

5.3 Mechanical attachment

The controller shall be attached in the system by four $\text{Ø}2.5\text{ mm}$ screws fitting $\text{Ø}2.6\text{ mm}$ screw holes, located as shown in Figure 5.1.

These holes are electrically connected together and each of these holes is also connected to GND through a 0 Ohm resistor component. If it is requested to have GND and

mechanical ground separated it is necessary to use insulating fixing material.

5.4 Connector J1

The controller is equipped with a 26 pin connector ref. HARWIN M80-5102642 for connection of the leads towards the cooler and the system.

J1 connector can be mated with the following connectors:

- NICOMATIC 222C26M16
- HARWIN M80-4602642

The J1 connector pinout is defined in § 7.3.

5.5 Leads length

The controller leads lengths are the following:

- Motor supply leads (2 × 2 wires): ~350 mm ±20mm
- Temperature sensors leads (4 shielded wires) : ~350 mm ±20mm
- Motor leads (8 wires): 220 mm ±20mm to PCB.
- Communication leads (3 wires): **Optional (not implemented on the standard cable)**



**COOLER DRIVE
Cable**

6. COOLER DRIVE ELECTRICAL INTERFACE

6.1 Input voltage

The DC input voltage of the controller is: +7.5V to +12V. The nominal input voltage is +9V. The maximum input current is 2A in steady state.

6.2 Motor control

6.2.1 Power supply

The controller generates a variable high DC voltage supply for the power stage from the low DC input voltage thanks to a controlled boost regulator.

6.2.2 Power stage

The power stage is a DMOS driver for 3 phases DC brushless motor. Its characteristics are the following:

	XD-2		
Temp (°C)	-40	22	71
Stability (mK)	600		
Pmax DC (W)	11.0	12	14
Preg DC (W)	2.5	4	6
Efficiency	0.70	0.78	0.82
Sigma TFPA (mK)	13	7	6

Note: these figures have to be considered only for information and depend of the cryostat.

6.2.3 Hall effect sensors

The controller supplies the Hall effect sensors with a fixed 6.1V DC voltage.

The current sinking is lower than 30 mA.

The motor excitation sequence and sensor output logic are given in table of Table 6.1.

Excitation step		1	2	3	4	5	6	1
Motor leads	A	+	+		-	-		+
	B	-		+	+		-	-
	C		-	-		+	+	
Sensor output	A	1	1	0	0	0	1	1
	B	0	1	1	1	0	0	0
	C	0	0	0	1	1	1	0

Table 6.1: Motor excitation sequence and hall effect sensor output logic

- 1 = high logic level.

- 0 = low logic level.
- + = the upper MOSFET of the bridge is ON.
- - = the lower MOSFET of bridge is ON.

6.3 COOLER DRIVE J1 connector electrical Pinout

The COOLER DRIVE controller pin-out is presented in Figure 6.2, in its complete definition with the communication leads for programming the controller at NCRIEO during Acceptance Test.



The default cable delivered with COOLER DRIVE does **not** implement the communication leads.

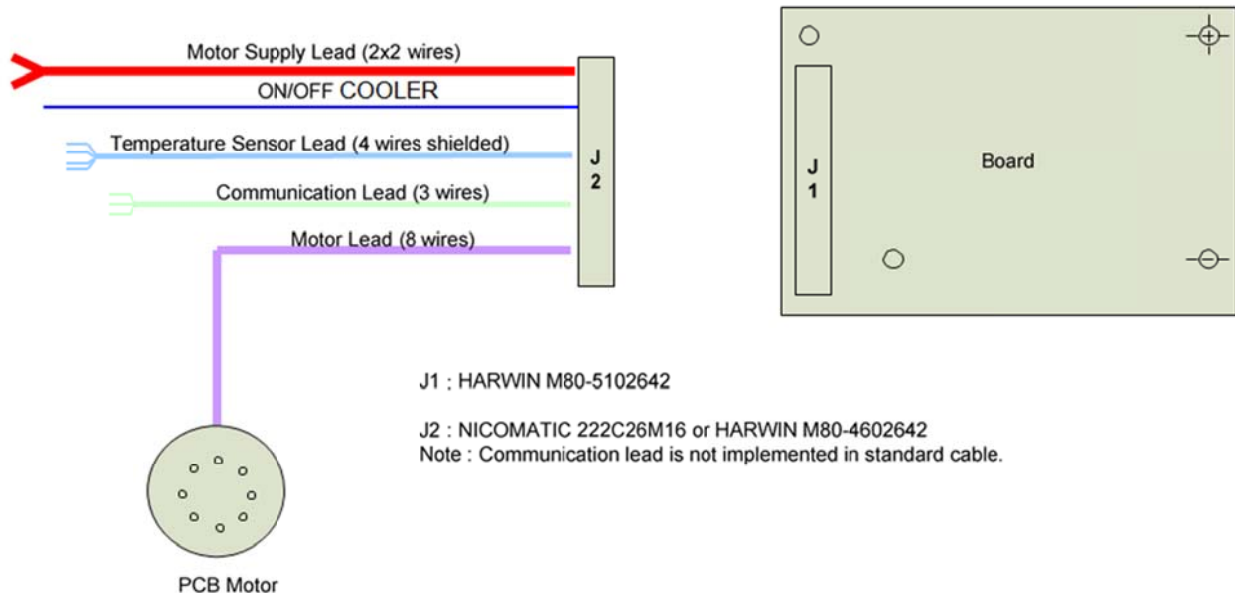


Figure 6.2: COOLER DRIVE electrical connections

The pin out of connector J1 is defined as follows :

J1 Pin number	Signal Name	Description
1, 14	ALIM+9V	Board positive power supply
2, 15	GND	Board power supply return
6	ON/OFF COOLER	On when Open, OFF when closed to GND
3	I+DT	Temperature sensor bias current input
4	I-DT	Temperature sensor bias current return
16	V+DT	Temperature sensor voltage + input
17	V-DT	Temperature sensor voltage - input
18	Chassis	Mechanical ground

J1 Pin number	Signal Name	Description
7	TX	Communication Link output (optional)
5	GND	GND (optional)
20	RX	Communication Link input (optional)
8	Not Connected	NCRIEO reserved DO NOT connect
21	Not Connected	
9	Not Connected	
22	Not Connected	
19	Not Connected	
10	+7V1	Hall effect sensor Supply +
23	0V	Hall effect sensor Supply return
11	IN1	Hall effect sensor position A
24	IN2	Hall effect sensor position B
12	IN3	Hall effect sensor position C
25	PHA	Motor phase A
13	PHB	Motor phase B
26	PHC	Motor phase C

6.3.1 Power supply leads

The four power supply leads are defined as follows:

J1-6	Motor Supply Lead	BLUE	MAF ON/OFF
J1-1		RED	Supply +
J1-14		RED	
J1-2		BLACK	Supply Return
J1-15		BLACK	

6.3.2 Temperature sensor leads

As stated in § 5.1, one of the two IDDCA temperature sensors must be connected to COOLER DRIVE for ensuring the FPA temperature regulation loop

If this connection is not performed, the cooler will not be able to regulate the Focal Plane Array temperature.

COOLER DRIVE controller is in charge of reading the temperature sensor bias with its own 100µA current source.

6.3.2.1 Case 1: direct connection to IDDCA

Connection to the chosen IDDCA temperature sensor is ensured by 2 wires :

- *Temperature sensor (+)* red wire must be connected to the Anode (base) pin DTA of one IDDCA temperature sensor,
- *Temperature sensor (-)* black wire must be connected to the cathode (emitter) pin DTK of the same IDDCA temperature sensor.

Connection between COOLER DRIVE connector and temperature sensor shall be made as follows:



6.3.2.2 Case 2: connection through the IDDCA C&CE

The IDDCA can be delivered with a NCRIEO FPA Command and Control Electronics (C&CE).

In that case, the NCRIEO C&CE provides the temperature sensor electrical interface through a 3 pin NICOMATIC connector referenced 101R03LS (Male Connector 1x3 pins Single row 90° SMT).

Then, the COOLER DRIVE leads shall be ended by a NICOMATIC connector referenced 102.S.03 (version crimp assembly for cable AWG=#24 - #28).

Others connector versions are possible (PCB/SMT, 90°, Straight...), see

NICOMATIC or HARWIN website for more information.

The NICOMATIC connector shall be according to the following pinout definition:

Pin	Name	Range	Function
1	DTA	0 to 5V	Anode of FPA Diode temperature sensor
2	GND	0 V	GND reference and shield
3	DTK	0 to 5V	Cathode of FPA diode temperature sensor

Figure 3 : temperature sensor Connector Pinout

6.3.3 Communication leads

The 3 communication leads are defined as follows:

J1-7	Communication Lead	TX (from board)
J1-5		GND
J1-20		RX (to board)

These signals are not available in the standard cable.



In case the focal plane temperature set point needs to be modified, the cable has to be completed by the customer to add these three signals, or a modified cable may be provided by NCRIEO. In any case, please contact NCRIEO (see § 5.3).

This communication is according to RS232 standard.

- Baud rate : 19200 baud.
- Coding : 8 bits without parity and 1 stop bit.

6.3.4 Motor leads, PCB connection to Cooler

The 8 cooler motor leads are defined as follows:

J1-10	Motor	BLACK	P7V1
J1-23		BLACK	RET_P7V1
J1-11		BLACK	IN1
J1-24		BLACK	IN2
J1-12		BLACK	IN3
J1-25		BLACK	PHA
J1-13		BLACK	PHB
J1-26		BLACK	PHC

The 8 motor leads are connected to a circular PCB.

This PCB is compliant with the cooler electrical interface defined by eight pins described in

Figure 6.5, with the following characteristics:

Cooler	XD-2 cooler
Pin diameter	$\varnothing 1\text{mm} \pm 0.03\text{mm}$
Pin length	$4\text{mm} \pm 0.2\text{mm}$
Pin angular position	at 45° on a $\varnothing 13.2\text{mm}$ diameter

The pinout of the three phase brushless motor is described by Figure 6.4 and Figure 6.5.

Pin number	Signal description
E1	motor lead B
E2	positive supply of the Hall effect sensor
E3	position sensor A
E4	position sensor C
E5	position sensor B
E6	ground
E7	motor lead A
E8	motor lead C

Figure 6.4: Pin out description of the three phases brushless motor

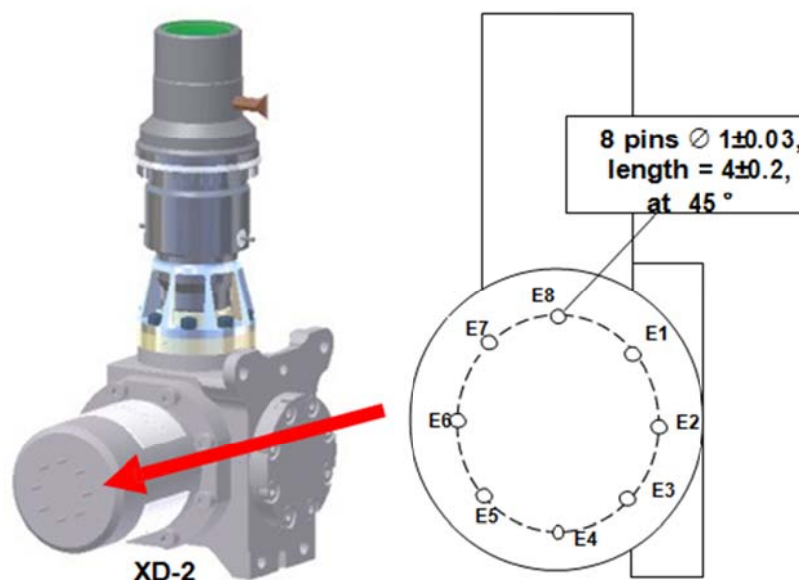


Figure 6.5: Pin out location of the three phases brushless motor

7. USER'S CAUTIONS

7.1 COOLER DRIVE connection on the cooler

The electrical connection is insured by a **direct clamp** of the PCB onto the microcooler pins.

The clamp of the PCB onto the cooler shall always be made with the **controller board OFF**.

The cooler pins are fragile and the force put on them must be minimized. A bent or twisted pin on the cooler cannot be inserted in the PCB COOLER DRIVE connector and can easily break if insertion is not performed with care. A strong effort on the pins could irreparably damage the cooler.

The mounting and dismounting of the controller PCB connection shall be performed with great care, and shall be minimized.



The Hall Effect sensors of the cooler electrical engine will be damaged by an inappropriate electrical connection. It is therefore mandatory to connect the controller PCB with great care in order to ensure a connection compliant with the electrical specifications.



The cooler pins location being symmetrical by 45° step, great care shall be taken when plugging the controller PCB connector in order to avoid an incorrect plug due to a 45° angle error.

Pin n°8 is the one closest to cold finger, as shown in Figure 6.5.

It is forbidden to solder the PCB on the cooler pins.

7.2 Cautions against electrostatic discharges (ESD)

Such as any electronic component, the COOLER DRIVE controller is sensitive to electrostatic discharges. The following cautions are therefore specified when mounting the controller in a system:

- Clean room with humidity level between 40% and 60%;
- Handling of the COOLER DRIVE controller on a static free work station equipped with grounded carpet;
- ground the electronic power supply before plugging it onto the COOLER DRIVE

controller;

7.3Cautions on electrical interface

Refer to § 4 for details on electrical interface to be respected.

7.4 Interchangeability

The COOLER DRIVE controller is interchangeable without any specific adjustment with all controllers of same reference.

8. ENVIRONMENTAL CONDITIONS

The design of COOLER DRIVE has been oriented to sustain hard environmental conditions (temperatures, vibrations), but no rugged module packaging is provided by default.

8.1 Temperature

The non operational COOLER DRIVE withstands without degradation the following conditions:

- Storage temperature, low: -54 °C, DIN ISO 9022-10-08-0 (MIL-STD-810E Method 502.3, Procedure I)
- Storage temperature, high: +85 °C , DIN ISO 9022-11-05-0 (MIL-STD-810E Method 501.3, Procedure I)

COOLER DRIVE is operational under the following conditions:

- Operating temperature, low: -46 °C , DIN ISO 9022-10-07-2 (MIL-STD-810E Method 502.3, Procedure II)
- Operating temperature, high: +71 °C , DIN ISO 9022-11-04-2 (MIL-STD-810E Method 501.3, Procedure II)

(End of the document)

Annex 7

Title: Drawings of MINI640 IDDCA

(with 15um pixel pitch and XD-2 Stirling cooler)

Reference: DRAW-MINI640-IDDCA-1801

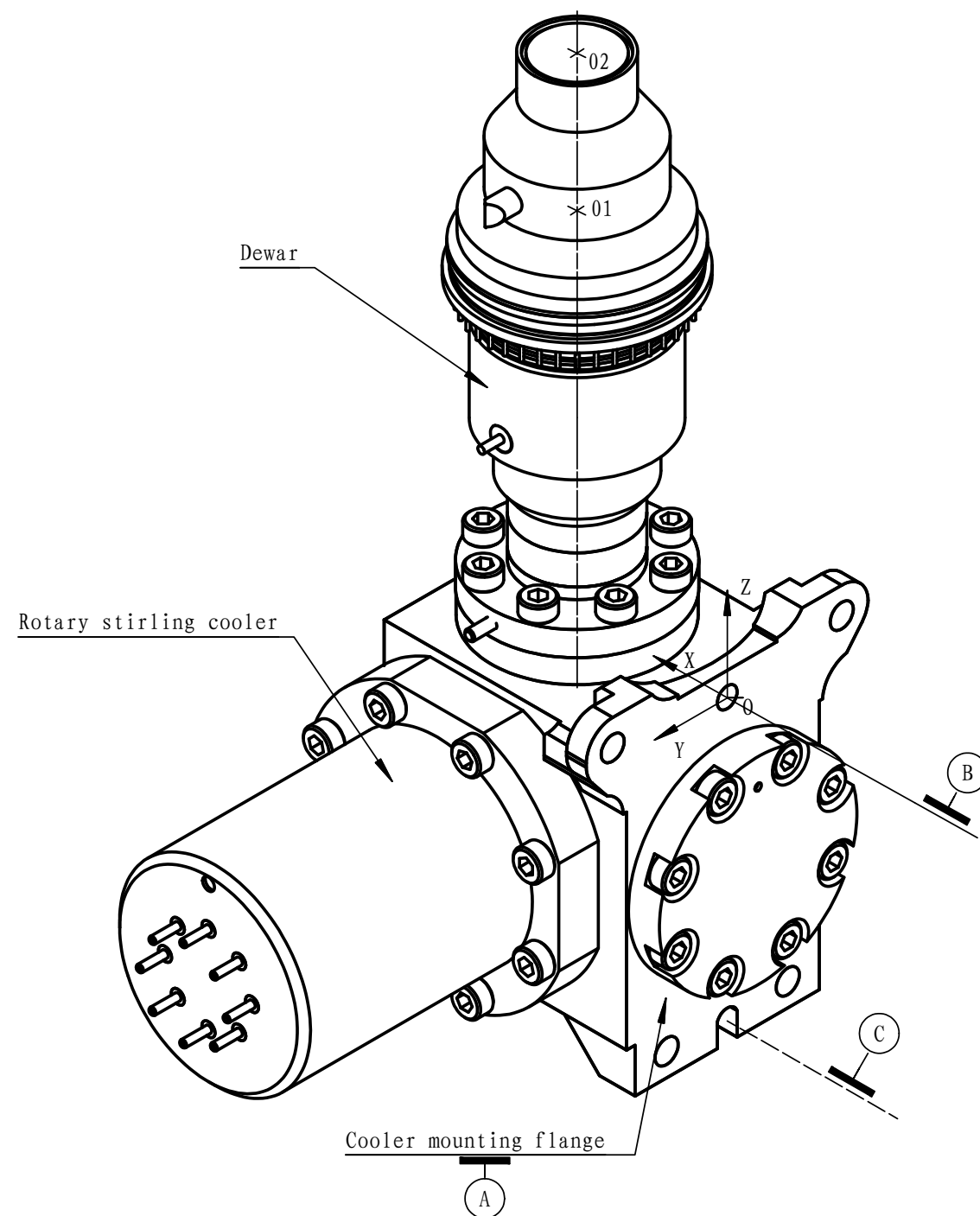
Issue 1 Revision 1

North China Research Institute of Electro-optics

(NCRIEO)

JAN, 2018

[illegible]



Note:

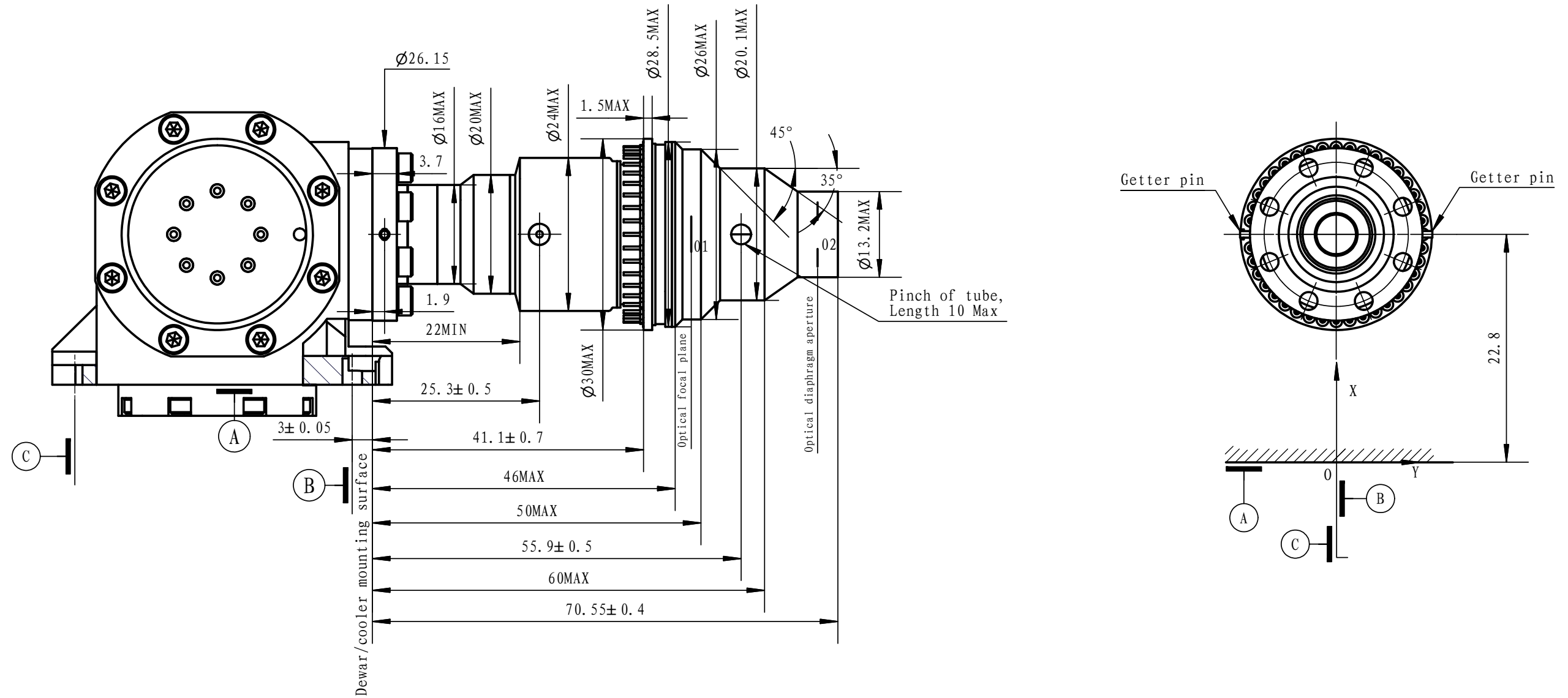
1-Mechanical references XY0Z are materialized by:

- YZ: Plane of cooler mounting flange;
- Z: Datum axis oriented by $\varnothing B$ and $\varnothing C$ centers;
- Y: Perpendicular to Z axis through $\varnothing B$ center;
- X: Normal to YZ plane;
- O: XYZ axis center.

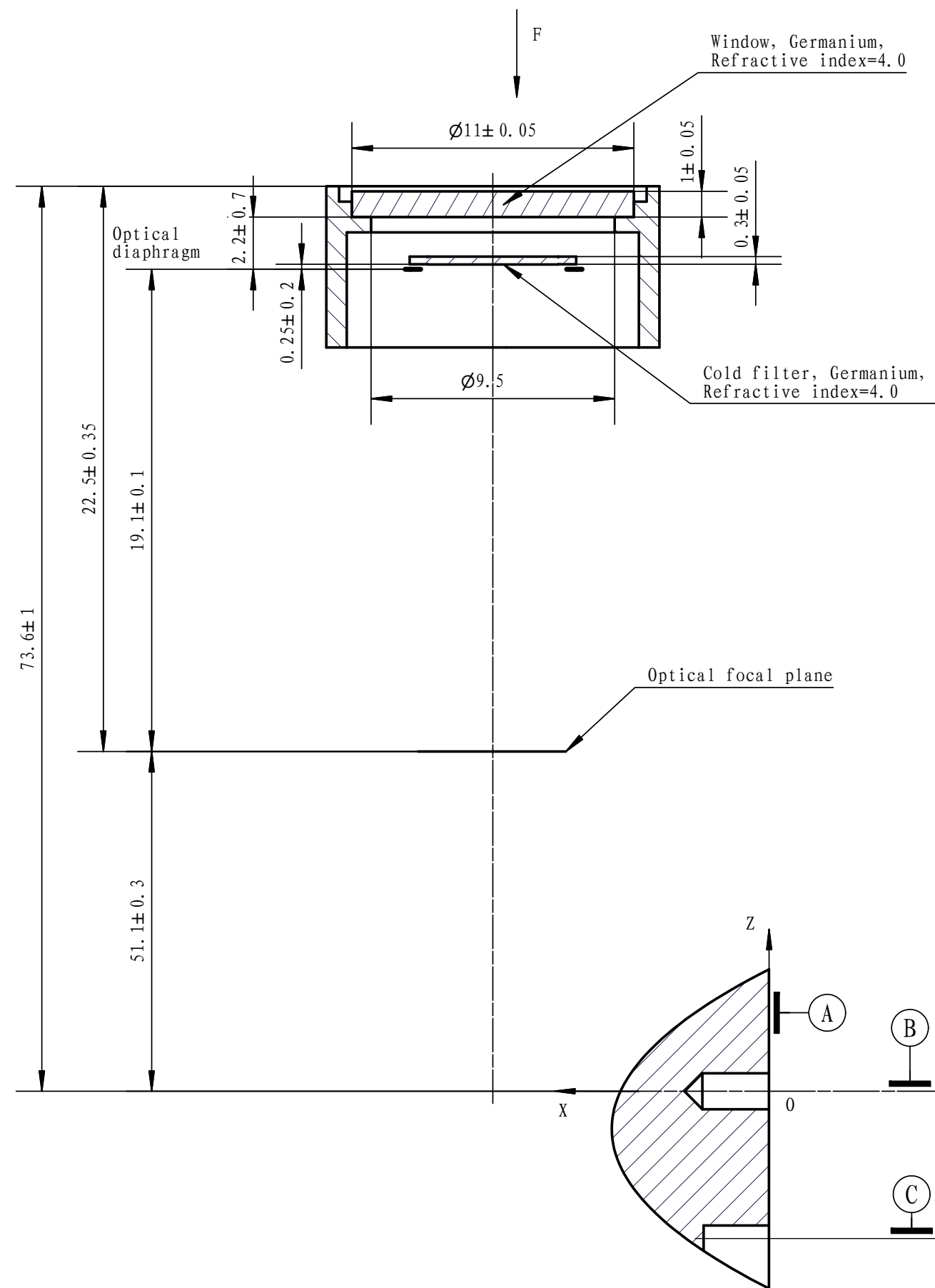
2-01: Optical focal plane center.

3-02: Optical diaphragm aperture center.

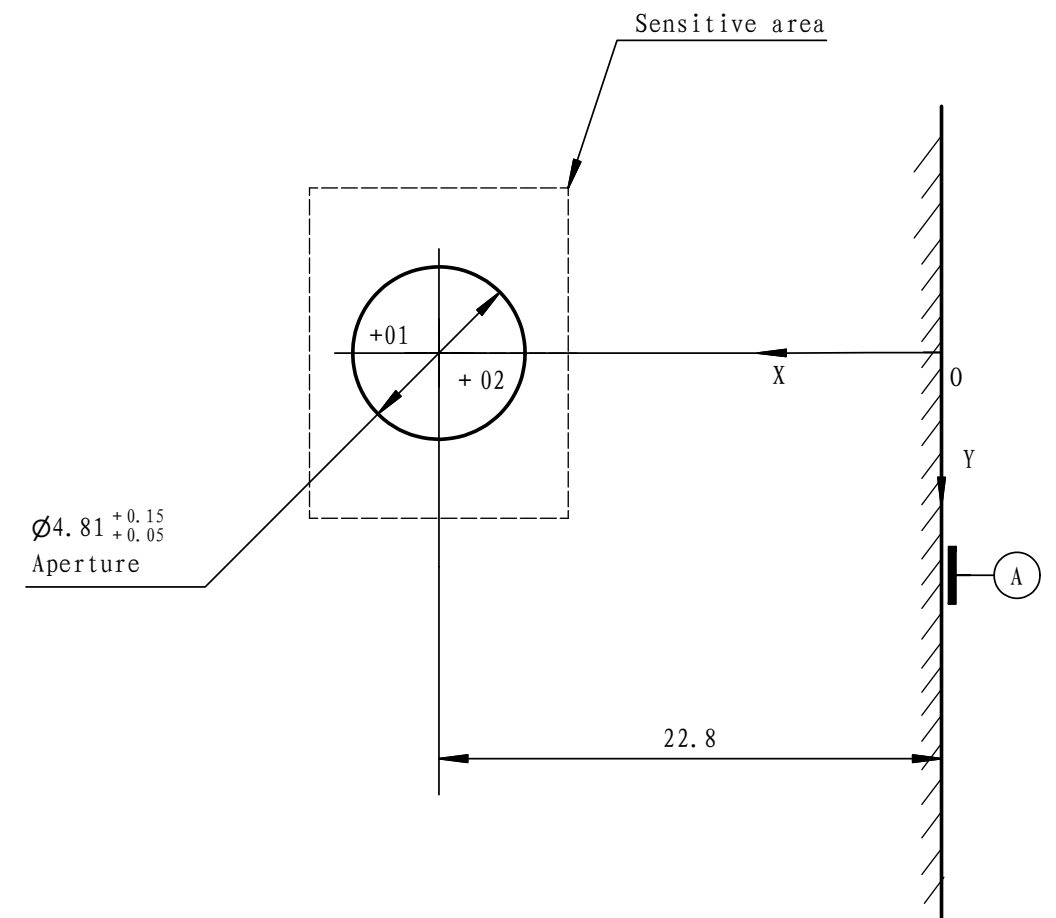
640 × 512-15 μm IDDCA Mechanical Interface	01/06	
	Scale	1.5: 1
2020102		



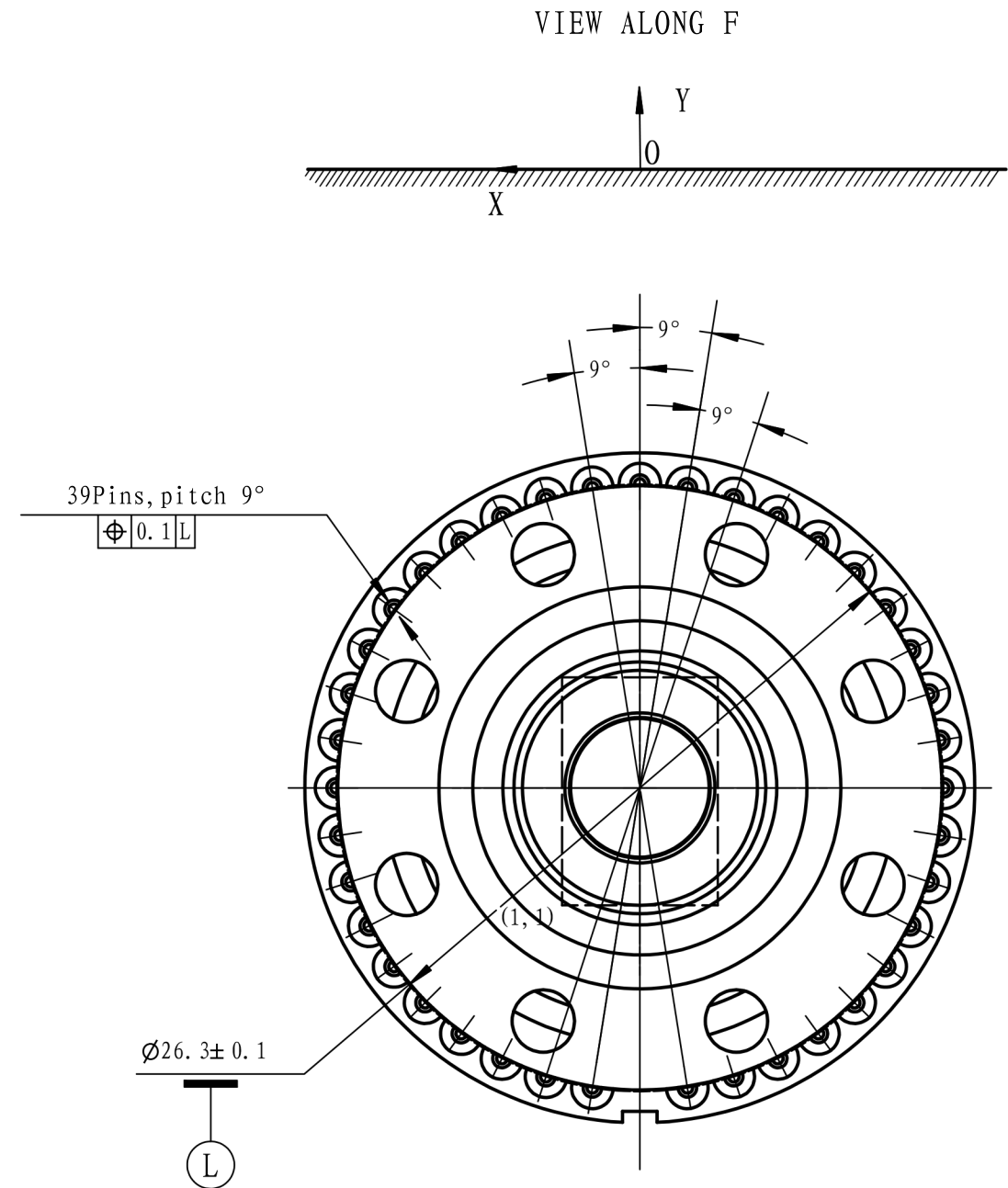
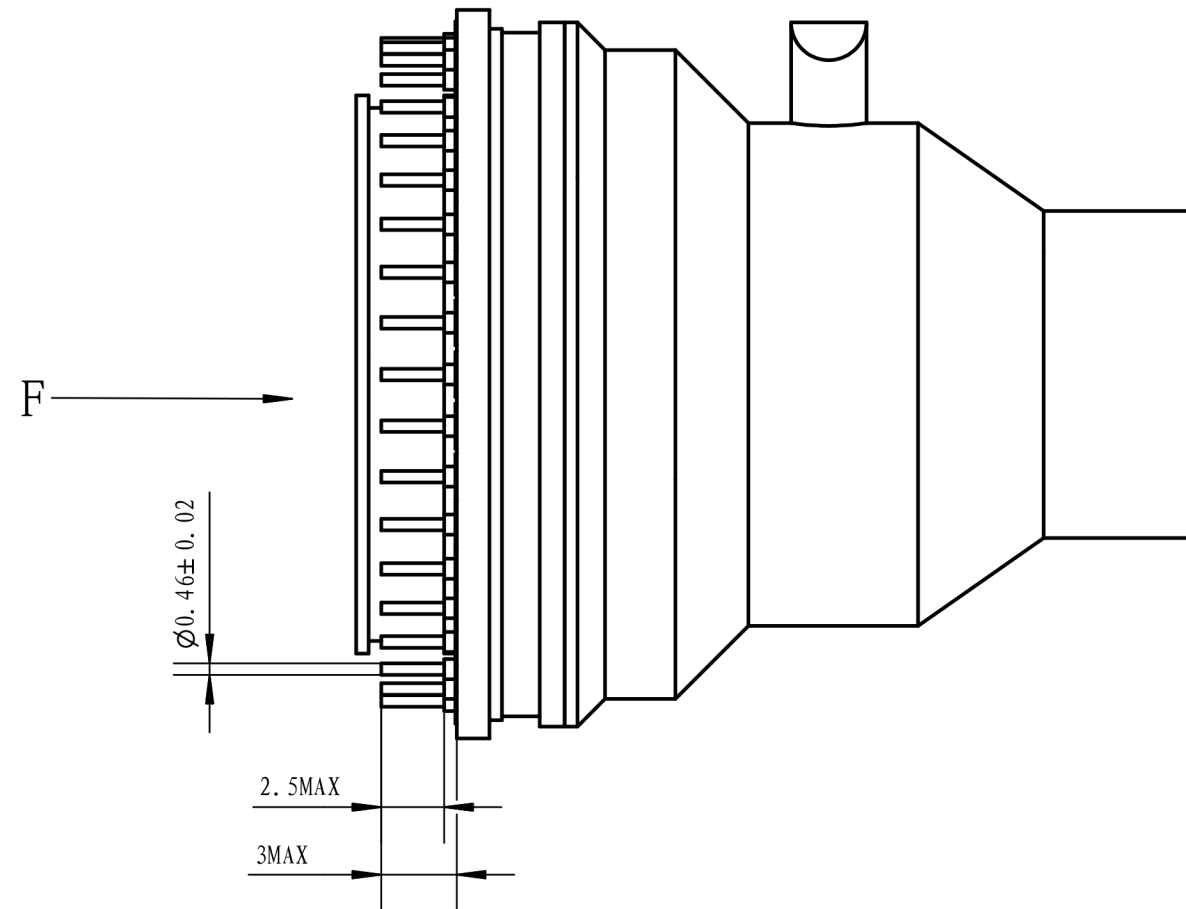
640 × 512-15 μm IDDCA Mechanical Interface	02/06	
	Scale	1.5: 1
2020102		



VIEW ALONG F (Optical Diaphragm Aperture)

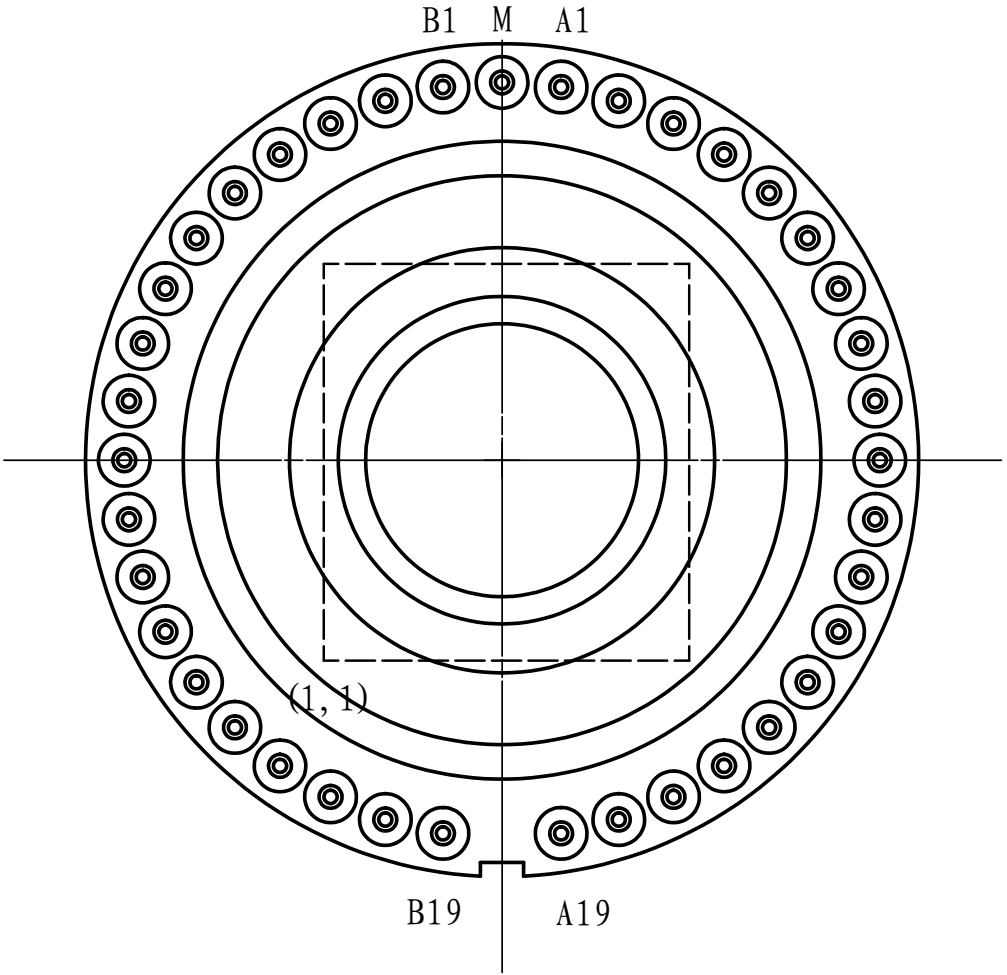


640 × 512-15 μm IDDCA Optical Interface	03/06	
	Scale	5: 1
2020102		



640 × 512-15 μm IDDCA Electrical Interface	04/06	
	Scale	4: 1
2020102		

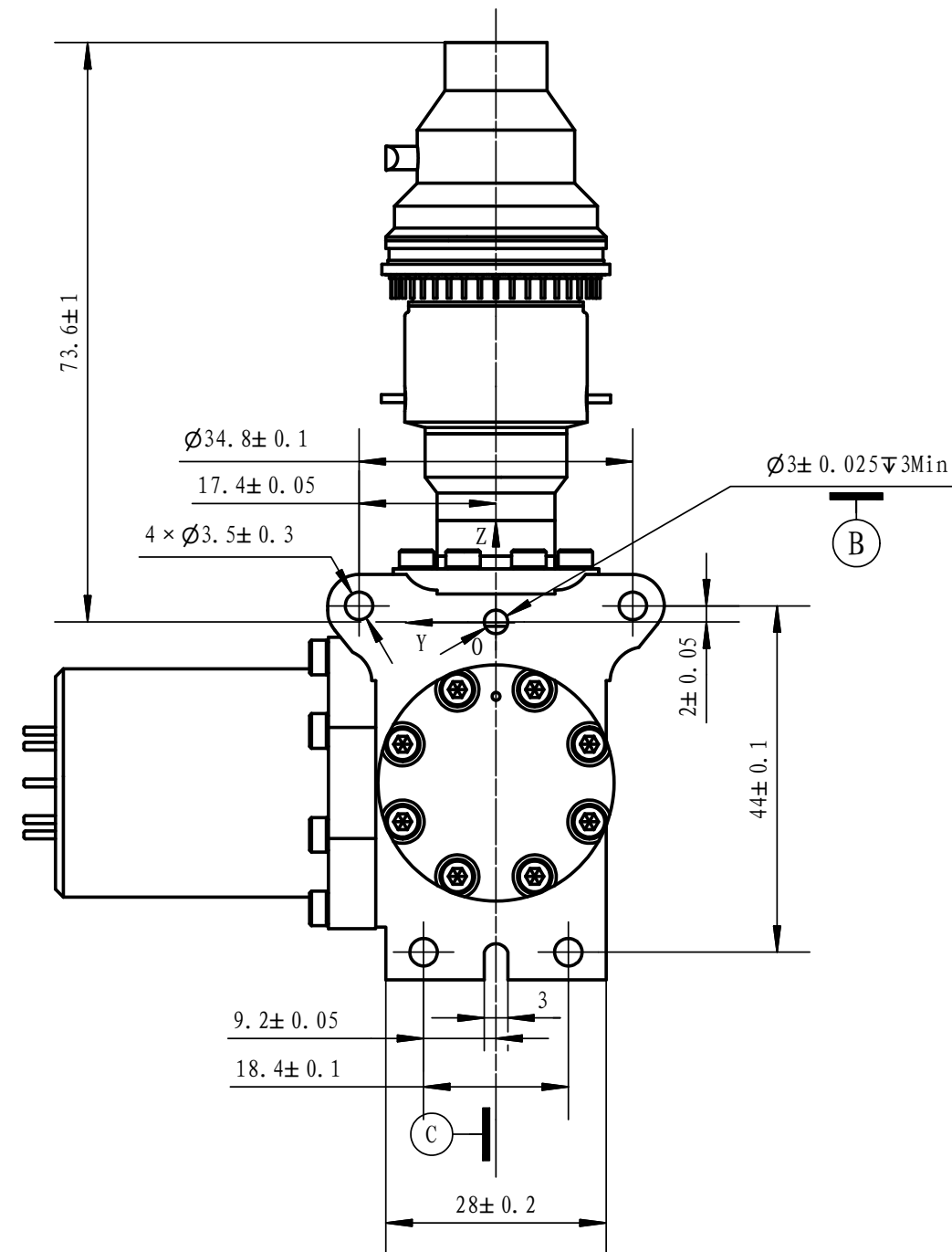
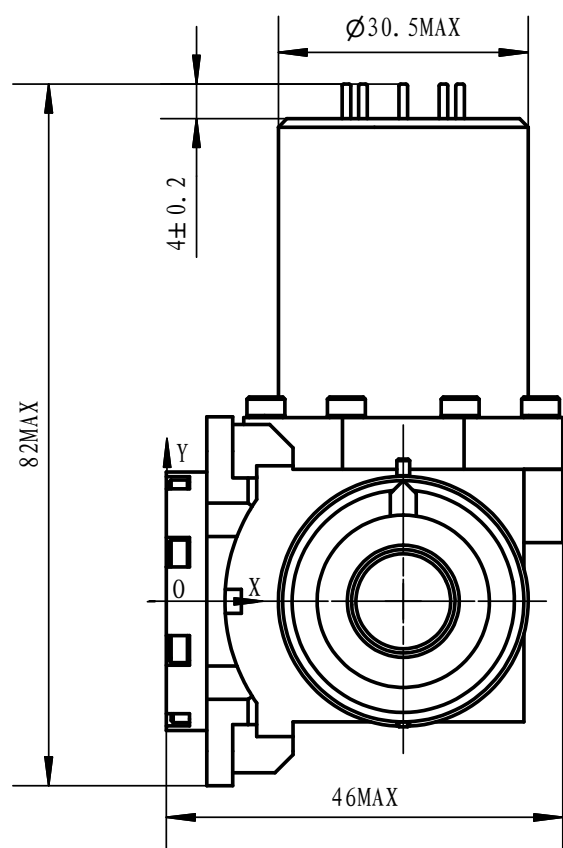
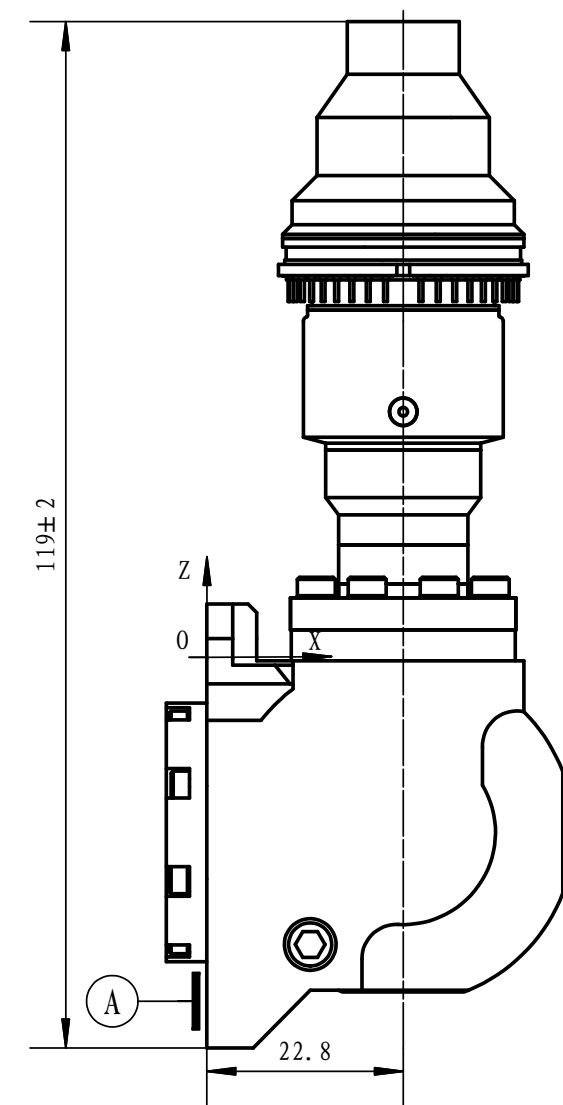
REAR VIEW OF DEWAR
(pins side)



PINS N°	FUNCTIONS	PINS N°	FUNCTIONS
A1	(*) NC 	B1	(*) NC 
A2	BIN	B2	(*) NC 
A3	VDD	B3	COLDSHIED GND 
A4	ERROR	B4	(*) DTK2-
A5	DATAVALID	B5	(*) DTA2+
A6	UPROW	B6	SUBPV
A7	UPCOL	B7	GND
A8	SIZEB	B8	VDD0
A9	SIZEA	B9	IBP
A10	INT	B10	OUT3
A11	MC	B11	OUT2
A12	SERCLR	B12	OUT1
A13	SERDAT	B13	OUT0
A14	ITR	B14	GND
A15	GNDD	B15	VR
A16	(*) DTK1-	B16	VDDA
A17	(*) DTA1+	B17	GPOL
A18	(*) NC 	B18	(*) NC 
A19	(*) NC 	B19	(*) NC 
M	(*) Dewar Ground 		

Note:
(*) Functions which are not connected to FPA processor
 To be connected to electrical ground
 To be connected to deware ground

640 × 512-15 μm IDDCA Electrical Interface	05/06	
	Scale	4: 1
2020102		



640×512-15 μm IDDCA Mechanical Interface	06/06	
	Scale	1.2: 1
2020102		